

# Quantum Mechanics Note by CHESTNUT

He Yingqiu

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# Preface

The notes begin with a review of Classical Mechanics and Classical Electrodynamics, providing a background and an introduction for comparison and mastery from an overview. For a more in-depth study of Classical Mechanics, you can refer to my other notes, which are also based on a video uploaded on Bilibili. I find the teaching in this video to be exceptional. Following this review, the notes progress to fundamental concepts such as representation and symbols.

Reference books:

- Modern Quantum Mechanics (Third Edition) by J.J. Sakurai and Jim Napolitano

Reference video:

- [Peking University - Quantum Mechanics Lecture \(Guo Hong\)](#)

Feel free to communicate with me about anything you would like. You can contact me at [heyinqiu02@gmail.com](mailto:heyinqiu02@gmail.com). Thanks again for visiting!

Yingqiu He 🍒  
Zhejiang, home  
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# Chapter 1

## Preliminaries

### 1.1 Classical Mechanics

#### Newtonian Mechanics

Euclidean space (physical space)

Cartesian coordinate:  $\vec{x}$

- Momentum force

$$\frac{d\vec{p}}{dt} = \vec{F}; \quad \vec{p} = m\vec{v}$$

- Angular momentum torque

$$\frac{d\vec{L}}{dt} = \vec{N}; \quad \vec{L} = \vec{x} \times \vec{p}$$

- Relation

$$\vec{N} = \vec{x} \times \vec{F}$$

*Proof.*  $\frac{d\vec{L}}{dt} = \frac{d}{dt}(\vec{x} \times \vec{p}) = \frac{d\vec{x}}{dt} \times \vec{p} + \vec{x} \times \frac{d\vec{p}}{dt} = \vec{v} \times m\vec{v} + \vec{x} \times \vec{F}$

□

Mass point - mass point system - rigid body

- Inertia of mass:  $m$ ; Inertia of rotation:  $mr^2$

#### Lagrangian Mechanics

$3N$  dependent coordinates  $x_1, \dots, x_{3N} \rightarrow D$  independent generalized coordinates  $q_1, \dots, q_D$ ,  $\partial q_i / \partial q_j = 0$  ( $i \neq j$ )

- Holonomic constraint  $f(x, t) \equiv 0$
- Non-holonomic constraint  $f(x, \dot{x}, t) \equiv 0$

#### Coordinates and Space

- Generalized coordinates  $q_\alpha(t)$  ( $\alpha = 1, \dots, D$ )  $D \leq 3N$
- in Configuration space (Abstract space)
- Functional

– e.g. Lagrangian:  $L(q_\alpha(t), \dot{q}_\alpha(t); t)$ , Hamiltonian:  $H(q_\alpha(t), p_\alpha(t); t)$ , Action:  $S[q(t)] = \int_{t_1}^{t_2} L dt$

– Dimensions of these functionals:

$$[L] = [H] = [\text{Energy}], [S] = [\text{Energy}] \cdot [\text{Time}] = [\text{Length}] \cdot [\text{Momentum}] = [\hbar]$$

Table 1.1: Comparison

|                       | Space               | Coordinates                | dimensional |
|-----------------------|---------------------|----------------------------|-------------|
| Newtonian Mechanics   | Physical Space      | $\vec{x}_\alpha(t)$        | 3N          |
| Lagrangian Mechanics  | Configuration Space | $q_\alpha(t)$              | D           |
| Hamiltonian Mechanics | Phase Space         | $q_\alpha(t), p_\alpha(t)$ | 2D          |

### Principle of Least Action

Consider  $f(q, \dot{q}, t)$ ,

- Differentiate:  $\Delta f = f(x_2) - f(x_1) = \frac{\partial f}{\partial q} \Delta q + \frac{\partial f}{\partial \dot{q}} \Delta \dot{q}$
- Differential:  $\Delta f \xrightarrow{\Delta x \rightarrow 0} df = \frac{\partial f}{\partial q} dq + \frac{\partial f}{\partial \dot{q}} d\dot{q}$
- Variation:  $df \xrightarrow{d \rightarrow \delta} \delta f = \frac{\partial f}{\partial q} \delta q + \frac{\partial f}{\partial \dot{q}} \delta \dot{q}$

Usually  $\delta q(t_1) = \delta q(t_2) = 0$ ,  $\delta t = 0$  and use  $\delta \dot{q} = \delta(dq/dt) = d(\delta q)/dt$ , principle of least action tell us

$$\begin{aligned}
0 \equiv \delta S &= \delta \int_{t_1}^{t_2} L(q, \dot{q}, t) dt = \int_{t_1}^{t_2} \left( \frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right) dt = \int_{t_1}^{t_2} \left\{ \frac{\partial L}{\partial q} \delta q + \frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}} \right) \delta q - \frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}} \right) \delta q \right\} dt \\
&= \int_{t_1}^{t_2} \left( \frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right) \delta q dt + \left. \frac{\partial L}{\partial \dot{q}} \delta q \right|_{t_1}^{t_2} \Rightarrow
\end{aligned}$$

Euler-Lagrangian's Equation:

$$\frac{\partial L}{\partial q_\alpha} = \frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}_\alpha} \right) \quad (\alpha = 1, \dots, D)$$

### Hamiltonian Mechanics

#### Hamilton's Canonical Transformations

$$\dot{q}_\alpha = \frac{\partial H}{\partial p_\alpha}, \quad \dot{p}_\alpha = -\frac{\partial H}{\partial q_\alpha}$$

Legendre's Transformation:

$$\begin{cases} p_\alpha = \frac{\partial L}{\partial \dot{q}_\alpha}, \\ H(q, p, t) = \sum_\alpha p_\alpha \dot{q}_\alpha - L(q, \dot{q}, t) \end{cases}$$

Methods of proof:

1. since  $H$  is  $\dot{q}$ -independent, and  $L$  is  $p$ -independent, then

$$\frac{\partial H}{\partial \dot{q}_\alpha} = p_\alpha - \frac{\partial L}{\partial \dot{q}_\alpha} \equiv 0, \quad \frac{\partial L}{\partial p_\alpha} = \dot{q}_\alpha - \frac{\partial H}{\partial p_\alpha} \equiv 0, \quad \text{and} \quad \dot{p}_\alpha = \frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}_\alpha} \right) = \frac{\partial L}{\partial q_\alpha} = -\frac{\partial H}{\partial q_\alpha}$$

2.  $dH = \frac{\partial H}{\partial t} dt + \frac{\partial H}{\partial q} dq + \frac{\partial H}{\partial p} dp$ , on the other hand  $dH = d(p\dot{q} - L(q, \dot{q}, t)) = \dot{q}dp - \dot{p}dq - \frac{\partial L}{\partial t} dt$

3.  $0 \equiv \delta S = \delta \int (p\dot{q} - H(q, p, t)) dt$  (under)

### Poisson's Bracket (Classical Canonical Commutator)

$\forall f(q, p, t)$ , then

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial q_\alpha} \dot{q}_\alpha + \frac{\partial f}{\partial p_\alpha} \dot{p}_\alpha = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial q_\alpha} \frac{\partial H}{\partial p_\alpha} - \frac{\partial f}{\partial p_\alpha} \frac{\partial H}{\partial q_\alpha} = \frac{\partial f}{\partial t} + [f, H]_{\text{PB}}$$

where

$$[f, H]_{\text{PB}} = \frac{\partial f}{\partial q_\alpha} \frac{\partial H}{\partial p_\alpha} - \frac{\partial f}{\partial p_\alpha} \frac{\partial H}{\partial q_\alpha}, \quad \text{or} \quad [A, B]_{\text{PB}} = \frac{\partial A}{\partial q_\alpha} \frac{\partial B}{\partial p_\alpha} - \frac{\partial A}{\partial p_\alpha} \frac{\partial B}{\partial q_\alpha}$$

- $[q_\alpha, p_\beta]_{\text{PB}} = \delta_{\alpha\beta}$ ,  $[q_\alpha, q_\beta]_{\text{PB}} = [p_\alpha, p_\beta]_{\text{PB}} \equiv 0$
- In QM, similarly, Heisenberg's equation and Ehrenfest's theorem are

$$\frac{d\hat{f}}{dt} = \frac{\partial \hat{f}}{\partial t} + \frac{1}{i\hbar} [\hat{f}, \hat{H}] \quad \frac{d}{dt} \langle \hat{f} \rangle = \left\langle \frac{\partial \hat{f}}{\partial t} \right\rangle + \frac{1}{i\hbar} \langle [\hat{f}, \hat{H}] \rangle$$

quantum canonical commutator  $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$

- if  $f(q, p, t) = q_\alpha$ , then  $\dot{q}_\alpha = \frac{\partial q_\alpha}{\partial t} + [q_\alpha, H]_{\text{PB}} = \frac{\partial H}{\partial p_\alpha}$ ; if  $f(q, p, t) = p_\alpha$ , then  $\dot{p}_\alpha = \frac{\partial p_\alpha}{\partial t} + [p_\alpha, H]_{\text{PB}} = -\frac{\partial H}{\partial q_\alpha}$

Properties:

1.  $[A, B]_{\text{PB}} = -[B, A]_{\text{PB}}$
2.  $[A + B, C]_{\text{PB}} = [A, C]_{\text{PB}} + [B, C]_{\text{PB}}$
3.  $[A, BC]_{\text{PB}} = [A, B]_{\text{PB}}C + B[A, C]_{\text{PB}}$
4.  $[A, B^n]_{\text{PB}} = n[A, B]_{\text{PB}}B^{n-1}$
5.  $[A, f(B)]_{\text{PB}} = [A, B]_{\text{PB}} \frac{\partial f(B)}{\partial B}$

### Hamilton-Jacobis Equation

We have discussed principle of least action with fixed  $q(t)$ , now we ask for  $q(t)$  not fixed ( $\delta q(t) \neq 0$ ,  $\delta q(t_0) = 0$ ), then

$$\delta S = \left. \frac{\partial L}{\partial \dot{q}} \delta q \right|_{t_0}^t = p \delta q = \frac{\partial S}{\partial q} \delta q \quad \Rightarrow \quad p = \frac{\partial S}{\partial q}$$

and

$$p\dot{q} - H = L = \frac{dS}{dt} = \frac{\partial S}{\partial t} + \frac{\partial S}{\partial q} \dot{q} = \frac{\partial S}{\partial t} + p\dot{q} \quad \Rightarrow \quad \frac{\partial S}{\partial t} + H(q, \frac{\partial S}{\partial q}, t) = 0$$

## 1.2 Classical Electrodynamics Mechanics

### Vector Analysis

$$\vec{v} = v_1 \vec{e}_1 + v_2 \vec{e}_2 + v_3 \vec{e}_3 = \sum_i v_i \vec{e}_i = v_i \vec{e}_i \quad (i = 1, 2, 3)$$

### Kronecker Tensor

$$\delta_{ij} = \begin{cases} 1, & (i = j) \\ 0, & (i \neq j) \end{cases}$$

- $\vec{e}_i \cdot \vec{e}_j = \delta_{ij}$

### Levi-Civite Tensor

$$\varepsilon_{ijk} = \begin{cases} +1, & (\text{even permutation}) \\ -1, & (\text{odd permutation}) \\ 0, & (\text{no permutation}) \end{cases}$$

- $\vec{e}_i \times \vec{e}_j = \varepsilon_{ijk} \vec{e}_k$
- $\vec{e}_i \cdot (\vec{e}_j \times \vec{e}_k) = \varepsilon_{ijk}$
- $\varepsilon_{ijk} \varepsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}$

**Vector Algebra**

- Dot product (Inner product):

$$\vec{A} \cdot \vec{B} = A_i \vec{e}_i \cdot B_i \vec{e}_j = (A_i B_j)(\vec{e}_i \times \vec{e}_j) = A_i B_j \delta_{ij} = A_i B_i$$

- Cross product:

$$\vec{A} \times \vec{B} = A_i \vec{e}_i \times B_j \vec{e}_j = (A_i B_j)(\vec{e}_i \times \vec{e}_j) = A_i B_j \varepsilon_{ijk} \vec{e}_k = (\vec{A} \times \vec{B})_k \vec{e}_k$$

$$(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C})$$

$$[L_i, L_j]_{PB} = \varepsilon_{ijk} L_k$$

*Proof.*  $\varepsilon_{iab} \varepsilon_{jcd} [x_a p_b, x_c p_d] = \varepsilon_{iab} \varepsilon_{jcd} (x_c p_b \delta_{ad} - x_a p_d \delta_{bc}) = (\delta_{bj} \delta_{ic} - \delta_{bc} \delta_{ij}) x_c p_b + (\delta_{ij} \delta_{ad} - \delta_{id} \delta_{aj}) x_a p_d = x_i p_j - x_j p_i$   $\square$

$$[L_i, \vec{L}^2]_{PB} \equiv 0$$

*Proof.*  $[L_i, L_j] L_j + L_j [L_i, L_j] = \varepsilon_{ijk} (L_j L_k + L_k L_j)$  (Anti-symmetric · Symmetric  $\equiv 0$ )  $\square$

$$[L_i, x_j] = \varepsilon_{ijk} x_k, [L_i, p_j] = \varepsilon_{ijk} p_k$$

**Vector Calculus**

$$(\vec{\nabla})_i = \frac{\partial}{\partial x_i} = \partial_i \quad (i = 1, 2, 3)$$

1. Gradient:

$$\vec{\nabla} \phi = (\partial_i \phi) \vec{e}_i = \frac{\partial \phi}{\partial x_1} \vec{e}_1 + \frac{\partial \phi}{\partial x_2} \vec{e}_2 + \frac{\partial \phi}{\partial x_3} \vec{e}_3$$

2. Divergence:

$$\vec{\nabla} \cdot \vec{A} = \partial_i A_i = \frac{\partial}{\partial x_1} A_1 + \frac{\partial}{\partial x_2} A_2 + \frac{\partial}{\partial x_3} A_3$$

3. Curl:

$$\vec{\nabla} \times \vec{A} = (\vec{e}_i \partial_i) \times \vec{e}_j A_j = \varepsilon_{ijk} \vec{e}_k \partial_i A_j = (\varepsilon_{ijk} \partial_i A_j) \vec{e}_k = (\vec{\nabla} \times \vec{A})_k \vec{e}_k$$

$$\bullet (\vec{\nabla} \times \vec{A})_i = \varepsilon_{ijk} \partial_j A_k$$

Example:

1.  $\vec{\nabla} \times \vec{\nabla} \phi \equiv 0$ ;  $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) \equiv 0$  curl-free for gradient field and divergence-free for curl field

*Proof.*  $(\vec{\nabla} \times \vec{\nabla} \phi)_i = \varepsilon_{ijk} \partial_j (\vec{\nabla} \phi)_k = \varepsilon_{ijk} \partial_j \partial_k \phi \equiv 0$  ( $\vec{E} = -\vec{\nabla} \phi$  static electric field)  
 $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = \partial_i \varepsilon_{ijk} \partial_j A_k = \varepsilon_{ijk} \partial_i \partial_j A_k \equiv 0$  ( $\vec{B} = \vec{\nabla} \times \vec{A}$ ,  $\vec{\nabla} \cdot \vec{B} \equiv 0$ )  $\square$

2.  $\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}$

*Proof.*  $[\vec{\nabla} \times (\vec{\nabla} \times \vec{A})]_i = \varepsilon_{ijk} \partial_j \varepsilon_{klm} \partial_l A_m = (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) \partial_j \partial_l A_m = \partial_j \partial_i A_j - \partial_j \partial_j A_i = \partial_i (\vec{\nabla} \cdot \vec{A}) - \nabla^2 A_i$   $\square$

3.  $\vec{v} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{v} \cdot \vec{A}) - (\vec{v} \cdot \vec{\nabla}) \vec{A}$

*Proof.*  $[\vec{v} \times (\vec{\nabla} \times \vec{A})]_i = \varepsilon_{ijk} v_j \varepsilon_{klm} \partial_l A_m = v_j \partial_l A_j - v_j \partial_j A_i = \partial_i (v_j A_j) - v_j \partial_j A_i$   $\square$

4.  $\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$

*Proof.*  $\vec{A} \cdot (\vec{B} \times \vec{C}) = A_i (\vec{B} \times \vec{C})_i = A_i \varepsilon_{ijk} B_j C_k = (\varepsilon_{jki} C_k A_i) B_j = (\varepsilon_{kij} A_i B_j) C_k$   $\square$



## Generalized Stokes Theorem

$$\int_{\Omega} d\omega = \int_{\partial\Omega} \omega$$

- Newton-Leibniz's formula:

$$\int_a^b df = f(b) - f(a)$$

- Green's theorem:

$$\oint_L f dx + g dy = \iint_S \left( \frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy$$

- Stokes's theorem:

$$\oint_L \vec{F} \cdot d\vec{l} = \iint_S (\vec{\nabla} \times \vec{F}) \cdot d\vec{S}$$

- Gauss's theorem:

$$\oiint_S \vec{F} \cdot d\vec{S} = \iiint_V (\vec{\nabla} \cdot \vec{F}) d^3\vec{x}$$

## Helmholtz's Theorem

$\forall \vec{F}$  continuous differentiable,  $\vec{F} = \vec{F}_{\perp} + \vec{F}_{\parallel}$

1.  $\vec{F}_{\perp}$ : transverse component,  $\vec{\nabla} \cdot \vec{F}_{\perp} = 0$  or equivalently  $\vec{F}_{\perp} = \vec{\nabla} \times \vec{A}$
2.  $\vec{F}_{\parallel}$ : longitudinal component,  $\vec{\nabla} \times \vec{F}_{\parallel} = 0$  or equivalently  $\vec{F}_{\parallel} = -\vec{\nabla}\phi$

## Experimental Laws and Maxwell's Equations

\*Gauss Units <sup>1</sup>

$$\vec{\nabla} \cdot \vec{E} = 4\pi\rho(\vec{x})$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{B} = \frac{4\pi}{c} \vec{j} + \frac{1}{c} \frac{\partial \vec{E}}{\partial t}$$

$$\vec{\nabla} \times \vec{E} = -\frac{1}{c} \frac{\partial \vec{B}}{\partial t}$$

1. Coulomb's law:

$$\vec{E}(\vec{x}) = \int_{V'} \frac{\rho(\vec{x}')(\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} d^3\vec{x}' = -\nabla\phi(\vec{x}), \quad \phi(\vec{x}) = \int_{V'} \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3\vec{x}'$$

2. Biot-Savart's law:

$$\vec{B}(\vec{x}) = \frac{1}{c} \int_{V'} \frac{\vec{j}(\vec{x}') \times (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3} d^3\vec{x}' = \vec{\nabla} \times \vec{A}(\vec{x}), \quad \vec{A}(\vec{x}) = \int_{V'} \frac{\vec{j}(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3\vec{x}'$$

3. Ampere's law of circuit:

$$\oint_L \vec{B} \cdot d\vec{l} = \frac{4\pi}{c} \iint_S \vec{j} \cdot d\vec{S}$$

4. Faraday's law in induction:

$$\oint_L \vec{E} \cdot d\vec{l} = -\frac{1}{c} \frac{d}{dt} \iint_S \vec{B} \cdot d\vec{S}$$

---

<sup>1</sup>  $\vec{\nabla} \left( \frac{1}{r} \right) = \frac{-\vec{r}}{r^3}$ ,  $\vec{\nabla} \cdot \vec{\nabla} \left( \frac{1}{r} \right) = \nabla^2 \left( \frac{1}{r} \right) = -4\pi\delta(\vec{x})$ ,  $\int_V \vec{\nabla} \cdot \frac{\vec{r}}{r^3} d^3\vec{r} = \oiint_S \frac{\vec{r}}{r^3} \cdot d\vec{S} = \oiint_S \frac{\vec{r}}{r^3} \cdot \vec{r}^0 r^2 \sin\theta d\theta d\phi = 4\pi$

5. Conversation of electric charge:

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{j} = 0, \quad \vec{j} = \rho \vec{v}$$

## Properties

Conversations, Invariance Covariance

### Invariance Gauge

$$\vec{E} = -\vec{\nabla}\phi - \frac{1}{c}\frac{\partial \vec{A}}{\partial t}, \quad \vec{B} = \vec{\nabla} \times \vec{A}, \text{ after Gauge transformation, we have } \vec{E}' = \vec{E}, \vec{B}' = \vec{B}$$

$$\phi \rightarrow \phi' = \phi + \vec{\nabla}\chi \quad \vec{A} \rightarrow \vec{A}' = \vec{A} - \frac{1}{c}\frac{\partial}{\partial t}\chi$$

1. Columb's gauge:  $\vec{\nabla} \cdot \vec{A} = 0$

2. Lorenz's gauge:  $\vec{\nabla} \cdot \vec{A} + \frac{1}{c}\frac{\partial}{\partial t}\phi = 0$

### Multiexpansion

$$\phi(\vec{x}) = \phi^{(0)} + \phi^{(1)} + \phi^{(2)} + \dots, \quad \vec{A}(\vec{x}) = \vec{A}^{(0)} + \vec{A}^{(1)} + \vec{A}^{(2)} + \dots \text{ monopole + dipole + quadrapole}$$

$$\vec{E} = \vec{E}^{(0)} + \vec{E}^{(1)} + \vec{E}^{(2)}, \quad \vec{B} = 0 + \vec{B}^{(1)} + \vec{B}^{(2)} + \dots$$

## Charged Particle in EM field

Newtonian Lorentz Equations:

$$\vec{f}_{\text{Lorentz}} = q\left(\vec{E} + \frac{\vec{v}}{c} \times \vec{B}\right) = m\ddot{\vec{x}}$$

### Lagrangian Mechanics

1. Euler-Lagrangian's equation:  $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_\alpha} = \frac{\partial L}{\partial q_\alpha}$

2. d'Alembert's virtual work principle:  $\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_\alpha}\right) - \frac{\partial T}{\partial q_\alpha} = Q_\alpha$  (generalized force)

If  $L = T - V$ , then take it into and find out  $V(q, \dot{q}, t)$

$$\frac{d}{dt}\left(\frac{\partial V}{\partial \dot{q}_\alpha}\right) - \frac{\partial V}{\partial q_\alpha} = Q_\alpha = (\vec{f}_{\text{Lorentz}})_\alpha = q\left[\vec{E}_\alpha + \left(\frac{\vec{v}}{c} \times \vec{B}\right)_\alpha\right]$$

Since  $E_\alpha = \left(-\vec{\nabla}\phi - \frac{1}{c}\frac{\partial}{\partial t}\vec{A}\right)_\alpha$ ,  $q\left(\frac{\vec{v}}{c} \times \vec{B}\right)_\alpha = \frac{q}{c}\varepsilon_{\alpha\beta\gamma}v_\beta(\varepsilon_{\gamma ab}\partial_a A_b) = \frac{q}{c}[v_\beta\partial_\alpha A_\beta - v_\beta\partial_\beta A_\alpha]$ , we have

$$Q_\alpha = -q\partial_\alpha\phi - \frac{q}{c}\partial_t A_\alpha + \frac{q}{c}\left[\partial_\alpha(\vec{v} \cdot \vec{A}) - (\vec{v} \cdot \vec{\nabla})A_\alpha\right] = -q\partial_\alpha\left[\phi - \frac{1}{c}(\vec{v} \cdot \vec{A})\right] - \frac{q}{c}\partial_t A_\alpha + (\vec{v} \cdot \vec{\nabla})A_\alpha$$

and  $\frac{d}{dt}\vec{A}(\vec{x}, t) = \frac{\partial \vec{A}}{\partial t} + \frac{d\vec{x}}{dt}\frac{\partial \vec{A}}{\partial \vec{x}} = \frac{\partial \vec{A}}{\partial t} + (\vec{v} \cdot \vec{\nabla})\vec{A}$ , therefore

$$Q_\alpha = -q\partial_\alpha\left[\phi - \frac{1}{c}(\vec{v} \cdot \vec{A})\right] - \frac{q}{c}\frac{d}{dt}A_\alpha$$

Construct an effective potential  $V(\vec{x}, \vec{\dot{x}}, t) = q\left[\phi - \frac{1}{c}(\vec{v} \cdot \vec{A})\right]$ , check:

$$\frac{d}{dt}\left(\frac{\partial V}{\partial \dot{\vec{x}}}\right) = -\frac{q}{c}\frac{d\vec{A}}{dt}, \quad \frac{\partial V}{\partial \vec{x}} = q\frac{\partial \phi}{\partial \vec{x}} - \frac{q}{c}(\vec{v} \cdot \vec{\nabla})\vec{A}$$

then  $L = T - V = \frac{m\dot{\vec{x}}^2}{2} - q\left[\phi - \frac{1}{c}(\vec{v} \cdot \vec{A})\right]$ , canonical momentum:  $\vec{p} = \frac{\partial L}{\partial \dot{\vec{x}}} = m\dot{\vec{x}} + \frac{q}{c}\vec{A}$

$$H = \vec{p} \cdot \dot{\vec{x}} - L = \frac{m\dot{\vec{x}}^2}{2} + q\phi = \frac{1}{2m}\left(\vec{p} - \frac{q}{c}\vec{A}\right)^2 + q\phi$$

# Chapter 2

## Introduction

### 2.1 Fundamental Postulates (Axiom) of Quantum Mechanics

*SOME IS*

#### Postulate 1: State

◇ *Description of the state of a system: At a fixed time  $t_0$ , the state of an isolated physical system is defined by specifying a ket  $|\psi(t_0)\rangle$  belonging to the state space  $\mathcal{E}$ .*

System  $\leftrightarrow$  State Vector    ket:  $|\psi\rangle$ ; bra:  $\langle\psi|$

Hermiltian Conjugate:  $(|\psi\rangle)^\dagger = \langle\psi|$     ( $\dagger$  : adjoint)

#### Postulate 2: Operator

◇ *Description of physical quantities: Every measurable physical quantity  $A$  is described by an operator  $\hat{A}$  acting in state space  $\mathcal{E}$ ; this operator is an observable.*

Physical quantity  $\leftrightarrow$  Observables  $\leftrightarrow$  Operator  $\hat{A}$  (Linear & Hermiltian)

#### Postulate 3: Measurement

◇ *The measurement of physical quantities: The only possible result of the measurement of a physical quantity  $A$  is one of the eigenvalues of the corresponding observable  $\hat{A}$*

Measurement outcome of  $\hat{A}$  (uncertain; certain). The eigenvalue equation is

$$\hat{A} |\psi_n\rangle = A_n |\psi_n\rangle$$

- $\hat{A}$  operator, q-number
- $A_n$  eigenvalue, c-number
- "eigen": "proper", "characteristic"

Probability of finding  $A_n$  when  $A$  is measured in  $|\psi\rangle$ :

$$\mathcal{P}(A_n) = |\langle\psi_n|\psi\rangle|^2$$

#### Postulate 4: Evolution

◇ *Time evolution of systems: The time evolution of the state vector  $|\psi(t)\rangle$  is governed by the Schrödinger's equation or Heisenberg's equation.*

### Schrödinger's Picture

(State: change; observable: no change) Schrödinger's equation:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle$$

- $\hat{H}$  is called Hamiltonian operator of the system, associated with the total energy of the system

$$\hat{H} = \hat{T} + \hat{V} = \frac{\hat{p}^2}{2m} + \hat{V}(\hat{x})$$

- $|\psi(t=0)\rangle = |\psi_0\rangle$

### Heisenberg's Picture

(State: no change; observable: change) Heisenberg's equation:

$$\frac{d\hat{F}}{dt} = \frac{\partial \hat{F}}{\partial t} + \frac{1}{i\hbar} [\hat{F}, \hat{H}]$$

### Postulate 5: Identical Particle and Symmetrization

$$\left. \begin{array}{l} \text{Symmetric} \\ \text{Anti-symmetric} \end{array} \right\} \text{State} \left\{ \begin{array}{l} \text{Boson (integer spin) Bose-Einstein Statistics} \\ \text{Fermion (half-integer spin) Femi-Diracs Statistics} \end{array} \right.$$

## 2.2 Vector and Operator

### 2.2.1 State Vector

Vector in Euclidean Space  $\leftrightarrow$  State Vector in Hilbert Space

- $|\psi\rangle$ : ket  $\in \mathbb{H} \leftrightarrow \begin{bmatrix} \psi_1 \\ \psi_2 \\ \vdots \end{bmatrix}$ ;  $\langle\psi| = (|\psi\rangle)^\dagger$ : bra  $\in \mathbb{H} \leftrightarrow [\psi_1^*, \psi_2^*, \dots]$
- Conjugate ( $\dagger$ ): Transverse ( $\top, \sim$ ) + Complex conjugate ( $*$ )  $\Rightarrow \langle\psi|\phi\rangle = \langle\phi|\psi\rangle^\dagger = \langle\phi|\psi\rangle^*$
- Orthonormality:  $\vec{e}_i \cdot \vec{e}_j = \delta_{ij} \leftrightarrow \langle u_m | u_n \rangle = \delta_{mn}$
- Completeness:  $\sum_i \vec{e}_i \cdot \vec{e}_j = \mathbb{1}, \leftrightarrow \sum_n |u_m\rangle \langle u_n| = \mathbb{1}$
- 1. Vector addition:  $|\psi_1\rangle + |\psi_2\rangle = |\psi_2\rangle + |\psi_1\rangle$
- 2. Scale multiplication:  $a(|\psi_1\rangle + |\psi_2\rangle) = a|\psi_1\rangle + a|\psi_2\rangle, ab|\psi\rangle = ba|\psi\rangle$
- 3. Inner product:  $\langle\psi|\phi\rangle = c \in \mathbb{C} \Rightarrow \langle\psi|\psi\rangle = r \in \mathbb{R} \geq 0, \langle\phi|\psi\rangle = (\langle\psi|\phi\rangle)^* = c^*$
- 4. Dyadic product of motion:  $(|\psi\rangle\langle\phi|)^\dagger = |\phi\rangle\langle\psi| = \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \end{bmatrix} \begin{bmatrix} \psi_1 & \psi_2 & \dots \end{bmatrix}$
- 5. Basis: for Hermiltian operator
  - (a) Discrete:  $\hat{A}|\psi_n\rangle = A_n|\psi_n\rangle \quad \langle\psi_n|\psi_m\rangle = \delta_{nm}, \sum_n |\psi_n\rangle\langle\psi_n| = \mathbb{1}$
  - (b) Continuous:  $\hat{A}|a\rangle = a|a\rangle \quad \langle a|a'\rangle = \delta(a-a'), \int da |a\rangle\langle a| = \mathbb{1}$

### 2.2.2 Operator

Consider

$$\hat{A}|\psi\rangle = |\phi\rangle$$

### Linear Operator

$\forall |\psi_1\rangle, |\psi_2\rangle, \forall a, b \in \mathbb{1}, \quad \text{if } \hat{A}(a|\psi_1\rangle + b|\psi_2\rangle) = a\hat{A}|\psi_1\rangle + b\hat{A}|\psi_2\rangle, \quad \text{then } \hat{A} \text{ is a linear operator,}$   
 otherwise  $\text{if } \hat{A}(a|\psi_1\rangle + b|\psi_2\rangle) = a^*\hat{A}|\psi_1\rangle + b^*\hat{A}|\psi_2\rangle, \quad \text{then } \hat{A} \text{ is a anti-linear operator.}$

Two kinds of operators in GM:

1. Observables: linear, Hermiltian

$$\hat{H}^\dagger = \hat{H}$$

2. Transformation: restricted to linear unitary transformation

$$\hat{U}^\dagger = \hat{U}^{-1} \quad \text{or} \quad \hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \mathbb{1}$$

### Rules

0.  $\forall |\psi\rangle, \text{ if } \hat{A}|\psi\rangle = \hat{B}|\psi\rangle, \text{ then } \hat{A} = \hat{B}$
1.  $\hat{A} + \hat{B} = \hat{B} + \hat{A}, \quad (\hat{A} + \hat{B}) + \hat{C} = \hat{A} + (\hat{B} + \hat{C}). \quad \forall |\psi\rangle, \hat{A}|\psi\rangle + \hat{B}|\psi\rangle = (\hat{A} + \hat{B})|\psi\rangle$
2.  $\hat{A}\hat{B} \neq \hat{B}\hat{A}, \text{ commutation: } [\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$
3. Adjoint (Hermiltian conjugate):  $\langle\phi|\hat{A}^\dagger = (\hat{A}|\phi\rangle)^\dagger$

### Properties

- Identity:  $\forall |\psi\rangle, \exists \mathbb{1}, \mathbb{1}|\psi\rangle = |\psi\rangle$
- Inverse:  $\hat{A}|\psi\rangle \rightarrow |\psi\rangle = \hat{A}^{-1}|\phi\rangle, (\hat{A}_1\hat{A}_2 \cdots \hat{A}_n)^{-1} = \hat{A}_n^{-1} \cdots \hat{A}_2^{-1}\hat{A}_1^{-1}$
- $(\hat{A}_1\hat{A}_2 \cdots \hat{A}_n)^\dagger = \hat{A}_n^\dagger \cdots \hat{A}_2^\dagger \hat{A}_1^\dagger$

## 2.3 Overview of Quantum Mechanics

Many "Two"s : POSE    P...M

1. Roles: State vector; Operator
2. Operator: Observables; Transformation. Hermiltian; Unitary
3. State: ket(vector); bra(dual vector)
4. Eigenspectrum: Discrete(bound state); Continuous(unbound state)
5. Properties: Wave particles duality
  - (a) Problems: Eigenvalues; Evolution
  - (b) Pictures: Schrödinger; Heisenberg
  - (c) Particles: Boson; Fermion
  - (d) Perturbation theories: non-degenerate; degenerate
  - (e) Angular momentum: Orbit; Spin
6. Mechanics: wave function; matrix form

## 2.4 Wave Mechanics and Matrix Mechanics

### 2.4.1 Wave Mechanics

Old wave function (WF) interpretation: matter wave / probabilistic wave

New wave function interpretation: projection of state vector in representation (basis)

- Discrete: wave function in  $\{|u_n\rangle\}$ -basis

$$|\psi\rangle = \mathbb{1} |\psi\rangle = \left( \sum_n |u_n\rangle \langle u_n| \right) |\psi\rangle = \sum_n c_n |u_n\rangle, \quad \text{WF: } c_n = \langle u_n | \psi \rangle$$

$$- \text{ energy-representation: } |\psi\rangle = \sum_n c_n |\psi_n\rangle, \quad c_n = \langle \psi_n | \psi \rangle, \quad \hat{A} |\psi_n\rangle = A_n |\psi_n\rangle$$

- Continuous: wave function in  $\{|a\rangle\}$ -basis

$$|\psi\rangle = \mathbb{1} |\psi\rangle = \left( \int da |a\rangle \langle a| \right) |\psi\rangle = \int \psi(a) |a\rangle da, \quad \text{WF: } \psi(a) = \langle a | \psi \rangle$$

- in  $\hat{x}$ -representation:

$$\hat{x} |\vec{x}\rangle = \vec{x} |\vec{x}\rangle, \quad |\psi\rangle = \int d^3 \vec{x} \psi(\vec{x}) |\vec{x}\rangle$$

$$* \text{ orthonormal: } \langle \vec{x} | \vec{x}' \rangle = \delta(\vec{x} - \vec{x}'); \quad \int d^3 |\vec{x}\rangle \langle \vec{x}| = \mathbb{1}$$

\* left ( $|x\rangle$ ) is eigenstate, right ( $\langle x|$ ) is eigenvalue

\* wave function:

- in  $\hat{p}$ -representation

$$\hat{p} |\vec{p}\rangle = \vec{p} |\vec{p}\rangle, \quad |\psi\rangle = \int d^3 \vec{p} \tilde{\psi}(\vec{p}) |\vec{p}\rangle$$

$$* \text{ orthonormal: } \langle \vec{p} | \vec{p}' \rangle = \delta(\vec{p} - \vec{p}'); \quad \int d^3 |\vec{p}\rangle \langle \vec{p}| = \mathbb{1}$$

Wave function in  $\hat{x}$  and  $\hat{p}$ -representation:

$$\psi(\vec{x}) = \langle \vec{x} | \psi \rangle, \quad \tilde{\psi}(\vec{p}) = \langle \vec{p} | \psi \rangle$$

Relation of  $\psi(\vec{x})$  and  $\tilde{\psi}(\vec{p})$ :

$$\begin{aligned} \tilde{\psi}(\vec{p}) &= \langle \vec{p} | \psi \rangle = \langle \vec{p} | \mathbb{1} | \psi \rangle = \left\langle \vec{p} \left| \left( \int d^3 \vec{x} |x\rangle \langle x| \right) \right| \psi \right\rangle = \int d^3 \vec{x} \langle \vec{p} | \vec{x} \rangle \psi(\vec{x}) \\ &= \int \psi(\vec{x}) \psi_{\vec{p}}^*(\vec{x}) d^3 \vec{x} \quad \text{where} \quad \psi_{\vec{p}}(\vec{x}) = \langle \vec{p} | \vec{x} \rangle = \left( \frac{1}{2\pi\hbar} \right)^{\frac{3}{2}} \exp \left\{ -i \frac{\vec{p} \cdot \vec{x}}{\hbar} \right\} \end{aligned}$$

IN  $\hat{x}$ -REPRESENTATION, the Schrödinger's equation is

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle, \quad \hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}) \quad \Rightarrow \quad i\hbar \frac{\partial}{\partial t} \psi(\vec{x}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{x}, t) + V(\vec{x}) \psi(\vec{x}, t)$$

$$* \psi(\vec{x}, t) = \langle \vec{x} | \psi(t) \rangle, \quad \hat{x} = \hat{x}, \quad \hat{p} = -i\hbar \vec{\nabla}, \quad \int d^3 x |\vec{x}\rangle \langle \vec{x}| = \mathbb{1}$$

same as IN  $\hat{p}$ -REPRESENTATION, the Schrödinger's equation is

$$\Rightarrow \quad i\hbar \frac{\partial}{\partial t} \tilde{\psi}(\vec{p}, t) = \frac{\hat{p}^2}{2m} \tilde{\psi}(\vec{p}, t) + V \left( i\hbar \frac{\partial}{\partial \vec{p}} \right) \tilde{\psi}(\vec{p}, t)$$

$$* \tilde{\psi}(\vec{p}, t) = \langle \vec{p} | \psi(t) \rangle, \quad \hat{p} = \hat{p}, \quad \hat{x} = i\hbar \frac{\partial}{\partial \vec{p}}, \quad \int d^3 p |\vec{p}\rangle \langle \vec{p}| = \mathbb{1}$$

**Ajoint (Hermiltian Conjugate)**

$$\langle \psi | \hat{A}^\dagger | \phi \rangle = \left( \langle \phi | \hat{A} | \psi \rangle \right)^* \quad \Rightarrow \quad \hat{A}^\dagger = \left( \hat{A}^\top \right)^*$$

*Proof.* Let  $|\psi\rangle = \sum_m c_m |u_m\rangle$ ,  $|\phi\rangle = \sum_n d_n |u_n\rangle$ , then  $\langle\psi|\hat{A}^\dagger|\phi\rangle = \sum_{m,n} d_n c_m^* \hat{A}_{mn}^\dagger$ , and

$$\left(\langle\phi|\hat{A}|\psi\rangle\right)^* = \left[\left(\sum_n d_n^* \langle u_n|\right) \hat{A} \left(\sum_m c_m |u_m\rangle\right)\right]^* = \left(\sum_{n,m} d_n^* c_m \langle u_n|\hat{A}|u_m\rangle\right)^* = \sum_{n,m} d_n c_m^* \hat{A}_{nm}^*$$

$$\text{Therefore, } \sum_{m,n} d_n c_m^* \hat{A}_{mn}^\dagger = \sum_{m,n} d_n c_m^* \hat{A}_{nm}^* \Rightarrow \left(\hat{A}^\dagger\right)_{mn} = \left(\hat{A}^*\right)_{nm} = \left(\hat{A}^\top\right)_{nm}^* \Rightarrow \quad \square$$

If  $\hat{A}^\dagger = \hat{A}$ , then

- Eigenvalues is  $\mathbb{R}$
- Eigenfunction is orthonormal and complete

*Proof.* Definition of adjoint:  $\left(\hat{A}^\dagger|\psi\rangle\right)^\dagger = \langle\psi|\hat{A}^\dagger \Leftrightarrow \langle\psi|\hat{A}^\dagger|\phi\rangle = \left(\langle\phi|\hat{A}|\psi\rangle\right)^*$

Let  $\hat{A}|\psi_n\rangle = A_n|\psi_n\rangle$ ,  $\hat{A}|\psi_m\rangle = A_m|\psi_m\rangle$  (non-degenerate, discrete, spectrum),

$$\langle\psi_n|\hat{A}|\psi_n\rangle = \left(\langle\psi_n|\hat{A}|\psi_n\rangle\right)^* = \left(\langle\psi_n|A_n|\psi_n\rangle\right)^* = A_n^* \langle\psi_n|\psi_n\rangle; \quad \langle\psi_n|\hat{A}|\psi_n\rangle = \langle A_n|\psi_n|A_n\rangle = A_n \langle\psi_n|\psi_n\rangle$$

Therefore  $A_n = A_n^* \Rightarrow A_n \in \mathbb{R}$ .

$$\begin{aligned} \langle\psi_n|\hat{A}^\dagger|\psi_m\rangle &= \left(\langle\psi_m|\hat{A}|\psi_n\rangle\right)^* = A_n^* \langle\psi_m|\psi_n\rangle^* = A_n \langle\psi_n|\psi_m\rangle \\ \langle\psi_n|\hat{A}|\psi_m\rangle &= A_m \langle\psi_n|\psi_m\rangle \end{aligned}$$

Therefore  $(A_n - A_m)\langle\psi_n|\psi_m\rangle = 0$ . Since  $n \neq m$ , so  $A_n \neq A_m$ ,  $\langle\psi_n|\psi_m\rangle \equiv 0 \Rightarrow |\psi_n\rangle \perp |\psi_m\rangle \quad \square$

Remark: degenerate  $\hat{A}|\psi_n^i\rangle = A_n|\psi_n^i\rangle$ ,  $i = 1, 2, \dots, g_n$  ( $g_n$ : degeneracy)

## 2.5 Matrix Mechanics

The state vector is

$$|\psi\rangle = \mathbb{1}|\psi\rangle = \sum_n c_n |u_n\rangle = \mathbb{1}$$

$$\bullet |u_n\rangle = [\dots \ 1 \ \dots]^\top, \quad \langle u_n| = [\dots \ 1 \ \dots] \Rightarrow |\psi\rangle \rightarrow [c_1 \ c_2 \ \dots \ c_n \ \dots]^\top$$

Observable is

$$\begin{aligned} \hat{A} &= \mathbb{1}\hat{A}\mathbb{1} = \left(\sum_n |u_n\rangle\langle u_n|\right) \hat{A} \left(\sum_m |u_m\rangle\langle u_m|\right) = \sum_{n,m} \langle u_n|\hat{A}|u_m\rangle (|u_n\rangle\langle u_m|) \\ &= \sum_{n,m} A_{nm} |u_n\rangle\langle u_m| = \begin{bmatrix} \cdot & \dots & \cdot & \cdot \\ \vdots & \ddots & \vdots & \vdots \\ \cdot & \dots & A_{nm} & \cdot \\ \cdot & \dots & \cdot & \cdot \end{bmatrix} \end{aligned}$$

## 2.6 Commutation Relation of Observables

**Commutation**

- $q$ -number:  $\hat{A}\hat{B} \neq \hat{B}\hat{A}$  (QM)
- $c$ -number:  $AB = BA$  (CM)

Definition of commutator:

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$



Compare it to Classical canonical commutator:

$$[A, B]_{\text{PB}} = \sum_{\alpha} \left( \frac{\partial A}{\partial q_{\alpha}} \frac{\partial B}{\partial p_{\alpha}} - \frac{\partial A}{\partial p_{\alpha}} \frac{\partial B}{\partial q_{\alpha}} \right)$$

The relation is

$$[A, B]_{\text{PB}} \leftrightarrow \left( \frac{1}{i\hbar} \right) [\hat{A}, \hat{B}]$$

### Properties of Commutation

1.  $[\hat{A}, \hat{B}] \neq 0$ , then it means uncertainty relation  $\Delta\hat{A} \cdot \Delta\hat{B} \geq \frac{|\langle [\hat{A}, \hat{B}] \rangle|}{2}$ ;  
otherwise  $[\hat{A}, \hat{B}] = 0$ , then it means common (simultaneous) eigenstates
2. Linearity:  $[a\hat{A}, b\hat{B} + c\hat{C}] = ab[\hat{A}, \hat{B}] + ac[\hat{A}, \hat{C}]$
3. Leibniz rule:  $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$ ,  $[\hat{A}\hat{B}, \hat{C}] = [\hat{A}, \hat{C}]\hat{B} + \hat{A}[\hat{B}, \hat{C}]$
4. Power related:  $[\hat{A}, \hat{B}^n] = n[\hat{A}, \hat{B}]\hat{B}^{n-1}$ , provided by  $[\hat{B}, [\hat{A}, \hat{B}]] = 0$

*Proof.* For  $n = 1$ ,  $[\hat{A}, \hat{B}] = [\hat{A}, \hat{B}^1] = 1[\hat{A}, \hat{B}]\hat{B}^{1-1}$ , suppose  $[\hat{A}, \hat{B}^n] = k[\hat{A}, \hat{B}]\hat{B}^{n-1}$ , then, when  $n = k + 1$ , one has  $[\hat{A}, \hat{B}^{k+1}] = [\hat{A}, \hat{B}^k \hat{B}] = [\hat{A}, \hat{B}^k] \hat{B} + \hat{B}^k [\hat{A}, \hat{B}] = k[\hat{A}, \hat{B}]\hat{B}^{k-1} \hat{B} + \hat{B}^k [\hat{A}, \hat{B}] = (k + 1)[\hat{A}, \hat{B}]\hat{B}^k \Rightarrow \square$

5. Function related:  $[\hat{A}, f(\hat{B})] = [\hat{A}, \hat{B}] \frac{\partial f}{\partial \hat{B}}$ , provided by  $[\hat{B}, [\hat{A}, \hat{B}]] = 0$

*Proof.*  $[\hat{A}, f(\hat{B})] = \left[ \hat{A}, \sum_n \frac{f^{(n)}(0)}{n!} \hat{B}^n \right] = \sum_n \frac{f^{(n)}(0)}{n!} [\hat{A}, \hat{B}^n] = \sum_n \frac{f^{(n)}(0)}{n!} n \hat{B}^{n-1} [\hat{A}, \hat{B}] \Rightarrow \square$

6. Jacobi's identity:

- (a) CM:  $[A, [B, C]]_{\text{PB}} + [B, [C, A]]_{\text{PB}} + [C, [A, B]]_{\text{PB}} = 0$
- (b) Vector analysis:  $\vec{A} \times (\vec{B} \times \vec{C}) + \vec{B} \times (\vec{C} \times \vec{A}) + \vec{C} \times (\vec{A} \times \vec{B}) = 0$
- (c) CED:  $F_{\mu\nu, \lambda} + F_{\nu\lambda, \mu} + F_{\lambda\mu, \nu} = 0$
- (d) QM:  $[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0$

### Fundamental Commutation Relation

Review the fundamental commutation relation in classical mechanics:  $[x_i, p_j]_{\text{PB}} = \delta_{ij}$ ,  $[x_i, x_j] = [p_i, p_j] \equiv 0$ , similarly

Fundamental commutation relation:

$$[\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij}, \quad [\hat{x}_i, \hat{x}_j] = [\hat{p}_i, \hat{p}_j] \equiv 0$$

*Proof.* In  $\hat{x}$ -representation, we have  $\hat{x} = \hat{x}$ ,  $\hat{p}_x = -i\hbar \partial/\partial x$ , then for  $\forall \psi(x)$

$$(\hat{x}\hat{p}_x - \hat{p}_x\hat{x})\psi(x) = \left[ x \left( -i\hbar \frac{\partial}{\partial x} \right) - \left( -i\hbar \frac{\partial}{\partial x} \right) x \right] \psi(x) = -i\hbar \left( x \frac{\partial}{\partial x} \psi(x) - \frac{\partial}{\partial x} x \psi(x) \right) = i\hbar \psi(x)$$

Therefore,  $[\hat{x}, \hat{p}_x] = i\hbar$ , since the equality is established for  $\forall \psi(x)$ .  $\square$

### Commutation Relation of Orbital Angular Momentum

According to fundamental commutation relation, we have

Commutation relation of orbital angular momentum:

$$\text{from } \hat{\vec{L}} = \hat{\vec{x}} \times \hat{\vec{p}} \Leftrightarrow \hat{L}_i = \varepsilon_{ijk} \hat{x}_j \hat{p}_k \quad \text{then} \quad [\hat{L}_i, \hat{L}_j] = i\hbar \varepsilon_{ijk} \hat{L}_k \Leftrightarrow \hat{\vec{L}} \times \hat{\vec{L}} = i\hbar \hat{\vec{L}}$$

$$1. [\hat{L}_i, \hat{L}_j] = i\hbar \varepsilon_{ijk} \hat{L}_k \Rightarrow \hat{\vec{L}} \times \hat{\vec{L}} = i\hbar \hat{\vec{L}}$$

$$\text{Proof. } [\hat{L}_i, \hat{L}_j] = [\varepsilon_{iab} \hat{x}_a \hat{p}_b, \varepsilon_{jmn} \hat{x}_m \hat{p}_n]$$

$$\begin{aligned} (\hat{\vec{L}} \times \hat{\vec{L}})_i &= \varepsilon_{ijk} \hat{L}_j \hat{L}_k = \frac{1}{2} \varepsilon_{ijk} \hat{L}_j \hat{L}_k + \frac{1}{2} \varepsilon_{ikj} \hat{L}_k \hat{L}_j = \frac{1}{2} \varepsilon_{ijk} (\hat{L}_j \hat{L}_k - \hat{L}_k \hat{L}_j) = \frac{1}{2} \varepsilon_{ijk} [\hat{L}_j, \hat{L}_k] \\ &= \frac{1}{2} \varepsilon_{ijk} i\hbar \varepsilon_{jkl} \hat{L}_l = \frac{i\hbar}{2} (\delta_{il} \delta_{jj} - \delta_{jl} \delta_{ij}) \hat{L}_l = \frac{i\hbar}{2} (2\delta_{il}) \hat{L}_l = i\hbar \hat{L}_i \end{aligned} \quad \square$$

$$2. \hat{\vec{L}} \times \hat{\vec{L}} = i\hbar \hat{\vec{L}} \Rightarrow [\hat{L}_i, \hat{L}_j] = i\hbar \varepsilon_{ijk} \hat{L}_k$$

$$\text{Proof. } (\hat{\vec{L}} \times \hat{\vec{L}})_i = i\hbar \hat{L}_i \rightarrow \varepsilon_{abi} \varepsilon_{ijk} \hat{L}_j \hat{L}_k = i\hbar \varepsilon_{abi} \hat{L}_i. \quad \text{l.h.s} = (\delta_{aj} \delta_{bk} - \delta_{ak} \delta_{bj}) \hat{L}_j \hat{L}_k = \hat{L}_a \hat{L}_b - \hat{L}_b \hat{L}_a = [\hat{L}_a, \hat{L}_b] \quad \square$$

$$3. [\hat{L}_i, \hat{L}_j] = i\hbar \varepsilon_{ijk} \hat{L}_k \Rightarrow [\hat{L}_z, \hat{L}^2] \equiv 0$$

$$\text{Proof. } [\hat{L}_z, \hat{L}^2] = [\hat{L}_z, \hat{L}_j \hat{L}_j] = i\hbar \varepsilon_{ijk} \hat{L}_k \hat{L}_j + i\hbar \varepsilon_{ijk} \hat{L}_j \hat{L}_k = i\hbar \varepsilon_{ijk} (\hat{L}_k \hat{L}_j + \hat{L}_j \hat{L}_k) \equiv 0 \quad \square$$

$$4. [\hat{L}_i, \hat{x}_j] = i\hbar \varepsilon_{ijk} \hat{x}_k \Leftrightarrow \hat{\vec{L}} \times \hat{\vec{x}} + \hat{\vec{x}} \times \hat{\vec{L}} = 2i\hbar \hat{\vec{x}}$$

$$\begin{aligned} \text{Proof. } [\hat{L}_i, \hat{x}_j] &= [\varepsilon_{iab} \hat{x}_a \hat{p}_b, \hat{x}_j] = \varepsilon_{iab} \hat{x}_a [\hat{p}_b, \hat{x}_j] = -i\hbar \varepsilon_{iaj} \hat{x}_a = i\hbar \varepsilon_{ijk} \hat{x}_k \Rightarrow \hat{L}_i \hat{x}_j = \hat{x}_j \hat{L}_i + i\hbar \varepsilon_{ijk} \hat{x}_k \\ (\hat{\vec{L}} \times \hat{\vec{x}} + \hat{\vec{x}} \times \hat{\vec{L}})_i &= \varepsilon_{ijk} (\hat{L}_j \hat{x}_k + \hat{x}_j \hat{L}_k) = \varepsilon_{ijk} (\hat{x}_k \hat{L}_j + i\hbar \varepsilon_{jkl} \hat{x}_l + \hat{x}_j \hat{L}_k) = \varepsilon_{ijk} (\hat{x}_k \hat{L}_j + \hat{x}_j \hat{L}_k) + i\hbar (\varepsilon_{ijk} \varepsilon_{jkl} \hat{x}_l) \\ &= 0 + i\hbar (\delta_{il} \delta_{jj} - \delta_{ij} \delta_{jl}) \hat{x}_l = i\hbar (2\delta_{il}) \hat{x}_l = 2i\hbar \hat{x}_i \end{aligned} \quad \square$$

$$5. [\hat{L}_i, \hat{p}_j] = i\hbar \varepsilon_{ijk} \hat{p}_k \Leftrightarrow \hat{\vec{L}} \times \hat{\vec{p}} + \hat{\vec{p}} \times \hat{\vec{L}} = 2i\hbar \hat{\vec{p}}$$

$$6. \forall \text{ vector } \vec{v} \Leftrightarrow i\hbar [\hat{L}_i, \hat{L}_j] = i\hbar \varepsilon_{ijk} \hat{v}_k \Leftrightarrow \hat{\vec{L}} \times \hat{\vec{v}} + \hat{\vec{v}} \times \hat{\vec{L}} = 2i\hbar \hat{\vec{v}}$$

$$7. \text{ For } V(\vec{x}) = V(r), r = \sqrt{x^2 + y^2 + z^2} \Rightarrow [\hat{L}_i, \hat{V}(r)] \equiv 0$$

$$\text{Proof. } [\hat{L}_i, V(r)] = [\hat{L}_i, \hat{x}_j] \frac{\partial V}{\partial x_j} = (i\hbar \varepsilon_{ijk} \hat{x}_k) \frac{dV}{dr} \frac{dx_j}{dr} = \left( \frac{i\hbar}{r} \frac{dV}{dr} \right) \varepsilon_{ijk} \hat{x}_k \hat{x}_j \equiv 0 \quad \square$$

$$8. \text{ For } \hat{T} = \frac{\hat{\vec{p}}^2}{2m} \Rightarrow [\hat{L}_i, \hat{T}] \equiv 0$$

$$\text{Proof. } [\hat{L}_i, \hat{\vec{p}}^2] = [\hat{L}_i, \hat{p}_j] \hat{p}_j + \hat{p}_j [\hat{L}_i, \hat{p}_j] = i\hbar \varepsilon_{ijk} (\hat{p}_k \hat{p}_j + \hat{p}_j \hat{p}_k) \equiv 0 \quad \square$$

$$9. \text{ For } \hat{H} = \hat{T} + V(r) \Rightarrow [\hat{L}_i, \hat{H}] \equiv 0, [\hat{\vec{L}}^2, \hat{H}] = 0$$

## 2.7 Eigenvalues Problem in QM

### 2.7.1 Observables

1. Position (canonical):

$$\hat{x} = \begin{cases} \hat{x}, & (\hat{x}\text{-representation}) \\ +i\hbar \frac{\partial}{\partial \hat{p}}, & (\hat{p}\text{-representation}) \end{cases}$$

2. Momentum (canonical)

$$\hat{p} = \begin{cases} -i\hbar \frac{\partial}{\partial \hat{x}}, & (\hat{x}\text{-representation}) \\ \hat{p}, & (\hat{p}\text{-representation}) \end{cases}$$

3. Orbital angular momentum

$$\hat{L} = \hat{x} \times \hat{p}, \quad \text{or} \quad \hat{L}_i = \varepsilon_{ijk} \hat{x}_j \hat{p}_k$$

$$[\hat{L}_i, \hat{L}_j] = i\hbar \varepsilon_{ijk} \hat{L}_k \quad \Rightarrow \quad [\hat{L}_i, \hat{L}^2] \equiv 0 \quad \text{and} \quad \hat{L} \times \hat{L} = i\hbar \hat{L}$$

Note:  $\hat{L} \times \hat{L} \neq 0$ , but  $\hat{L} = \hat{x} \times \hat{p} = -\hat{p} \times \hat{x}$ ,  $(\hat{x} \times \hat{p})_i = \varepsilon_{ijk} \hat{x}_k \hat{p}_j$ ,  $\hat{x}_k \hat{p}_j + \hat{p}_j \hat{x}_k = [\hat{x}_k, \hat{p}_j] = i\hbar \delta_{kj} = 0$

Consider vector  $\hat{v}$ , then  $\hat{L} \times \hat{v} + \hat{v} \times \hat{L} = 2i\hbar \hat{v}$ ,  $[\hat{L}_i, \hat{v}_j] = i\hbar \varepsilon_{ijk} \hat{v}_k$

4. Intrinsic angular momentum

$$[\hat{S}_i, \hat{S}_j] = i\hbar \varepsilon_{ijk} \hat{S}_k \quad \Leftrightarrow \quad \hat{S} \times \hat{S} = i\hbar \hat{S}$$

$$[\hat{L}_i, \hat{S}_j] = 0$$

5. Hamiltonian

$$\hat{H} = \hat{T} + \hat{V} = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

The kinetic energy only depend on  $\hat{p}$ , and the potential only depend on  $\hat{x}$

6. Energy

$$\hat{E} = i\hbar \frac{\partial}{\partial t} \quad \text{or} \quad \hat{E} = i\hbar \frac{d}{dt}$$

Note: in non-relativistic QM, time  $t$  is not observable

Schrödinger's equation:  $\hat{E}\psi(\vec{x}, t) = \hat{H}\psi(\vec{x}, t)$ ,  $\psi(\vec{x}, t) = e^{i(\vec{p} \cdot \vec{x} - Et)/\hbar} \Rightarrow \hat{H}\psi(\vec{x}) = E\psi(\vec{x})$

### 2.7.2 Eigenvalues and Eigenstates

#### Position

Eigenvalue equation is

$$\hat{x} |\vec{x}'\rangle = \vec{x}' |\vec{x}'\rangle$$

- Orthonormalism:  $\langle \vec{x}' | \vec{x}'' \rangle = \delta(\vec{x}' - \vec{x}'')$
- Resolution identity:  $\int d\vec{x} |\vec{x}\rangle \langle \vec{x}| = \mathbb{1}$

Multiply both sides with  $\int |\vec{x}\rangle \langle \vec{x}| d\vec{x}$ , then

$$\text{l.h.s} = \int |\vec{x}\rangle \langle \vec{x} | \hat{x} | \vec{x}' \rangle = \text{r.h.s} = \int |\vec{x}\rangle \vec{x}' \langle \vec{x} | \vec{x}' \rangle \Rightarrow \langle \vec{x} | \hat{x} | \vec{x}' \rangle = \vec{x}' \delta(\vec{x} - \vec{x}')$$

In  $\hat{x}$ -representation, wave function of  $|\vec{x}'\rangle$  is <sup>1</sup>

$$\langle \vec{x}' | \vec{x} \rangle = \delta(\vec{x} - \vec{x}') \quad \text{and} \quad \hat{x} \delta(\vec{x} - \vec{x}') = \vec{x}' \delta(\vec{x} - \vec{x}')$$

---

<sup>1</sup>  $f(x)\delta(x-a) = f(a)\delta(x-a)$

### Momentum

Eigenvalue equation is

$$\hat{\vec{p}} |\vec{p}'\rangle = \vec{p}' |\vec{p}'\rangle$$

- Orthonormalism:  $\langle \vec{p}' | \vec{p}'' \rangle = \delta(\vec{p}' - \vec{p}'')$
- Revolution identity:  $\int d\vec{p} |\vec{p}\rangle \langle \vec{p}| = \mathbf{1}$

In  $\hat{\vec{p}}$ -representation, wave function of  $|\vec{p}\rangle$  is

$$\langle \vec{p}' | \vec{p} \rangle = \delta(\vec{p} - \vec{p}') \quad \text{and} \quad \hat{\vec{p}} \delta(\vec{p} - \vec{p}') = \vec{p}' \delta(\vec{p} - \vec{p}')$$

Note: "Spectrum" - set of eigenvalues of observable is eigenspectrum for this condition is continuous spectrum

### Relation of between wave function in two representation

The state and wave function are

$$|\psi\rangle = \mathbf{1} |\psi\rangle = \begin{cases} \text{in } \hat{x}\text{-rep:} & \int d\vec{x} |\vec{x}\rangle \langle \vec{x}| \psi\rangle = \int d\vec{x} |\vec{x}\rangle \psi(\vec{x}) \\ \text{in } \hat{\vec{p}}\text{-rep:} & \int d\vec{p} |\vec{p}\rangle \langle \vec{p}| \psi\rangle = \int d\vec{p} |\vec{p}\rangle \tilde{\psi}(\vec{p}) \end{cases}, \quad \begin{cases} \psi(\vec{x}) = \langle \vec{x} | \psi \rangle \\ \tilde{\psi}(\vec{p}) = \langle \vec{p} | \psi \rangle \end{cases}$$

Steps of asking relation of  $\psi(\vec{x})$  and  $\tilde{\psi}(\vec{p})$ :

1. Multiply  $\int d\vec{x} |\vec{x}\rangle \langle \vec{x}|$  to two rep of  $|\psi\rangle$ , then  $\psi(\vec{x}) = \langle \vec{x} | \vec{p} \rangle \tilde{\psi}(\vec{p}) = \psi_{\vec{p}}(\vec{x}) \tilde{\psi}(\vec{p})$ , hence, we need to ask  $\psi_{\vec{p}}(\vec{x})$
2. Multiply  $\int d\vec{x} |\vec{x}\rangle \langle \vec{x}|$  to both side of  $\hat{\vec{p}} |\vec{p}\rangle = \vec{p} |\vec{p}\rangle$ , it can be considered as changing  $\hat{\vec{p}}$ -rep to  $\hat{x}$ -rep, therefore

$$\langle \vec{x} | \hat{\vec{p}} | \vec{p} \rangle = \vec{p} \langle \vec{x} | \vec{p} \rangle \Rightarrow -i\hbar \frac{\partial}{\partial \vec{x}} \psi_{\vec{p}}(\vec{x}) = \vec{p} \psi_{\vec{p}}(\vec{x}) \Rightarrow \psi_{\vec{p}}(\vec{x}) = N e^{i\vec{p} \cdot \vec{x} / \hbar}$$

Note: for 1-D we have  $-i\hbar d\psi_p(x)/dx = p\psi_p(x) \Rightarrow \psi_p(x) = N e^{ipx/\hbar}$ . Then, normalization is needed to ask  $N$ .

3. Consider orthonormalism, take into the terms <sup>2</sup>

$$\begin{aligned} \int d\vec{x} \psi_{\vec{p}'}^*(\vec{x}) \psi_{\vec{p}}(\vec{x}) &= \int d\vec{x} \langle \vec{p}' | \vec{x} \rangle \langle \vec{x} | \vec{p} \rangle = \left\langle \vec{p}' \left| \left( \int d\vec{x} |\vec{x}\rangle \langle \vec{x}| \right) \right| \vec{p} \right\rangle = \langle \vec{p}' | \vec{p} \rangle = \delta(\vec{p} - \vec{p}') \\ &= \int d\vec{x} |N|^2 e^{i(\vec{p} - \vec{p}') \cdot \vec{x} / \hbar} = |N|^2 (2\pi)^3 \delta\left(\frac{\vec{p} - \vec{p}'}{\hbar}\right) = |N|^2 (2\pi\hbar)^3 \delta(\vec{p} - \vec{p}') \end{aligned}$$

Therefore,  $|N|^2 = \left(\frac{1}{2\pi\hbar}\right)^3$ , and we finished.

The relation between  $\psi(\vec{x})$  and  $\tilde{\psi}(\vec{p})$

$$\begin{aligned} \psi(\vec{x}) &= \psi_{\vec{p}}(\vec{x}) \tilde{\psi}(\vec{p}) \Rightarrow \psi_{\vec{p}}(\vec{x}) = \langle \vec{x} | \vec{p} \rangle = \left(\frac{1}{2\pi\hbar}\right)^{\frac{3}{2}} e^{i\vec{p} \cdot \vec{x} / \hbar}, \\ \tilde{\psi}(\vec{p}) &= \psi_{\vec{x}}(\vec{p}) \psi(\vec{x}) \Rightarrow \psi_{\vec{x}}(\vec{p}) = \langle \vec{p} | \vec{x} \rangle = \left(\frac{1}{2\pi\hbar}\right)^{\frac{3}{2}} e^{-i\vec{p} \cdot \vec{x} / \hbar} \end{aligned}$$

$$\frac{2 \int_{-\infty}^{+\infty} e^{\pm i k x} dx}{(2\pi)^3 \delta(\vec{x})} = 2\pi \delta(x), \quad \int_{-\infty}^{+\infty} e^{\pm i \vec{k} \cdot \vec{x}} d\vec{k} = \int_{-\infty}^{+\infty} e^{\pm i k_x x} dk_x \int_{-\infty}^{+\infty} e^{\pm i k_y y} dk_y \int_{-\infty}^{+\infty} e^{\pm i k_z z} dk_z = (2\pi)^3 \delta(x) \delta(y) \delta(z) =$$

## 2.8 Uncertainty Principle

### 2.8.1 Generalized Uncertainty Principle

1. Heisenberg's uncertainty principle

$$\Delta \hat{x} \cdot \Delta \hat{p} \geq \frac{\hbar}{2}$$

2. Generalized uncertainty principle

$$\text{when } [\hat{A}, \hat{B}] \neq 0, \quad \text{then } \Delta \hat{A} \cdot \Delta \hat{B} \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$$

Expectation value:

$$\langle \hat{O} \rangle_\psi = \langle \psi | \hat{O} | \psi \rangle$$

Deviation: (mean square / root deviation)

$$\Delta \hat{A} = \sqrt{\Delta A^2} = \left\langle \left( \hat{A} - \langle \hat{A} \rangle \right)^2 \right\rangle^{\frac{1}{2}} = \left( \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2 \right)^{\frac{1}{2}}$$

### Proof of Generalized Uncertainty Principle

Deviation operator:

$$\hat{\sigma}_A = \hat{A} - \langle \hat{A} \rangle, \quad \hat{\sigma}_B = \hat{B} - \langle \hat{B} \rangle$$

Properties of  $\hat{\sigma}_A, \hat{\sigma}_B$

1. Hermitz:  $\hat{\sigma}_A^\dagger = \hat{\sigma}_A \Rightarrow \hat{\sigma}_A^\dagger \hat{\sigma}_A = \hat{\sigma}_A^2$
2. Commutator:  $[\hat{\sigma}_A, \hat{\sigma}_B] = [\hat{A}, \hat{B}]$
3.  $\hat{\sigma}_A \hat{\sigma}_B = \frac{1}{2} [\hat{\sigma}_A, \hat{\sigma}_B] + \frac{1}{2} \{\hat{\sigma}_A, \hat{\sigma}_B\}$
4. Expectation value of  $[\hat{\sigma}_A, \hat{\sigma}_B]$  and  $\{\hat{\sigma}_A, \hat{\sigma}_B\}$

$$(a) [\hat{\sigma}_A, \hat{\sigma}_B]^\dagger = (\hat{\sigma}_A \hat{\sigma}_B - \hat{\sigma}_B \hat{\sigma}_A)^\dagger = \hat{\sigma}_B \hat{\sigma}_A - \hat{\sigma}_A \hat{\sigma}_B = -[\hat{\sigma}_A, \hat{\sigma}_B] \quad (\text{skew-hermitian})^3$$

$$(b) \{\hat{\sigma}_A, \hat{\sigma}_B\}^\dagger = (\hat{\sigma}_A \hat{\sigma}_B + \hat{\sigma}_B \hat{\sigma}_A)^\dagger = \{\hat{\sigma}_A, \hat{\sigma}_B\} \quad (\text{hermitian})$$

$$\text{Therefore, } |\langle \hat{\sigma}_A \hat{\sigma}_B \rangle| = \frac{1}{2} |\langle [\hat{\sigma}_A, \hat{\sigma}_B] + \{\hat{\sigma}_A, \hat{\sigma}_B\} \rangle| = \frac{1}{2} |\langle [\hat{\sigma}_A, \hat{\sigma}_B] \rangle + \langle \{\hat{\sigma}_A, \hat{\sigma}_B\} \rangle| \geq \frac{1}{2} |\langle [\hat{\sigma}_A, \hat{\sigma}_B] \rangle| = \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$$

We need to prove  $\langle \sigma_A^2 \rangle \langle \sigma_B^2 \rangle \geq \frac{1}{4} |\langle [\hat{\sigma}_A, \hat{\sigma}_B] \rangle|^2$

Cauchy-Schwarz Inequality:

1.  $\left( \sum_n a_n^2 \right) \left( \sum_n b_n^2 \right) \geq \left( \sum_n a_n b_n \right)^2$
2.  $u = |\vec{u}|, v = |\vec{v}| \rightarrow uv \geq \vec{u} \cdot \vec{v}; |\vec{u}| |\vec{v}| \geq |\vec{u}| |\vec{v}| \cos \theta; |\vec{u}| + |\vec{v}| \geq |\vec{u} + \vec{v}| \Rightarrow \vec{u}^2 \cdot \vec{v}^2 \geq (\vec{u} \cdot \vec{v})^2$
3.  $u^2 \leftrightarrow \langle u|u \rangle, v^2 \leftrightarrow \langle v|v \rangle, \vec{u} \cdot \vec{v} \leftrightarrow \langle u|v \rangle \Rightarrow \langle u|u \rangle \langle v|v \rangle \geq (\langle u|v \rangle)^2$  equality iff  $|u\rangle = C|v\rangle$

$$\begin{aligned} \langle \sigma_A^2 \rangle \langle \sigma_B^2 \rangle &= \langle \psi | \hat{\sigma}_A^2 | \psi \rangle \langle \psi | \hat{\sigma}_B^2 | \psi \rangle = \langle \psi | \hat{\sigma}_A^\dagger \hat{\sigma}_A | \psi \rangle \langle \psi | \hat{\sigma}_B^\dagger \hat{\sigma}_B | \psi \rangle = \langle \psi_A | \psi_A \rangle \langle \psi_B | \psi_B \rangle \\ &\geq |\langle \psi_A | \psi_B \rangle|^2 = \left| \langle \psi | \hat{\sigma}_A^\dagger \hat{\sigma}_B | \psi \rangle \right|^2 = |\langle \psi | \hat{\sigma}_A \hat{\sigma}_B | \psi \rangle|^2 \end{aligned}$$

Since  $\Delta \hat{A} = \langle \hat{\sigma}_A^2 \rangle^{\frac{1}{2}}$ , therefore

$$\Delta \hat{A} \cdot \Delta \hat{B} \geq \sqrt{|\langle \hat{\sigma}_A \hat{\sigma}_B \rangle|^2} \geq \sqrt{\left( \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle| \right)^2} = \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$$

---

<sup>3</sup>Hermitian:  $\hat{A}^\dagger = \hat{A}, \langle \hat{A} \rangle = C \in \mathbb{R}$ ; skew-hermitian:  $\hat{A}^\dagger = -\hat{A}, \langle \hat{A} \rangle = C \in \mathbb{I}$ .

### 2.8.2 Minimum Uncertainty Wavepacket

$$\exists |\psi\rangle \Rightarrow \Delta \hat{A} \cdot \Delta \hat{B} = \frac{1}{2} \left| \langle [\hat{A}, \hat{B}] \rangle \right| \quad (\text{minimum uncertainty state})$$

**Proof with  $\hat{x}, \hat{p}$**

Let  $\hat{\sigma}_x = \hat{x} - \langle \hat{x} \rangle$ ,  $\hat{\sigma}_p = \hat{p} - \langle \hat{p} \rangle$ ,

$$|\alpha\rangle = (\hat{\sigma}_x + i\lambda\hat{\sigma}_p)|\psi\rangle$$

satisfy MUW ( $\lambda \in \mathbb{R}$ ). Therefore,  $\langle \alpha|\alpha \rangle \geq 0$  equality iff  $|\alpha\rangle = \vec{0}$ , take into the term,

$$\langle \alpha|\alpha \rangle = \langle \psi|(\hat{\sigma}_x - i\lambda\hat{\sigma}_p)(\hat{\sigma}_x + i\lambda\hat{\sigma}_p)|\psi\rangle = \langle \hat{\sigma}_p^2 \rangle \lambda^2 - \hbar\lambda + \langle \hat{\sigma}_x^2 \rangle \geq 0$$

Therefore,  $\Delta = (-\hbar)^2 - 4\langle \hat{\sigma}_p^2 \rangle \langle \hat{\sigma}_x^2 \rangle \leq 0 \Rightarrow \langle \hat{\sigma}_x^2 \rangle \langle \hat{\sigma}_p^2 \rangle \geq \frac{\hbar^2}{4} \Rightarrow \Delta \hat{x} \Delta \hat{p} \geq \frac{\hbar}{2}$  (HUP).

When  $\Delta = 0$ ,  $\Delta \hat{x} \Delta \hat{p} = \frac{\hbar}{2}$ ,  $\lambda = \frac{\hbar}{2\langle \hat{\sigma}_p^2 \rangle} = \frac{2\Delta \hat{x}^2}{\hbar} \Rightarrow \lambda \hbar = \Delta \hat{x}^2$

#### Wave function in MUS

Since  $|\alpha\rangle \equiv 0$ ,  $\langle x|\alpha\rangle = 0$ , and in  $\hat{x}$ -rep,  $\hat{x} = x$ ,  $\hat{p} = -i\hbar d/dx$ , then

$$\langle x|(\hat{\sigma}_x + i\lambda\hat{\sigma}_p)|\psi\rangle = \left[ (\hat{x} - \langle \hat{x} \rangle) + \lambda\hbar \frac{d}{dx} - i\lambda\langle \hat{p} \rangle \right] \psi(x) = 0$$

Consider  $\psi(x) = u(x - \langle x \rangle) \exp\{i\langle \hat{p} \rangle x/\hbar\}$ , let  $q = x - \langle x \rangle$ , the equation become

$$\left[ q + \lambda\hbar \frac{d}{dq} \right] u(q) = 0 \Rightarrow \int \frac{du(q)}{u(q)} = \int -\frac{q}{\lambda\hbar} dq \Rightarrow u(q) = A \exp\left\{ -\frac{q^2}{2\lambda\hbar} \right\}$$

Since  $\lambda\hbar = \Delta \hat{x}^2$ , then the wave function is a Gauss function

$$\psi(x) = A \exp\{i\langle \hat{p} \rangle x/\hbar\} \exp\left\{ -\frac{1}{2} \left( \frac{x - \langle x \rangle}{\Delta x} \right)^2 \right\} \Rightarrow |\psi(x)|^2 \sim \exp\left\{ -\left( \frac{x - \langle x \rangle}{\Delta x} \right)^2 \right\}$$

Remarks:

1. Uncertainty principle always holds, provided that  $[\hat{A}, \hat{B}] \neq 0$ , conjugate observables  $[\hat{A}, \hat{B}] \sim [\hbar]$  (dimension)
2.  $\exists |\psi\rangle$  (Gauss classical state)  $\Rightarrow$  Minimum uncertainty state

### 2.8.3 Energy-Time Uncertainty Principle

$$\Delta E \cdot \Delta T \geq \frac{\hbar}{2}$$

However, time is not observable.

*Proof.* Since Ehrenfest theorem (Eq: 4.1), then  $\Delta \hat{A} \cdot \hat{H} \geq \frac{1}{2} \left| \langle [\hat{A}, \hat{H}] \rangle \right| = \frac{\hbar}{2} \left| \frac{d}{dt} \langle \hat{A} \rangle \right|$ ,  $(\partial \hat{A} / \partial t \equiv 0, [\hat{A}, \hat{H}] \neq 0)$

let  $\frac{\Delta \hat{A}}{|d\langle \hat{A} \rangle / dt|} = \Delta T$  and  $\Delta \hat{H} = \Delta E \Rightarrow$

□

## Chapter 3

# Theory of Angular Momentum

### 3.1 Orbital Angular Momentum

#### 3.1.1 Eigenvalue Problem of AM

We have already known that  $[\hat{\vec{L}}, \hat{L}_i] = 0$ , which means  $\hat{\vec{L}}, \hat{L}_i$  are compatible observables, and they have common eigenstates. Suppose z-axis as quantum axis which is the  $\vec{B}$  direction, then

$$[\hat{\vec{L}}^2, \hat{L}_z] \equiv 0$$

In the later note, we can get

$$\hat{\vec{L}}^2 |lm\rangle = \hbar^2 l(l+1) |lm\rangle, \quad \hat{L}_z |lm\rangle = \hbar m |lm\rangle$$

where  $|lm\rangle$  is the eigenstate, e.g. for a fixed  $l$ ,  $m = -l, -l+1, \dots, l-1, l$ , there are  $|l, -l\rangle, |l, -l+1\rangle, \dots, |l, l-1\rangle, |l, l\rangle$ ,  $(2l+1)$  eigenstates. We call one eigenvalue corresponding to multi-eigenstates as degeneracy.

- Orthonormality (ON):  $\langle lm | l'm' \rangle = \delta_{ll'} \delta_{mm'}$
- Resolution of identity completeness (RI):  $\sum_{l,m} |lm\rangle \langle lm| = \mathbb{1}_l$

#### 3.1.2 Ladder Operator

Define the ladder operator as

$$\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y$$

- Raising operator:  $\hat{L}_+$ ; Lowing operator:  $\hat{L}_-$
- $\hat{L}_x = \frac{\hat{L}_+ + \hat{L}_-}{2}, \quad \hat{L}_y = \frac{\hat{L}_+ - \hat{L}_-}{2i}$
- $[\hat{L}_{\pm}, \hat{\vec{L}}^2] = 0$
- $[\hat{L}_{\pm}, \hat{L}_z] = -i\hbar \hat{L}_y \pm i \cdot i\hbar \hat{L}_x = \mp \hbar \hat{L}_{\pm} \Rightarrow \hat{L}_z \hat{L}_{\pm} = \hat{L}_{\pm} \hat{L}_z \pm \hbar \hat{L}_{\pm}$

Then, we find

$$\hat{L}_z \hat{L}_{\pm} |lm\rangle = (\hat{L}_{\pm} \hat{L}_z \pm \hbar \hat{L}_{\pm}) |lm\rangle = \hbar m \hat{L}_{\pm} |lm\rangle \pm \hbar \hat{L}_{\pm} |lm\rangle = \hbar(m \pm 1) \hat{L}_{\pm} |lm\rangle$$

which means  $\hat{L}_{\pm} |lm\rangle$  is the eigenstate of  $\hat{L}_z$ , corresponding to eigenvalue of  $\hbar(m \pm 1)$ , therefore, we can assume

$$\hat{L}_{\pm} |l, m\rangle = C_{\pm} |l, m \pm 1\rangle \quad \text{where} \quad C_{\pm} = \hbar \sqrt{l(l+1) - m_l(m_l \pm 1)}$$

Quantum numbers (qn) Summary:

- Principal qn:  $n \leftrightarrow \hat{H}$
- Orbital angular qn:  $l \leftrightarrow \vec{\hat{L}}^2$ ; Spin angular qn:  $s \leftrightarrow \vec{\hat{S}}^2$
- Magnetic qn:  $m_l \leftrightarrow \hat{L}_z$ ; spin magnetic qn:  $m_s \leftrightarrow \hat{S}_z$

## 3.2 Spin and Pauli Matrices

### 3.2.1 The Stern-Gerlach Experiment

- Quantumization of spatial orientation of angular momentum
- Intrinsic angular momentum (Spin)
- SGE
  1. Silver neutral atom beam: no Lorentz force
  2. Magnetic field: Inhomogeneous,  $\vec{F} = -\vec{\nabla}W_m$ ,  $W_m = -\vec{\mu} \cdot \vec{B} \Rightarrow \vec{F} = \vec{\mu} \vec{\nabla} \cdot \vec{B}$ ,  $\vec{\mu} \propto \vec{L}$
  3. Deflection: two vector, up(50%), down (50%)
  4. Further:  $\vec{v}$  is small,  $\vec{p}$  is small,  $\vec{L} = \vec{x} \times \vec{p}$  is small,  $\vec{\mu}$  is small, so intrinsic freedom?

NEED insert picture and the caption.

Significances of spin:

1. Spin is an intrinsic angular momentum
2. Spin is an eigenstate to distinguish two families
  - (a) Spin is integer:  $s = 0, 1, 2, \dots$ ,  
bosons: Bose-Einstein statistics; permutation  $\psi_s(\dots x_i, \dots, x_j, \dots) = \psi_s(\dots x_j, \dots, x_i, \dots)$  unchanged
  - (b) Spin is half-integer:  $s = 1/2, 3/2, \dots$   
Fermion: Fermi-Dirac statistics; permutation  $\psi_s(\dots x_i, \dots, x_j, \dots) = -\psi_s(\dots x_j, \dots, x_i, \dots)$  opposite spin

### 3.2.2 Spin Angular Momentum

Similar to orbital angular momentum, we have

$$[\vec{\hat{S}}^2, \hat{S}_i] \equiv 0 \Leftrightarrow [\hat{S}_i, \hat{S}_j] = i\hbar \varepsilon_{ijk} \hat{S}_k \Leftrightarrow \vec{\hat{S}} \times \vec{\hat{S}} = i\hbar \vec{\hat{S}}$$

Eigen function is with common eigenstates  $|s, m_s\rangle$

$$\begin{cases} \vec{\hat{S}}^2 |s, m_s\rangle = \hbar^2 s(s+1) |s, m_s\rangle \\ \hat{S} |s, m_s\rangle = \hbar m_s |s, m_s\rangle \end{cases}$$

#### Spin 1/2

Consider  $s = \frac{1}{2}$ , it stands for electron, proton, neutron. Then  $s(s+1) = \frac{3}{4}$ ,  $m_s = \pm \frac{1}{2}$ , and define two states as

$$\begin{array}{ll} \text{UP:} & \left| s = \frac{1}{2}, m_s = +\frac{1}{2} \right\rangle = |+\rangle = |\uparrow\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \text{DOWN:} & \left| s = \frac{1}{2}, m_s = -\frac{1}{2} \right\rangle = |-\rangle = |\downarrow\rangle = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{array} \Rightarrow \begin{cases} |0\rangle\langle 0| + |1\rangle\langle 1| = \mathbb{1}_2 \\ \langle 0|0\rangle = \langle 1|1\rangle = 1 \\ \langle 0|1\rangle = \langle 1|0\rangle = 0 \end{cases}$$



Therefore, we have the eigen function of in  $\hat{S}_z$ -representation, and define Pauli matrix as

$$\begin{aligned} \hat{S}_z |0\rangle &= +\frac{\hbar}{2} |0\rangle \\ \hat{S}_z |1\rangle &= -\frac{\hbar}{2} |1\rangle \end{aligned} \Rightarrow \hat{S} = \frac{\hbar}{2} \hat{\sigma} \Rightarrow \begin{aligned} \hat{\sigma}_z |0\rangle &= +1 |0\rangle \\ \hat{\sigma}_z |1\rangle &= -1 |1\rangle \end{aligned}$$

Discussion:

$$1. \hat{\sigma}_i^2 = \mathbb{1}_2 \quad (i = 1, 2, 3), \quad \hat{S}^2 = \frac{3}{4}\hbar^2 \mathbb{1}_2 \Rightarrow \hat{\sigma}^2 = 3\mathbb{1}_2$$

$$2. \hat{\sigma}_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\text{Proof. } \hat{\sigma}_z = \mathbb{1}_2 \cdot \hat{\sigma}_z \cdot \mathbb{1}_2 = (|0\rangle\langle 0| + |1\rangle\langle 1|)\hat{\sigma}_z(|0\rangle\langle 0| + |1\rangle\langle 1|) = |0\rangle\langle 0| - |1\rangle\langle 1| \quad \square$$

$$3. [\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\hat{\sigma}_k$$

$$\text{Proof. } [\hat{S}_i, \hat{S}_j] = [\hbar\hat{\sigma}_i/2, \hbar\hat{\sigma}_j/2] = i\hbar\varepsilon_{ijk}\hat{S}_k = i\hbar\varepsilon_{ijk}(\hbar\hat{\sigma}_k/2) \quad \square$$

$$4. \{\hat{\sigma}_i, \hat{\sigma}_j\} = 2\delta_{ij}\mathbb{1}_2$$

$$\begin{aligned} \text{Proof. } \hat{\sigma}_i(\hat{\sigma}_i\hat{\sigma}_j - \hat{\sigma}_j\hat{\sigma}_i) &= \hat{\sigma}_i 2i\varepsilon_{ijk}\hat{\sigma}_k \Rightarrow \hat{\sigma}_j - \hat{\sigma}_i\hat{\sigma}_j\hat{\sigma}_i = 2i\varepsilon_{ijk}\hat{\sigma}_i\hat{\sigma}_k \text{ minus }^1 \\ (\hat{\sigma}_j\hat{\sigma}_i - \hat{\sigma}_i\hat{\sigma}_j)\hat{\sigma}_i &= 2i\varepsilon_{jik}\hat{\sigma}_k\hat{\sigma}_i \Rightarrow \hat{\sigma}_j - \hat{\sigma}_i\hat{\sigma}_j\hat{\sigma}_i = -2i\varepsilon_{ijk}\hat{\sigma}_k\hat{\sigma}_i \Rightarrow 0 = \varepsilon_{ijk}(\hat{\sigma}_k\hat{\sigma}_i + \hat{\sigma}_i\hat{\sigma}_k) \end{aligned} \quad \square$$

$$5. \hat{\sigma}_i\hat{\sigma}_j = \delta_{ij}\mathbb{1}_2 + i\varepsilon_{ijk}\hat{\sigma}_k$$

Similar to the orbitla angular momentum ladder operator, in spin

$$\hat{S}_{\pm} = \hat{S}_x \pm i\hat{S}_y, \quad \hat{S}_{\pm} |s, m_s\rangle = \hbar\sqrt{s(s+1) - m_s(m_s \pm 1)} |s, m_s \pm 1\rangle$$

Apply the ladder operator to each case of spin 1/2,

$$\hat{S}_+ |1\rangle = \hbar |0\rangle, \quad \hat{S}_- |0\rangle = \hbar |1\rangle$$

Use the relation, we can also define Pauli ladder operator by /

$$\begin{aligned} (|0\rangle\langle 1|) |1\rangle &= |0\rangle \quad \leftrightarrow \quad \hat{S}_+ = \hbar |0\rangle\langle 0| = \hbar\hat{\sigma}_+ \\ (|1\rangle\langle 0|) |0\rangle &= |1\rangle \quad \leftrightarrow \quad \hat{S}_- = \hbar |1\rangle\langle 0| = \hbar\hat{\sigma}_- \end{aligned}$$

Therefore, we have

$$\hat{S}_{\pm} = \hbar\hat{\sigma}_{\pm}, \quad \hat{\sigma}_{\pm} = \frac{1}{2}(\hat{\sigma}_x \pm i\hat{\sigma}_y), \quad \begin{cases} \hat{\sigma}_+ = |0\rangle\langle 1| \\ \hat{\sigma}_- = |1\rangle\langle 0| \end{cases}, \quad \begin{cases} \hat{\sigma}_x = \hat{\sigma}_+ + \hat{\sigma}_- \\ \hat{\sigma}_y = (-i)(\hat{\sigma}_+ - \hat{\sigma}_-) \end{cases}$$

Pauli matrix ( $\hat{S}_z$ -representation):

$$\hat{\sigma}_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \hat{\sigma}_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \hat{\sigma}_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \hat{\sigma}_0 = \mathbb{1}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Properties of Pauli matrix:

$$1. \hat{\sigma}_i = \begin{bmatrix} \delta_{i3} & \delta_{i1} - i\delta_{i2} \\ \delta_{i1} + i\delta_{i2} & \delta_{i3} \end{bmatrix}$$

$$2. \text{Tr}(\hat{\sigma}_i) = 0$$

$$3. \det(\hat{\sigma}_i) = -1$$

---

<sup>1</sup> $i$ : no Einstein convention

$$4. \left( \hat{\sigma} \cdot \hat{A} \right) \left( \hat{\sigma} \cdot \hat{B} \right) = \left( \hat{A} \cdot \hat{B} \right) \mathbb{1}_2 + i \hat{\sigma} \cdot \left( \hat{A} \times \hat{B} \right)$$

$$\text{e.g. } \left( \hat{\sigma} \cdot \hat{p} \right)^2 = \hat{p}^2 \cdot \mathbb{1}_2, \quad \left[ \hat{\sigma} \cdot \left( \hat{p} - \frac{q}{c} \hat{A} \right) \right]^2 = \left( \hat{\sigma} \cdot \hat{\Pi} \right)^2 = \hat{\Pi} \cdot \mathbb{1} + i \hat{\sigma} \cdot \left( \hat{\Pi} \times \hat{\Pi} \right)$$

### 3.3 Algebra Method in QM

Summary of orbital angular momentum eigenvalue problem:

$$\begin{aligned} [\hat{x}_i, \hat{p}_j] &= i\hbar \delta_{ij}, \quad [\hat{x}_i, \hat{x}_j] = [\hat{p}_i, \hat{p}_j] \equiv 0, \quad \hat{L} = \hat{x} \times \hat{p} \quad \text{or} \quad \hat{L}_i = \varepsilon_{ijk} \hat{x}_j \hat{p}_k \\ \Rightarrow \quad [\hat{L}_i, \hat{L}_j] &= i\hbar \varepsilon_{ijk} \hat{L}_k \quad \Leftrightarrow \quad \hat{L} \times \hat{L} = i\hbar \hat{L} \\ \Rightarrow \quad [\hat{L}^2, \hat{L}_i] &\equiv 0 \quad \Rightarrow \quad \begin{cases} \hat{L}^2 |l, m_l\rangle = \hbar^2 l(l+1) |l, m_l\rangle \\ \hat{L}_z |l, m_l\rangle = \hbar m_l |l, m_l\rangle \end{cases} \quad (m_l = -l, -l+1, \dots, l-1, l) \end{aligned}$$

Generalize to intrinsic angular momentum  $\Leftrightarrow$  SGE

$$\begin{aligned} [\hat{S}_i, \hat{S}_j] &= i\hbar \varepsilon_{ijk} \hat{S}_k \quad \Leftrightarrow \quad \vec{S} \times \vec{S} = i\hbar \vec{S} \quad \Rightarrow \quad [\hat{S}^2, \hat{S}_i] \equiv 0 \\ \Rightarrow \quad \begin{cases} \hat{S}^2 |s, m_s\rangle = \hbar^2 s(s+1) |s, m_s\rangle \\ \hat{S}_z |s, m_s\rangle = \hbar m_s |s, m_s\rangle \end{cases} \end{aligned}$$

SEG

Generalize to arbitrary angular momentum

$$\hat{J} = \hat{L} + \hat{S}$$

Commutation:

$$1. \quad [\hat{J}_i, \hat{J}_j] = i\hbar \varepsilon_{ijk} \hat{J}_k$$

$$\text{Proof. } [\hat{J}_i, \hat{J}_j] = [\hat{L}_i + \hat{S}_i, \hat{L}_j + \hat{S}_j] = [\hat{L}_i, \hat{S}_j] + [\hat{S}_i, \hat{L}_j] = i\hbar \varepsilon_{ijk} (\hat{L}_k + \hat{S}_k) = i\hbar \varepsilon_{ijk} \hat{J}_k \quad [\hat{L}_i, \hat{L}_j] \equiv 0 \quad \square$$

$$2. \quad [\hat{J}^2, \hat{J}_i] \equiv 0 \quad (i = 1, 2, 3)$$

$$3. \quad [\hat{J}^2, \hat{J}_{\pm}] = 0 \quad \text{define ladder operator } \hat{J}_{\pm} = \hat{J}_x \pm i\hat{J}_y$$

$$\text{Proof. } [\hat{J}_z, \hat{J}_{\pm}] = [\hat{J}_z, \hat{J}_x \pm i\hat{J}_y] = i\hbar \hat{J}_y \pm i(-i\hbar) \hat{J}_x = \pm(\hat{J}_x \pm i\hbar \hat{J}_y) \quad \square$$

$$4. \quad [\hat{J}_z, \hat{J}_{\pm}] = \pm \hbar \hat{J}_{\pm}$$

$$5. \quad \hat{J}_+ \hat{J}_- = \hat{J}^2 - \hat{J}_z^2 + \hbar \hat{J}_z$$

$$6. \quad \hat{J}_{\pm}^{\dagger} = \hat{J}_{\mp}$$

Next, ask for the eigenvalue problem. Assume

$$\begin{cases} \hat{J}^2 |\lambda, m\rangle = \hbar^2 \lambda |\lambda, m\rangle \\ \hat{J}_z |\lambda, m\rangle = \hbar m |\lambda, m\rangle \end{cases}$$

Since  $[\hat{J}^2, \hat{J}_{\pm}] \equiv 0$ , then

$$\hat{J}^2 (\hat{J}_{\pm} |\lambda, m\rangle) = \hat{J}_{\pm} \hat{J}^2 |\lambda, m\rangle = \hbar^2 \lambda (\hat{J}_{\pm} |\lambda, m\rangle)$$

Since  $\hat{J}_z \hat{J}_{\pm} = \hat{J}_{\pm} \hat{J}_z \pm \hbar \hat{J}_{\pm}$ , then

$$\hat{J}_z (\hat{J}_{\pm} |\lambda, m\rangle) = (\hat{J}_{\pm} \hat{J}_z \pm \hbar \hat{J}_{\pm}) |\lambda, m\rangle = (m \pm 1) \hbar (\hat{J}_{\pm} |\lambda, m\rangle)$$

which means  $\hat{J}_\pm |\lambda, m\rangle$  is eigen state of  $\hat{J}_z$  with eigen value of  $(m \pm 1)\hbar$ . Therefore,

$$\hat{J}_\pm |\lambda, m\rangle = C_\pm |\lambda, m \pm 1\rangle \Rightarrow \hat{J}_\pm^n |\lambda, m\rangle = P_\pm |\lambda, m \pm n\rangle$$

And

$$\left. \begin{aligned} \langle \hat{J}^2 \rangle &= \langle \lambda, m | \hat{J}^2 | \lambda, m \rangle = \lambda \hbar^2 & \langle \hat{J}^2 \rangle &\geq \langle \hat{J}_z^2 \rangle \\ \langle \hat{J}_z^2 \rangle &= \langle \lambda, m | \hat{J}_z^2 | \lambda, m \rangle = m^2 \hbar^2 & \lambda &\geq m^2 \end{aligned} \right\}$$

Therefore,  $\exists m_0$  (minimal),  $\exists N$  (integer),  $m_0 + N$  is the maximal, let

$$\hat{J}_- |\lambda, m_0\rangle \equiv 0, \quad \hat{J}_+ |\lambda, m_0 + N\rangle \equiv 0$$

Since  $\hat{J}_\mp \hat{J}_\pm = \hat{J}^2 - \hat{J}_z^2 \mp \hbar \hat{J}_z$ ,

$$0 \equiv \hat{J}_+ (\hat{J}_- |\lambda, m_0\rangle) = \hbar^2 (\lambda - m_0^2 + m_0), \quad 0 \equiv \hat{J}_- (\hat{J}_+ |\lambda, m_0 + N\rangle) = \hbar^2 (\lambda - (m_0 + N)^2 - (m_0 + N))$$

Minus each other, we got  $(1 + N)(2m_0 + N) = 0$ , then

$$m_0 = -\frac{N}{2}, \quad m_0 = \frac{N}{2}, \quad \lambda = \frac{N}{2} \left( \frac{N}{2} + 1 \right)$$

let  $j = \frac{N}{2}$ , therefore

$$\lambda = j(j+1), \quad m_j = -j, -j+1, \dots, j-1, j$$

Eigenvalue equations:

$$\begin{cases} \hat{J}^2 |j, m_j\rangle = \hbar^2 j(j+1) |j, m_j\rangle & (j \geq 0) \\ \hat{J}_z |j, m_j\rangle = \hbar m_j |j, m_j\rangle & (m_j = -j, -j+1, \dots, +j) \end{cases}$$

Next, ask for  $C_\pm$

$$\hbar^2 (j(j+1) - m_j^2 \mp m_j) \langle j, m_j | j, m_j \rangle = \langle j, m_j | \hat{J}^2 - \hat{J}_z^2 \mp \hbar \hat{J}_z | j, m_j \rangle = \langle j, m_j | \hat{J}_\mp \hat{J}_\pm | j, m_j \rangle = |C_\pm|^2 \langle j, m_j | j, m_j \rangle$$

For simplicity,  $C_\pm \in \mathbb{R}$ , therefore

$$C_\pm = \hbar \sqrt{j(j+1) - m_j(m_j \pm 1)}$$

Ladder operation:

$$\hat{J}_\pm |j, m_j\rangle = \hbar \sqrt{j(j+1) - m_j(m_j \pm 1)} |j, m_j \pm 1\rangle$$

Remark:  $\hat{J}_+ |j, j\rangle \equiv 0, \hat{J}_- |j, -j\rangle \equiv 0$

- $j$ : azimuthal quantum number
- $m_j$ : magnetic quantum number

Consider

$$H = \frac{\vec{p}^2}{2m} + V(r) + \xi(r) \vec{L} \cdot \vec{S}$$

- $\xi(r) \vec{L} \cdot \vec{S} \rightarrow \vec{L} - \vec{S}$  coupling  $\leftrightarrow$  fine structure
- $V(r) \rightarrow$  central potential

Since  $[\hat{L}_i, \hat{p}_j] = i\hbar \varepsilon_{ijk} \hat{p}_k$ ,

$$[\hat{L}_i, \hat{p}^2] \equiv 0, [\hat{S}_i, \hat{p}^2] \equiv 0, [\hat{J}_i, \hat{p}^2] \equiv 0, [\hat{L}_i, \hat{V}(r)] \equiv 0, [\hat{S}_i, \hat{V}(r)] \equiv 0, [\hat{J}_i, \hat{V}(r)] \equiv 0$$

and

$$[\hat{L}_i, \hat{\vec{L}} \cdot \hat{\vec{S}}] = [\hat{L}_i, \hat{L}_j \hat{S}_j] = i\hbar \varepsilon_{ijk} \hat{L}_k \hat{S}_j \neq 0, \quad [\hat{L}^2, \hat{\vec{L}} \cdot \hat{\vec{S}}] = [\hat{L}_i \hat{L}_i, \hat{L}_j \hat{S}_j] = i\hbar \varepsilon_{ijk} (\hat{L}_k \hat{L}_i + \hat{L}_i \hat{L}_k) \hat{S}_j \equiv 0$$

Similar to  $\hat{S}$  and  $\hat{J}$

$$[\hat{S}_i, \hat{L} \cdot \hat{S}] \neq 0, [\hat{L}^2, \hat{L} \cdot \hat{S}] = 0, \quad [\hat{J}_i, \hat{L} \cdot \hat{S}] = 0, \quad [\hat{J} \cdot \hat{S}] = 0$$

Therefore,  $(\hat{L}^2, \hat{S}^2, \hat{J}^2, \hat{J}_i)$  commutes with  $\hat{H}$ .

### 3.4 Addition of Angular Momenta

#### 3.4.1 Basis

Consider  $\hat{J}_1$  and  $\hat{J}_2$ , then  $\hat{J} = \hat{J}_1 + \hat{J}_2$ ,  $\hat{J}_z = \hat{J}_{1z} + \hat{J}_{2z}$ , eigenvalue equations are (same as  $\hat{J}_2$ )

$$\begin{cases} \hat{J}_1^2 |j_1, m_1\rangle = \hbar^2 j_1(j_1 + 1) |j_1, m_1\rangle \\ \hat{J}_{1z} |j_1, m_1\rangle = \hbar m_1 |j_1, m_1\rangle \end{cases} \quad \begin{cases} \langle j_1 m_1 | j'_1 m'_1 \rangle = \delta_{j_1 j'_1} \delta_{m_1 m'_1} & \text{(ON)} \\ \sum_{m_1} |j_1 m_1\rangle \langle j_1 m_1| = \mathbb{1}_{2j_1+1} & \text{(RI)} \end{cases}$$

Commutation:  $(i, j, k = x, y, z; \alpha, \beta = 1, 2)$

$$1. [\hat{J}_i, \hat{J}_j] = i\hbar \varepsilon_{ijk} \hat{J}_k, \quad [\hat{J}_{\alpha i}, \hat{J}_{\beta j}] = i\hbar \delta_{\alpha\beta} \varepsilon_{ijk} \hat{J}_{\alpha k}$$

$$\text{Proof. } [\hat{J}_i, \hat{J}_j] = [\hat{J}_{1i} + \hat{J}_{2i}, \hat{J}_{1j} + \hat{J}_{2j}] = i\hbar \varepsilon_{ijk} \hat{J}_{1k} + i\hbar \varepsilon_{ijk} \hat{J}_{2k} \quad \square$$

$$2. [\hat{J}^2, \hat{J}_i] = 0, \quad [\hat{J}^2, \hat{J}_i^2] = 0, \quad [\hat{J}^2_\alpha, \hat{J}_{\alpha i}] = 0, \quad [\hat{J}^2, \hat{J}_{\alpha i}] \neq 0$$

$$\text{Proof. } [\hat{J}^2, \hat{J}_{\alpha i}] = [\hat{J}_1^2 + \hat{J}_2^2 + 2\hat{J}_1 \cdot \hat{J}_2, \hat{J}_{\alpha i}] = 2[\hat{J}_{1j} \hat{J}_{2j}, \hat{J}_{\alpha i}] = 2i\hbar \varepsilon_{jik} \hat{J}_{1k} \hat{J}_{2j} \neq 0 \quad (\alpha = 1) \quad \square$$

Therefore,  $\hat{J}_{1z}, \hat{J}_{2z}$  don't commute with  $\hat{J}^2$ ;  $\hat{J}_1, \hat{J}_2$  commute with  $\hat{J}^2$ .

Complete set of commuting observables (CSCO):  $(\hat{J}_1^2, \hat{J}_2^2, \hat{J}_z, \hat{J}^2)$

Related good quantum number (GQN):  $(j_1, j_2, m, j)$

#### 3.4.2 Two Representation

##### Uncoupled Basis

Tensor product:  $\otimes$

$$|j_1 m_1\rangle \otimes |j_2 m_2\rangle = |j_1 m_1 j_2 m_2\rangle$$

- $(\vec{J}_1 + \vec{J}_2) |j_1 j_2 m_1 m_2\rangle = (\vec{J}_1 + \vec{J}_2) |j_1 m_1\rangle \otimes |j_2 m_2\rangle = (\vec{J}_1 |j_1 m_1\rangle) \otimes |j_2 m_2\rangle + |j_1 m_1\rangle \otimes (\vec{J}_2 |j_2 m_2\rangle)$
- ON:  $\langle j_1 m_1 j_2 m_2 | j'_1 m'_1 j'_2 m'_2 \rangle = \delta_{j_1 j'_1} \delta_{j_2 j'_2} \delta_{m_1 m'_1} \delta_{m_2 m'_2}$ , for fixed  $j_1, j_2$ ,  $\langle j_1 m_1 j_2 m_2 | j'_1 m'_1 j'_2 m'_2 \rangle = \delta_{m_1 m'_1} \delta_{m_2 m'_2}$
- RI:  $\sum_{m_1, m_2} |j_1 m_1 j_2 m_2\rangle \langle j_1 m_1 j_2 m_2| = \mathbb{1}_{(2j_1+1)(2j_2+1)}$
- Dimension of tensor product:  $\dim |j_1 m_1\rangle = 2j_1 + 1$ ,  $\dim |j_1 m_1\rangle \otimes |j_2 m_2\rangle = (2j_1 + 1)(2j_2 + 1)$

##### Coupled Basis

$$|j_1 j_2 j m\rangle = |j m\rangle, \quad \text{for fixed } j_1, j_2$$

since  $-j_1 \leq m_1 \leq j_1$ ,  $-j_2 \leq m_2 \leq j_2$ ,  $-j \leq m \leq j$ , and  $-j_1 \leq m - m_2 \leq j_1$ ,  $-j_2 \leq m - m_1 \leq j_2$ , choose  $m_1 = j_1$ ,  $m_2 = j_2$ ,  $m = j$ , then  $j_2 - j_1 \leq j \leq j_1 + j_2$ ,  $j_1 - j_2 \leq j \leq j_1 + j_2$ , therefore

$$|j_2 - j_1| \leq j \leq j_1 + j_2, \quad m = -j, -j + 1, \dots, j - 1, j$$

- Dimensionality:  $\dim |j m\rangle = \sum_{j=|j_1-j_2|}^{j_1+j_2} (2j + 1)$

### 3.4.3 Clebsch Gordan Coefficient

$$\begin{aligned} |j_1 j_2 j m\rangle &= |j m\rangle = \mathbb{1}_{(2j_1+1)(2j_2+1)} \cdot |j m\rangle = \left[ \sum_{m_1=-j_1}^{j_1} \sum_{m_2=-j_2}^{j_2} |j_1 m_1 j_2 m_2\rangle \langle j_1 m_2 j_2 m_2| \right] |j m\rangle \\ &= \sum_{m_1=-j_1}^{j_1} \sum_{m_2=-j_2}^{j_2} (\langle j_1 m_1 j_2 m_2 | j m \rangle) |j_1 m_1 j_2 m_2\rangle \end{aligned}$$

Define Clebsch Gordan coefficient  $C_{m_1 m_2}^j$ :

$$|j_1 j_2 j m\rangle = \sum_{m_1+m_2=m} C_{m_1 m_2}^j |j_1 m_1 j_2 m_2\rangle$$

*Proof.*  $\hat{J}_z |j m\rangle = m\hbar |j m\rangle = (\hat{J}_{1z} + \hat{J}_{2z}) \sum_{m_1 m_2} C_{m_1 m_2}^j |j_1 m_2 j_2 m_2\rangle = \sum_{m_1 m_2} C_{m_1 m_2}^j (m_1 + m_2)\hbar |j_1 m_2 j_2 m_2\rangle$

Then  $m_1 + m_2 = m \Rightarrow \sum_{m_1+m_2=m}$

□

#### An example of Spin 1/2 $\times$ Spin 1/2

Consider spins of two electron, we have  $j_1 = \frac{1}{2}, m_1 = \pm \frac{1}{2}$  and  $j_2 = \frac{1}{2}, m_2 = \pm \frac{1}{2}$

$$\begin{aligned} \{|j_1 m_1\rangle \otimes |j_2 m_2\rangle\} &= \{|m_1\rangle \otimes |m_2\rangle\} = \left\{ \left| -\frac{1}{2} \right\rangle \otimes \left| -\frac{1}{2} \right\rangle, \left| -\frac{1}{2} \right\rangle \otimes \left| \frac{1}{2} \right\rangle, \left| \frac{1}{2} \right\rangle \otimes \left| -\frac{1}{2} \right\rangle, \left| \frac{1}{2} \right\rangle \otimes \left| \frac{1}{2} \right\rangle \right\} \\ &= \{|1\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |0\rangle \otimes |0\rangle\} \\ &= \{|11\rangle, |10\rangle, |01\rangle, |00\rangle\} \end{aligned}$$

Since  $|j_2 - j_1| \leq j \leq j_1 + j_2$ , therefore  $j = 0, 1$  and  $m = 0, -1, 1, \prod_j (2j+1) = (2j_1+1)(2j_2+1) = 4$

Note:  $C_{m_1 m_2}^j; m_1 + m_2 = m; |+\frac{1}{2}\rangle = |0\rangle = |\uparrow\rangle = |+\rangle$

1. Singlet:  $j = 0, m = 0$

$$|j, m\rangle = |00\rangle = C_{-\frac{1}{2} \frac{1}{2}}^0 \left| -\frac{1}{2} \frac{1}{2} \right\rangle + C_{\frac{1}{2} -\frac{1}{2}}^0 \left| \frac{1}{2} -\frac{1}{2} \right\rangle = \frac{1}{\sqrt{2}} |1\rangle_1 \otimes |0\rangle_2 - \frac{1}{\sqrt{2}} |0\rangle_1 \otimes |1\rangle_2 = \frac{1}{\sqrt{2}} (|10\rangle - |01\rangle)$$

2. Triplet:  $j = 1, m = 0, \pm 1$

$$\begin{aligned} |1 - 1\rangle &= C_{-\frac{1}{2} -\frac{1}{2}}^1 \left| -\frac{1}{2} -\frac{1}{2} \right\rangle = |1\rangle_1 \otimes |1\rangle_2 = |11\rangle \\ |10\rangle &= C_{-\frac{1}{2} \frac{1}{2}}^1 \left| -\frac{1}{2} \frac{1}{2} \right\rangle + C_{\frac{1}{2} -\frac{1}{2}}^1 \left| \frac{1}{2} -\frac{1}{2} \right\rangle = \frac{1}{\sqrt{2}} |1\rangle_1 \otimes |0\rangle_2 + \frac{1}{\sqrt{2}} |0\rangle_1 \otimes |1\rangle_2 = \frac{1}{\sqrt{2}} (|10\rangle + |01\rangle) \\ |1 + 1\rangle &= C_{\frac{1}{2} \frac{1}{2}}^1 \left| \frac{1}{2} \frac{1}{2} \right\rangle = |0\rangle_1 \otimes |0\rangle_2 = |00\rangle \end{aligned}$$

Entangled state: cannot written tensor product state.

# Chapter 4

## Quantum Dynamics and Symmetry

### 4.1 The Schrödinger's Equation

#### 4.1.1 Wave Function

Schrödinger's Equation (SE):

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle, \quad \text{where} \quad \hat{H} = \frac{\hat{p}^2}{2m} + V(\vec{x})$$

Remarks:

1. Pure state
2. Closed system:  $\langle \psi(t) | \psi(t) \rangle = 1 \xrightarrow{\vec{x}\text{-representation}} \int |\psi(\vec{x}, t)|^2 d^3\vec{x} = 1$
3. Non-relativistic Quantum Mechanics
4. Initial value problem (IVP):  $|\psi(t = t_0)\rangle = |\psi(t_0)\rangle$ ; stationary SE:  $\psi(\vec{x}, t = t_0) = \psi(\vec{x}, t_0)$ , with normalization.
5. Schrödinger's picture:  $\frac{d\hat{A}}{dt} \equiv 0$ ,  $\hat{A}$ : observables

In  $\vec{x}$ -representation, we have

$$\hat{p} = i\hbar \vec{\nabla}, \quad \langle \vec{x} | \psi(t) \rangle = \psi(\vec{x}, t)$$

Therefore, the wave function in  $\vec{x}$ -representation is

$$i\hbar \frac{\partial}{\partial t} \psi(\vec{x}, t) = -\left( \frac{\hbar}{2m} \nabla^2 + V(\vec{x}) \right) \psi(\vec{x}, t)$$

$$\text{Generally, } \frac{\partial \hat{A}}{\partial t} \equiv 0, \quad \frac{\partial \hat{H}}{\partial t} \equiv 0$$

#### 4.1.2 Time Evolution Operator in SE

Operator  $\hat{U}$

$$|\psi(t)\rangle = \hat{U}(t, t_0) |\psi(t_0)\rangle$$

then take it into SE, we have

$$i\hbar \frac{d}{dt} [\hat{U}(t, t_0) |\psi(t_0)\rangle] = \hat{H} [\hat{U}(t, t_0) |\psi(t_0)\rangle] \Rightarrow i\hbar \frac{d}{dt} \hat{U}(t, t_0) = \hat{H} \hat{U}(t, t_0) \Rightarrow \frac{d\hat{U}(t, t_0)}{\hat{U}(t, t_0)} = -\frac{i}{\hbar} \hat{H}(t, t_0) dt$$

If  $\partial \hat{H} / \partial t \equiv 0$ , which means a conservation system, then

$$\hat{U}(t, t_0) = \exp \left\{ -\frac{i}{\hbar} \hat{H}(t - t_0) \right\}$$

Properties of  $\hat{U}(t, t_0)$

1.  $\hat{U}^\dagger(t, t_0) = \hat{U}^{-1}(t, t_0) = \hat{U}(t_0, t)$  or  $\hat{U}^\dagger(t, t_0)\hat{U}(t_0, t) = 1$
2.  $\hat{U}(t, t_1)\hat{U}(t_1, t_0) = \hat{U}(t, t_0) \Rightarrow$  Chain rule:  $\hat{U}(t, t_n)\hat{U}(t_n, t_{n-1}) \cdots \hat{U}(t_2, t_1)\hat{U}(t_1, t_0) \equiv \hat{U}(t, t_0)$
3.  $\hat{U}(t, t_0) = \hat{U}(t - t_0) = \hat{U}(\tau)$ ,  $(\tau = t - t_0)$  and  $\hat{U}(t_0, t_0) = \mathbb{1}$  and Dyson series: If  $\partial\hat{U}/\partial t \neq 0$ , then  

$$\hat{U}(t, t_0) = \mathbb{1} - \frac{i}{\hbar} \int \hat{U}(t', t_0) dt' = \mathbb{1} - \frac{i}{\hbar} \int \left[ \mathbb{1} - \frac{i}{\hbar} \int \hat{U}(t_1, t_0) \right] dt_1 = \cdots$$

### 4.1.3 Continuity Equation

If  $V^*(\vec{x}) = V(\vec{x})$ , which means potential  $\in \mathfrak{R}$ ,

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \psi(\vec{x}, t) &= -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{x}, t) + V(\vec{x})\psi(\vec{x}, t) \\ -i\hbar \frac{\partial}{\partial t} \psi^*(\vec{x}, t) &= -\frac{\hbar^2}{2m} \nabla^2 \psi^*(\vec{x}, t) + V(\vec{x})\psi^*(\vec{x}, t) \end{aligned}$$

then we multiply  $\psi^*$  ( $\psi$  for second one) in each side of the equation, and move  $i\hbar$  ( $-i\hbar$  for second one) to the other side. Then, sum up the two equations of each sides, we got

$$\begin{aligned} \text{l.h.s} &= \psi^* \frac{\partial}{\partial t} \psi + \psi \frac{\partial}{\partial t} \psi^* = \frac{\partial}{\partial t} (\psi^* \psi) \\ \text{r.h.s} &= -\frac{\hbar}{2mi} [\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*] = -\frac{\hbar}{2mi} \vec{\nabla} \cdot [\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*] \end{aligned}$$

Therefore,

$$\frac{\partial}{\partial t} (\psi^* \psi) + \frac{\hbar}{2mi} \vec{\nabla} \cdot [\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*] = 0$$

we can rewrite the equation as

Continuity equation:

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{j} \equiv 0$$

define Probability density:

$$\rho(\vec{x}, t) = \psi^*(\vec{x}, t)\psi(\vec{x}, t) = |\psi(\vec{x}, t)|^2$$

Probability current density:

$$\vec{j} = \frac{\hbar}{2mi} (\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*) = \frac{1}{m} \text{Re}(\psi^* \frac{\hbar}{i} \vec{\nabla} \psi)$$

- If  $Q = \int_V \rho(\vec{x}, t) d^2 \vec{x}$ , the continuity equation is  $\frac{dQ}{dt} \equiv 0$

### 4.1.4 Ehrenfest's Theorem

From  $\frac{d}{dt} |\psi(t)\rangle = \frac{\hat{H}}{i\hbar} |\psi(t)\rangle$ ,  $\frac{d}{dt} \langle \psi(t)| = -\frac{\hat{H}}{i\hbar} \langle \psi(t)|$ ,  $\langle \hat{O} \rangle = \langle \psi(t) | \hat{O} | \psi(t) \rangle$ , we have

$$\begin{aligned} \frac{d}{dt} \langle \hat{O} \rangle(t) &= \left( \frac{d}{dt} \langle \psi(t) | \right) \hat{O} | \psi(t) \rangle + \left\langle \psi(t) \left| \frac{d}{dt} \hat{O} \right| \psi(t) \right\rangle + \langle \psi(t) | \hat{O} \left( \frac{d}{dt} | \psi(t) \rangle \right) \\ &= -\frac{1}{i\hbar} \langle \psi(t) | \hat{H} \hat{O} | \psi(t) \rangle + \left\langle \psi(t) \left| \frac{d}{dt} \hat{O} \right| \psi(t) \right\rangle + \frac{1}{i\hbar} \langle \psi(t) | \hat{O} \hat{H} | \psi(t) \rangle \\ &= \left\langle \frac{\partial}{\partial t} \hat{O} \right\rangle + \frac{1}{i\hbar} \langle [\hat{O}, \hat{H}] \rangle \end{aligned}$$

- Conserved quantity (constant of motion):  $\partial \hat{O} / \partial t \equiv 0$ ;  $[\hat{O}, \hat{H}] \equiv 0$

- Classical mechanics:

$$\frac{d}{dt}O = \frac{\partial O}{\partial t} + [O, H]_{\text{PB}} \rightarrow m \frac{d^2}{dt^2}x = -V'(x)$$

- Quantum mechanics:

$$\frac{d}{dt}\hat{O}_H = \left( \frac{\partial \hat{O}_s}{\partial t} \right)_H + \frac{1}{i\hbar} [\hat{O}_H, \hat{H}]$$

Ehrenfest's theorem:

$$\frac{d}{dt} \langle \hat{O} \rangle(t) = \left\langle \frac{\partial}{\partial t} \hat{O} \right\rangle + \frac{1}{i\hbar} \langle [\hat{O}, \hat{H}] \rangle \quad (4.1)$$

Example: for 1-dim,  $\hat{x}$ ,  $\hat{p}$ ,  $\hat{H} = \frac{\hat{p}^2}{2m} + V(x)$ , then

$$\begin{aligned} \frac{d}{dt} \langle \hat{x} \rangle &= \frac{1}{i\hbar} \langle [\hat{x}, \hat{H}] \rangle = \frac{1}{i\hbar} \left\langle [\hat{x}, \frac{\hat{p}^2}{2m}] \right\rangle = \frac{\langle \hat{p} \rangle}{m} \\ \frac{d}{dt} \langle \hat{p} \rangle &= \frac{1}{i\hbar} \langle [\hat{p}, \hat{H}] \rangle = \frac{1}{i\hbar} \langle [\hat{p}, V(x)] \rangle = \frac{1}{i\hbar} \langle [\hat{p}, \hat{x}] V'(x) \rangle = -\langle V'(x) \rangle \end{aligned}$$

for 3-dim,  $\vec{F} = -\vec{\nabla}V(x)$ , consider 2nd differentiation,

$$\frac{d^2}{dt^2} \langle \hat{x} \rangle = \frac{1}{m} \frac{d}{dt} \langle \hat{p} \rangle = -\frac{1}{m} \langle V'(x) \rangle \Rightarrow m \frac{d^2}{dt^2} \langle \hat{x} \rangle = -\langle V'(x) \rangle$$

QM:  $\langle V'(x) \rangle$  v.s. CM  $V'(x)|_{x=\langle x \rangle}$ ,

assume  $V(x) = \sum_n V_n \hat{x}^n$ , then  $V'(x) = \sum_n V_n n \hat{x}^{n-1}$ , which means  $\langle V'(x) \rangle = \sum_n V_n \langle \hat{x}^{n-1} \rangle$  v.s.  $V'(x)|_{x=\langle x \rangle} = \sum_n V_n \langle \hat{x} \rangle^{n-1}$ ,

then compare  $\langle \hat{x}^{n-1} \rangle$  v.s.  $\langle \hat{x} \rangle^{n-1}$

- $n = 0$ :  $V(x) = \text{const} \Rightarrow \vec{F} = 0 \Rightarrow \text{CM} \sim \text{QM}$
- $n = 1$ :  $\langle \hat{x}^{n-1} \rangle = 1$  v.s.  $\langle \hat{x} \rangle^0 = 1 \Rightarrow \text{CM} \sim \text{QM}, V(x) = x$
- $n = 2$ :  $\langle \hat{x} \rangle$  v.s.  $\langle \hat{x} \rangle \Rightarrow \text{CM} \sim \text{QM}, V(x) = x^2$
- $n \geq 3$ :  $\langle \hat{x}^3 - 1 \rangle$  v.s.  $\langle \hat{x} \rangle^2 \Rightarrow \text{CM} \not\sim \text{QM},$

$$(\text{QM}) \langle V'(x) \rangle = \int \psi^*(x, t) V'(x) \psi(x, t) dx = V'(\langle x \rangle) \int \psi^* \psi dx = V'(\langle x \rangle) = V'(x)|_{x=\langle x \rangle} (\text{CM})$$

## 4.2 The Schrödinger Versus the Heisenberg Picture

In Schrödinger picture, the observables don't change with time

$$i\hbar \frac{d}{dt} |\psi(t)\rangle_S = \hat{H}_S |\psi(t)\rangle_S, \quad \frac{d}{dt} \hat{A}_S(t) \equiv 0$$

In Heisenberg Picture, the state vector don't change with time

$$\frac{d}{dt} \hat{F}_H = \left( \frac{\partial}{\partial t} \hat{F}_S \right)_H + \frac{1}{i\hbar} [\hat{F}_H, \hat{H}], \quad \frac{d}{dt} |\psi(t)\rangle_H \equiv 0$$

### Time Evolution Operator

To prove  $S \Leftrightarrow H$ , we have

$$|\psi(t)\rangle_S = \hat{U}(t) |\psi(t)\rangle_H, \quad \hat{U}(t) = \exp \left\{ -\frac{i}{\hbar} \hat{H} t \right\}$$

Since  ${}_S \langle \psi(t) | \psi(t) \rangle_S = {}_H \langle \psi(t) | \psi(t) \rangle_H \Rightarrow {}_H \langle \psi(t) | \hat{U}^\dagger \hat{U} | \psi(t) \rangle_H = {}_H \langle \psi(t) | \psi(t) \rangle_H,$



$\hat{F}_S, \hat{F}_H$  are observables under Schrödinger and Heisenberg picture, similarly

$${}_S \langle \psi(t) | \hat{F}_S | \psi(t) \rangle = {}_H \langle \psi(t) | \hat{F}_H | \psi(t) \rangle \Rightarrow {}_H \langle \psi(t) | \hat{U}^\dagger \hat{F}_S \hat{U} | \psi(t) \rangle = {}_H \langle \psi(t) | \hat{F}_H | \psi(t) \rangle$$

Therefore,

$$\hat{F}_H = \hat{U}^\dagger \hat{F}_S \hat{U} \quad \text{or} \quad \hat{F}_S = \hat{U} \hat{F}_H \hat{U}^\dagger$$

Remind  $d\hat{F}_S/dt \equiv 0$ ,  $d/dt \hat{U} = (-i\hat{H}/\hbar)\hat{U}$ ,  $\hat{H}_H = \hat{U}^\dagger \hat{H}_S \hat{U} = \hat{H}_S$ , then

$$\begin{aligned} \frac{d}{dt} \hat{F}_H &= \frac{d}{dt} (\hat{U}^\dagger \hat{F}_S \hat{U}) = \frac{i}{\hbar} \hat{U}^\dagger \hat{F}_S \hat{U} + \hat{U}^\dagger \left( \frac{\partial \hat{F}_S}{\partial t} \right) \hat{U} - \frac{i}{\hbar} \hat{F}_S \hat{H} \hat{U} \\ &= \left( \frac{\partial \hat{F}_S}{\partial t} \right)_H + \frac{1}{i\hbar} (\hat{U}^\dagger \hat{F}_S \hat{U} \hat{U}^\dagger \hat{H} \hat{U} - \hat{U}^\dagger \hat{H} \hat{U} \hat{U}^\dagger \hat{F}_S \hat{U}) \\ &= \left( \frac{\partial \hat{F}_S}{\partial t} \right)_H + \frac{1}{i\hbar} [\hat{F}_H, \hat{H}_H] \\ \frac{d}{dt} |\psi(t)\rangle_H &= \frac{d}{dt} \hat{U}^\dagger |\psi(t)\rangle_S = \left( \frac{d\hat{U}^\dagger}{dt} \right) |\psi(t)\rangle_S + \hat{U}^\dagger \frac{d}{dt} |\psi(t)\rangle_S \\ &= \frac{i}{\hbar} \hat{U}^\dagger \hat{H}_S |\psi(t)\rangle_S + \frac{1}{i\hbar} \hat{U}^\dagger \hat{H}_S |\psi(t)\rangle_S \equiv 0 \end{aligned}$$

### 4.3 Symmetry Transformation

Different to symmetric transformation

Wigner's Theorem

- Unitary: Space translation, space reflection, time translation, permutation
- Anti-unitary: time reversal

#### State Vector

Transfer  $|\psi\rangle$

$$\begin{aligned} |\psi\rangle &\xrightarrow{\hat{U}} |\psi'\rangle = \hat{U} |\psi\rangle \\ |\phi\rangle &\xrightarrow{\hat{U}} |\phi'\rangle = \hat{U} |\phi\rangle \end{aligned}$$

Invariance:

$$\langle \psi | \phi \rangle \xrightarrow{\hat{U}} \langle \psi' | \phi' \rangle = \langle \psi | \phi \rangle$$

Inner product invariance:

$$\langle \psi | \hat{U}^\dagger \hat{U} | \phi \rangle = \langle \psi | \phi \rangle \Rightarrow \hat{U}^\dagger \hat{U} = \mathbb{1} \quad \text{or} \quad \hat{U}^\dagger = \hat{U}^{-1}$$

#### Operator (observables)

$$\langle \psi | \hat{A} | \phi \rangle \xrightarrow{\hat{U}} \langle \psi' | \hat{A}' | \phi' \rangle = \langle \psi | \hat{A} | \phi \rangle$$

Inner product invariance:

$$\langle \psi | \hat{U}^\dagger \hat{A}' \hat{U} | \phi \rangle = \langle \psi | \hat{A} | \phi \rangle \Rightarrow \hat{U}^\dagger \hat{A}' \hat{U} = \hat{A} \quad \text{or} \quad \hat{A}' = \hat{U} \hat{A} \hat{U}^\dagger$$

Linear unitary transformation:

$$\begin{aligned} |\psi'\rangle = \hat{U} |\psi\rangle &\Leftrightarrow \text{Invariance of } \langle \psi | \phi \rangle \\ \hat{A}' = \hat{U} \hat{A} \hat{U}^\dagger &\Leftrightarrow \text{Invariance of } \langle \psi | \hat{A} | \phi \rangle \end{aligned}$$

**Hamilton**

If  $\hat{H} \xrightarrow{\hat{U}} \hat{H}' = \hat{U}\hat{H}\hat{U}^\dagger$ , then one yields  $\hat{U}\hat{H}\hat{U}^\dagger \hat{H} \Rightarrow \hat{U}\hat{H} = \hat{H}\hat{U} \Rightarrow [\hat{U}, \hat{H}] \equiv 0$

**4.4 Symmetry & Conservation in QM**

## 1. Discrete Symmetry

## (a) Space Reflection

$$|\vec{x}\rangle \xrightarrow{\hat{\pi}} |\vec{x}'\rangle = \hat{\pi}|\vec{x}\rangle = |-\vec{x}\rangle \Rightarrow \hat{x} \rightarrow \hat{x}' = \hat{\pi}\hat{x}\hat{\pi}^\dagger = -\hat{x}, \quad \{\hat{x}, \hat{\pi}\} = 0$$

If  $[\hat{\pi}, \hat{H}] = 0$ , then  $\hat{\pi}$  is a conserved quantity, which called parity, and  $\hat{\pi}^\dagger = \hat{\pi}$ .

## (b) Pomutation (transposition)

$$\hat{T}_{ij} |\cdots u_i, \cdots, u_j \cdots\rangle = |\cdots u_j, \cdots, u_i \cdots\rangle = \pm |\cdots u_i, \cdots, u_j \cdots\rangle$$

## 2. Continuous Symmetry

In CM:

- (a) Space translation  $\leftrightarrow$  Conservation of momentum
- (b) Time translation  $\leftrightarrow$  Conservation of energy
- (c) Space rotation  $\leftrightarrow$  Conservation of angular momentum

In QM: Infinitesimal transformation ( $\hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \mathbb{1}$ )

$$\hat{U}(\varepsilon) = \mathbb{1} - i\varepsilon\hat{F} \quad (\varepsilon \ll 1)$$

$$\text{Then } (\mathbb{1} + i\varepsilon\hat{F}^\dagger)\mathbb{1} - i\varepsilon\hat{F} = \mathbb{1} \Rightarrow \mathbb{1} + i\varepsilon(\hat{F}^\dagger - \hat{F}) + \mathcal{O}(\varepsilon^2) = \mathbb{1}$$

- $\hat{U}$ : group
- $\hat{F}$ : generator/observable (Hermitian)
- $\varepsilon/a$ : parameter (like space translation quantity)

Let  $N\varepsilon = a$ ,  $N \gg 1$ ,

$$\underbrace{\hat{U}(\varepsilon) \cdots \hat{U}}_N = \hat{U}^N(\varepsilon) = (\mathbb{1} - i\varepsilon\hat{F})^N = \left[ \mathbb{1} - \frac{1}{N}(-ia\hat{F}) \right]^N \xrightarrow{N \rightarrow \infty} e^{-ia\hat{F}}$$

Summary:

$$\text{Lie Group: } \hat{U}(a) = e^{-ia\hat{F}}$$

$$\text{Lie Algebra: } \hat{u}(\varepsilon) = \mathbb{1} - i\varepsilon\hat{F}$$

**Nother's Theorem****Space Translation**

Define

$$\hat{T}(a)|x+a\rangle \Leftrightarrow \langle x|\hat{T}^\dagger(a) = \langle x+a|, \quad \hat{T}(-a)|x-a\rangle \Leftrightarrow \langle x|\hat{T}^\dagger(-a) = \langle x-a|$$

Since  $\langle x+a|y+a\rangle = \delta_{(x-y)} \Rightarrow \langle x'|y'\rangle = \langle x|\hat{T}^\dagger(a)\hat{T}(a)|y\rangle = \langle x|y\rangle$ , then  $\hat{T}^\dagger(a)\hat{T}(a) = \mathbb{1}$  or  $\hat{T}^\dagger(a) = \hat{T}^{-1}(a)$ ,

Since  $\hat{T}(-a)\hat{T}(a)|x\rangle = \hat{T}(-a)|x+a\rangle = |x\rangle \Rightarrow \hat{T}(-a) = \hat{T}^{-1}(a)$ , therefore

$$\hat{T}^\dagger(a) = \hat{T}^{-1}(a) = \hat{T}(-a)$$

Consider  $\hat{x}\hat{T}(a)|x\rangle = \hat{x}|x+a\rangle = (x+a)|x+a\rangle$ ,  $\hat{T}(a)\hat{x}|x\rangle = \hat{T}(a)x|x\rangle = x\hat{T}(a)|x\rangle = x|x+a\rangle$ , then

$$\left[\hat{x}, \hat{T}(a)\right]|x\rangle = a|x+a\rangle = a\hat{T}(a)|x\rangle \Rightarrow \left[\hat{x}, \hat{T}(a)\right] = a\hat{T}(a)$$

so that  $\hat{x}\hat{T}(a) = \hat{T}(a)\hat{x} + a\hat{T}(a) \Rightarrow \hat{x} = \hat{T}(a)\hat{x}\hat{T}^\dagger(a) + a\mathbb{1}$ , therefore

$$\hat{x}' = \hat{T}(a)\hat{x}\hat{T}^\dagger(a) = \hat{x} - a \cdot \mathbb{1}$$

Taylor series: compare to  $\hat{F}$

$$\hat{T}(a) = \hat{T}(0) + a \left. \frac{d\hat{T}(a)}{da} \right|_{a=0} + \dots = \mathbb{1} + a \left. \frac{d\hat{T}(a)}{da} \right|_{a=0} + \dots = \mathbb{1} - ia\hat{k}$$

So that  $\hat{k} = i a \left. \frac{d\hat{T}(a)}{da} \right|_{a=0}$ , then

$$\frac{d\hat{T}(a)}{da} = \lim_{\epsilon \rightarrow 0} \frac{\hat{T}(a+\epsilon) - \hat{T}(a)}{\epsilon} = \lim_{\epsilon \rightarrow 0} \frac{\hat{T}(\epsilon)\hat{T}(a) - \hat{T}(a)}{\epsilon} = \lim_{\epsilon \rightarrow 0} \frac{\hat{T}(\epsilon) - \mathbb{1}}{\epsilon} \cdot \hat{T}(a) = -i\hat{k}\hat{T}(a)$$

solve the differntiate function, got

$$\hat{T}(a) = \hat{T}(0) \exp\{-ia\hat{k}\} = \exp\{-ia\hat{k}\}$$

Since  $[\hat{x}, \hat{T}(a)] = a\hat{T}(a)$ , take it into and got  $[\hat{x}, \hat{k}] = i$ . Since  $[\hat{x}, \hat{p}] = i\hbar$ , therefore  $\hat{k} = \frac{\hat{p}}{\hbar}$ , and

$$\hat{T}(a) = \exp\left\{\frac{i\vec{a} \cdot \hat{\vec{p}}}{\hbar}\right\}$$

if  $[\hat{T}(a), \hat{H}] = 0$ , then  $[\hat{p}, \hat{H}] = 0 \Rightarrow \hat{p}$  is conserved.

### Time Translation

$$\hat{U}(\tau) = \exp\left\{\frac{i\tau\hat{H}}{\hbar}\right\} \Leftrightarrow [\hat{H}, \hat{H}] \equiv 0 \Rightarrow \text{Energy is conserved.}$$

### Space Rotation

- Rotation axis:  $\hat{\mathbf{n}}$
- Rotation angle:  $\theta$

Rodrigues rotation formula:

$$\vec{x}' = \vec{x} \cos \theta + (1 - \cos \theta)(\vec{x} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}} + (\hat{\mathbf{n}} \times \vec{x}) \sin \theta$$

*Proof.*  $\vec{x} = \vec{x}_{\parallel} + \vec{x}_{\perp}$ ,  $\vec{x}_{\parallel} = (\vec{x} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$ ,  $\vec{x}_{\perp} = \vec{x} - \vec{x}_{\parallel} = \vec{x} - (\vec{x} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$ ,  
 $\vec{x}' = \vec{x}'_{\parallel} + \vec{x}'_{\perp}$ ,  $\vec{x}'_{\parallel} = \vec{x}_{\parallel} = (\vec{x} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$ ,  $\vec{x}'_{\perp} = \vec{x}_{\perp} \cos \theta + (\hat{\mathbf{n}} \times \vec{x}_{\perp}) \sin \theta$ , take into  $\square$

when  $\theta \ll 1$ ,  $\cos \theta \sim 1$ ,  $\sin \theta \sim \theta$ , then  $\vec{x}' = \vec{x} + \theta(\hat{\mathbf{n}} \times \vec{x}) = \vec{x} + \delta \vec{x}$ ,  $\delta \vec{x} = \vec{x}' - \vec{x} = \theta(\hat{\mathbf{n}} \times \vec{x})$ , similarly

$$\begin{aligned} \hat{U}(\theta) &= \mathbb{1} - i\delta \vec{x} \cdot \frac{\vec{p}}{\hbar} = \mathbb{1} - i\theta(\hat{\mathbf{n}} \times \vec{x}) \cdot \frac{\vec{p}}{\hbar} = \mathbb{1} - i\theta(\vec{x} \times \vec{p}) \cdot \frac{\hat{\mathbf{n}}}{\hbar} = \mathbb{1} - i\theta(\vec{L} \cdot \hat{\mathbf{n}})/\hbar \\ &\Rightarrow \hat{U}(\hat{\mathbf{n}}, \theta) = \exp\left\{-i\theta(\hat{\mathbf{n}} \cdot \hat{\vec{L}})/\hbar\right\} \end{aligned}$$

If  $[\hat{U}(\hat{\mathbf{n}}, \theta), \hat{H}] \equiv 0$ , then  $[\hat{\vec{L}}, \hat{H}] \equiv 0 \Rightarrow \hat{\vec{L}}$  is conserved.

**Summary (QM)**

1. Space translation

$$\begin{cases} \hat{T}(\vec{a}) = \exp\left\{-i\vec{a} \cdot \hat{\vec{p}}/\hbar\right\} \\ \hat{t}(\vec{\varepsilon}) = \mathbb{1} - i\vec{\varepsilon} \cdot \hat{\vec{p}}/\hbar \end{cases} \quad \Rightarrow \quad \hat{\vec{p}} \text{ is conserved.}$$

2. Time translation

$$\begin{cases} \hat{U}(\tau) = \exp\left\{+i\tau\hat{H}/\hbar\right\} \\ \hat{u}(\varepsilon) = \mathbb{1} - i\varepsilon\hat{H}/\hbar \end{cases} \quad \Rightarrow \quad E \text{ is conserved.}$$

3. Space rotation

$$\begin{cases} \hat{U}(\hat{\mathbf{n}}, \theta) = \exp\left\{-i\theta\left(\hat{\mathbf{n}} \cdot \hat{\vec{L}}\right)/\hbar\right\} \\ \hat{u}(\hat{\mathbf{n}}, \varepsilon) = \mathbb{1} - i\varepsilon\left(\hat{\mathbf{n}} \cdot \hat{\vec{L}}\right)/\hbar \end{cases} \quad \Rightarrow \quad \hat{\vec{L}} \text{ is conserved.}$$

## Chapter 5

# The One-dimensional Stationary Schrödinger's Equations

### 5.1 Stationary Schrödinger's Equation

Take a general discussion first,

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle, \quad \text{if } \frac{\partial \hat{H}}{\partial t} \equiv 0, \quad \text{then } |\psi(t)\rangle = e^{i\hat{H}t/\hbar} |\psi(0)\rangle$$

Stationary Schrödinger's equation (SSE) or Time-independent Schrödinger's equation or Energy eigenvalue equation:

$$\hat{H} |\psi(0)\rangle = E |\psi(0)\rangle, \quad \text{or at a moment } \hat{H} |\psi(t)\rangle = E |\psi(t)\rangle$$

- For  $\forall \hat{A}$  observables, the expectation is

$$\langle \psi(t) | \hat{A} | \psi(t) \rangle = \langle \psi(0) | e^{iEt/\hbar} \hat{A} e^{-iEt/\hbar} | \psi(0) \rangle = \langle \psi(0) | \hat{A} | \psi(0) \rangle$$

Therefore,  $\langle \hat{A} \rangle(t) = \langle \hat{A} \rangle(t=0)$ , e.g.  $\langle \hat{H} \rangle(t) = \langle \hat{H} \rangle(t=0)$

- The probability of one of eigenvalues  $A_k$  of  $\hat{A}$

$$p(A_k; t) = \left| \langle u_k | \psi(t) \rangle \right|^2 = \left| \langle u_k | e^{-iEt/\hbar} | \psi(0) \rangle \right|^2 = \left| \langle u_k | \psi(0) \rangle \right|^2 = p(A_k; t=0)$$

- Consider  $\hat{H} = E$ , then  $|\psi(t)\rangle = e^{-iEt/\hbar} |\psi(0)\rangle$ , the wave function is

$$\psi(\vec{x}, t) = \langle \vec{x} | \psi(t) \rangle = \langle \vec{x} | e^{-iEt/\hbar} | \psi(0) \rangle = e^{-iEt/\hbar} \psi(\vec{x}, t=0)$$

Take it into the Schrödinger's equation, we can get the stationary Schrödinger's equation

$$\left[ -\frac{\hbar^2}{2m} \nabla^2 + V(\vec{x}) \right] \psi(\vec{x}) = E \psi(\vec{x})$$

- \* Remember that  $\frac{\partial \hat{H}}{\partial t} \equiv 0 \Leftrightarrow \frac{\partial V}{\partial t} \equiv 0$  here

- \* Consider  $\rho(\vec{x}, t) = \psi^*(\vec{x}, t) \psi(\vec{x}, t) = e^{iEt/\hbar} \psi^*(\vec{x}) e^{-iEt/\hbar} \psi(\vec{x}) = |\psi(\vec{x})|^2 \Rightarrow \frac{\partial \rho}{\partial t} = 0$ ,  
since continuity equation, we have  $\vec{\nabla} \cdot \vec{j} \equiv 0$

**Comparison: conserved quantity v.s. stationary state**

1. (a) If  $[\hat{A}, \hat{H}] \equiv 0$ , then  $\hat{A}$  is conserved quantity  
(b)  $\forall \hat{A}$ , if in stationary state, then  $\langle \hat{A} \rangle(t) = \langle \hat{A} \rangle(t=0)$

2. If  $\hat{A}$ ,  $\hat{B}$  are conserved quantities, and  $[\hat{A}, \hat{B}] \neq 0$ , then the system is mostly degenerate;

If  $[\hat{A}, \hat{H}] = [\hat{B}, \hat{H}] \equiv 0$ ,  $[\hat{A}, \hat{B}] \neq 0$ , then degenerate happen,

e.g.  $[\hat{L}_x, \hat{H}] = [\hat{L}_y, \hat{H}] \equiv 0$ ,  $[\hat{L}_x, \hat{L}_y] \neq 0$ , then degenerate,  $|lm_l\rangle$  has  $(2l+1)$  degeneracy, except  $|l=0, m=0\rangle$  \* The relation between symmetry, conservation and degeneracy.

3. More on Ehrenfest's theorem:

$$\frac{d}{dt} \langle \hat{x} \cdot \hat{p} \rangle = \frac{1}{m} \langle \hat{p}^2 \rangle - \langle \hat{x} \cdot \vec{\nabla} V \rangle$$

*Proof.*  $\frac{d}{dt} \langle \hat{x} \cdot \hat{p} \rangle = \frac{1}{i\hbar} \langle [\hat{x} \cdot \hat{p}, \hat{H}] \rangle = \frac{1}{i\hbar} \left\langle \left[ \hat{x} \cdot \hat{p}, \frac{\hat{p}^2}{2m} + V(\hat{x}) \right] \right\rangle = \frac{1}{i\hbar} \left\langle \frac{1}{2m} [\hat{x}_i, \hat{p}_j] 2\hat{p}_j \hat{p}_i + \hat{x}_i [\hat{p}_i, \hat{x}_i] \frac{\partial V}{\partial x_i} \right\rangle$  □

If  $|\psi(t)\rangle$  is stationary state, then  $\frac{d}{dt} \langle \hat{x} \cdot \hat{p} \rangle \equiv 0$ , hence  $\frac{1}{m} \langle \hat{p}^2 \rangle = \langle \hat{x} \cdot \vec{\nabla} V \rangle$ , since  $\hat{T} = \frac{\hat{p}^2}{2m}$ ,

$$\text{Virial Theorem: } 2\langle \hat{T} \rangle = \langle \hat{x} \cdot \vec{\nabla} V \rangle$$

e.g. If  $V(\vec{x}) = V(a\vec{x}) = a^n V(\vec{x}) \Rightarrow 2\langle T \rangle = n\langle \hat{V} \rangle$

Harmonic oscillation:  $V(x) = \frac{1}{2}m\omega^2 x^2 \Rightarrow \langle T \rangle = \langle \hat{V} \rangle$ ; Coulomb's potential:  $V(r) = -\frac{e^2}{r} \Rightarrow \langle T \rangle = -\frac{\langle \hat{V} \rangle}{2}$

## Stationary Schrödinger's Equation in 1D

One-dimensional stationary Schrödinger's equation:

$$\left[ -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \right] \psi(x) = E\psi(x) \Rightarrow \psi''(x) = -\frac{2m}{\hbar^2} [E - V(x)]\psi(x)$$

- $\psi(x)$  is dependent on  $V(x)$
- $\psi$  is continuous integrable, and

$$\int_{-\infty}^{+\infty} |\psi(x)|^2 dx < \infty$$

- Three choices of  $V(x)$ 
  - $V(x)$  is continuous, then  $\psi(x), \psi'(x), \psi''(x)$  are continuous;
  - $V(x)$  has a number of discontinuous, and  $V(x)$  contains no derivatives of  $\delta$ -function, then  $\psi''$  maybe contain  $\delta(x)$ , but  $\psi'(x), \psi(x)$  don't contain  $\delta(x)$ , so  $\psi(x), \psi'(x)$  are continuous;
  - $V(x)$  contains hard walls (e.g. infinite square wall), then  $\psi(x) = 0$  at boundary point, and beyond the wall.
- Definition for solving SSE, we usually define afterwards

$$k^2 = \frac{2m}{\hbar^2} E, \quad \text{and} \quad \kappa^2 = -\frac{2m}{\hbar^2} E, \quad \text{and} \quad k'^2 = \frac{2m}{\hbar^2} [E - V(x)]$$

## 5.2 Potential Well Problem

### 5.2.1 Infinite Square Well

Consider

$$V(x) = \begin{cases} 0, & -a/2 < x < a/2 \\ \infty, & |x| \geq a/2 \end{cases}$$

Beacasue of hard wall,  $\psi(x) \xrightarrow{x \rightarrow \infty} 0$ , and

$$\psi(x) = 0 \quad \text{for} \quad |x| \geq a/2$$

Inside the wall,  $V(x) = 0$ , and use Eq.(??), SSE becomes

$$\psi''(x) = -k^2\psi(x) \quad \text{where} \quad k^2 = \frac{2mE}{\hbar^2}$$

The solution of the equation with two coefficient is

$$\psi(x) = Ae^{ikx} + Be^{-ikx}$$

Consider boundary condition:  $\psi(x)$  is continuous, so

$$\left. \begin{array}{l} \text{for } \psi\left(\frac{a}{2}\right) = 0, \quad Ae^{ika/2} + Be^{-ika/2} = 0 \\ \text{for } \psi\left(-\frac{a}{2}\right) = 0, \quad Ae^{-ika/2} + Be^{ika/2} = 0 \end{array} \right\} \quad k_na = n\pi, \quad \text{and} \quad A = (-1)^{n+1}B, \quad (n = 1, 2, \dots)$$

Typical bound state  $\Leftrightarrow$  discret energy spectrum

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2 \pi^2}{2ma^2} n^2 = E_g n^2$$

- Ground state energy:

$$E_g = \frac{\hbar^2 \pi^2}{2ma^2}$$

After normalization, we have the wave function inside the wall

$$\psi_n(x) = \begin{cases} \sqrt{\frac{2}{a}} \cos\left(\frac{n\pi}{a}x\right), & \text{even } n \\ \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right), & \text{odd } n \end{cases}$$

We can find out that since  $V(x) = V(-x)$  which is even function, therefore  $\psi_n(x)$  has the property of even function when  $n$  is even (odd function when  $n$  is odd). So if  $\hat{H}$  is even function,  $\psi(x)$  has explicit parity.

\*Node theorem:  $\psi(x) = 0$  points, except  $x = \pm \frac{a}{2}$  boundary points, there are  $(n-1)$  nodes for  $\psi_n(x)$ .

### 5.2.2 Finite Square Well (need learn again)

Consider potential is

$$V(x) = \begin{cases} -V_0, & -a/2 < x < a/2 \\ 0, & |x| \geq a/2 \end{cases}$$

Since it is boundary state, we can know the result is discrete energy spectrum. And the SSE is

Boundary condition:  $\psi(x)$ ,  $\psi'(x)$  are both continuous.

$$\xi^2 = ka^2, \quad \eta = \kappa a^2, \quad z_0^2 = \frac{2mV_0a^2}{\hbar^2}$$

$$\text{Even solution: } \xi^2 + \eta^2 = z_0^2, \quad \xi = \eta \tan \eta$$

$$\text{Odd solution: } \xi^2 + \eta^2 = z_0^2, \quad \xi = -\eta \cot \eta$$

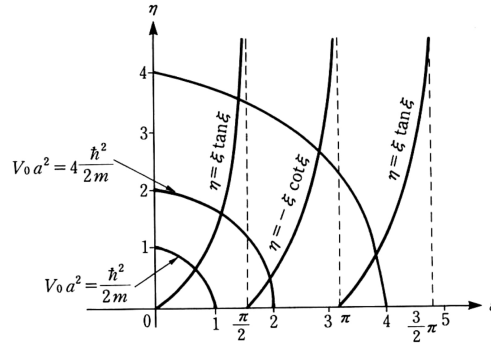


Figure 5.1: Finite square well

### 5.2.3 Harmonic Oscillation

#### Classical Harmonic Oscillation

In Classical Mechanics, the Hamiltonian of harmonic oscillation is

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

If  $E = H$ , then we can get an ellipse equation

$$\frac{x^2}{\left(\sqrt{\frac{2E}{m\omega^2}}\right)^2} + \frac{p^2}{(\sqrt{2mE})^2} = 1 \quad \Rightarrow \quad S = \pi \left( \sqrt{\frac{2E}{m\omega^2}} \right) (\sqrt{2mE}) = \frac{2\pi E}{\omega}$$

If  $E = n\hbar\omega$ , then

$$S = \frac{2\pi n\hbar\omega}{\omega} = hn, \quad n = 1, 2, 3, \dots$$

If  $E = (n + 1/2)\hbar\omega$ ,  $n = 1, 2, 3, \dots$ , then (WKB approximation)

$$S = \frac{2\pi(n + \frac{1}{2})\hbar\omega}{\omega} = h\left(n + \frac{1}{2}\right)$$

From these we can also have Sommerfeld Quantization:

$$S = \oint p dx \quad \Rightarrow \quad \oint p dq = hn \quad \text{or} \quad \oint p dq = h\left(n + \frac{1}{2}\right)$$

Since  $a^2 + b^2 = (a + ib)(a - ib)$ , when  $E = \hbar\omega$ , we have

$$\frac{x}{\sqrt{\frac{2E}{m\omega^2}}} + i\frac{p}{\sqrt{2mE}} \rightarrow \alpha \equiv \sqrt{\frac{m\omega}{2\hbar}}x + i\frac{p}{\sqrt{2m\hbar\omega}} \quad \Rightarrow \quad \alpha\alpha^* = 1$$

#### Quantum Harmonic Oscillation

Similarly, in QM, we can have  $\alpha \rightarrow \hat{a}$ ,  $\alpha^* \rightarrow \hat{a}^\dagger$ , i.e.

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\hat{x} + i\frac{\hat{p}}{\sqrt{2m\hbar\omega}}, \quad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\hat{x} - i\frac{\hat{p}}{\sqrt{2m\hbar\omega}}$$

Since  $[\hat{x}, \hat{p}] = i\hbar$ , therefore we got the commutation relation between  $\hat{a}$  and  $\hat{a}^\dagger$ :

$$[\hat{a}, \hat{a}^\dagger] = 1$$

- $\hat{a}$ : annihilation operator
- $\hat{a}^\dagger$ : creation operator



In QM, we have the Hamiltonian of harmonic oscillation:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)$$

Define occupation number operator as:

$$\hat{N} = \hat{a}^\dagger\hat{a}, \quad \rightarrow \quad \begin{cases} [\hat{N}, \hat{a}] = -\hat{a} & \Rightarrow \quad \hat{N}\hat{a} = \hat{a}\hat{N} - \hat{a} \\ [\hat{N}, \hat{a}^\dagger] = \hat{a}^\dagger & \Rightarrow \quad \hat{N}\hat{a}^\dagger = \hat{a}^\dagger\hat{N} + \hat{a}^\dagger \end{cases}$$

Let  $\hat{N}|\lambda\rangle = \lambda|\lambda\rangle$ , we have

$$\begin{aligned} \hat{N}(\hat{a}|\lambda\rangle) &= (\hat{a}\hat{N} - \hat{a})|\lambda\rangle = \lambda\hat{a}|\lambda\rangle - \hat{a}|\lambda\rangle = (\lambda - 1)(\hat{a}|\lambda\rangle) \Rightarrow \hat{a}|\lambda\rangle = c_-|\lambda - 1\rangle \\ \langle\lambda|\hat{a}^\dagger &= c_-^*\langle\lambda - 1| \end{aligned}$$

Then, ask for normalization

$$\begin{aligned} \langle\lambda|\hat{a}^\dagger\hat{a}|\lambda\rangle &= |c_-|^2\langle\lambda - 1|\lambda - 1\rangle = |c_-|^2 \\ &= \langle\lambda|\hat{N}|\lambda\rangle = \lambda\langle\lambda|\lambda\rangle = \lambda \Rightarrow |c_-|^2 = \lambda \rightarrow c_- = \sqrt{\lambda} \end{aligned}$$

Therefore, we can get series of state

$$\hat{a}|\lambda\rangle = \sqrt{\lambda - 1} \Rightarrow \hat{a}|0\rangle = \sqrt{0}|0 - 1\rangle = 0 \Rightarrow \hat{a}|0\rangle = 0, \hat{a}|1\rangle = |0\rangle, \hat{a}|2\rangle = \sqrt{2}|1\rangle, \dots$$

We can assume  $|\lambda\rangle = |n\rangle$ , therefore

$$\hat{N}|n\rangle = n|n\rangle, \quad \hat{a}|n\rangle = \sqrt{n}|n - 1\rangle \quad (n = 0, 1, 2, \dots)$$

Similarly, for creation operator we have

$$\hat{N}\hat{a}^\dagger|n\rangle = (\hat{a}^\dagger\hat{N} + \hat{a}^\dagger)|n\rangle = (n + 1)\hat{a}^\dagger|n\rangle \Rightarrow \hat{a}^\dagger|n\rangle = c_+|n + 1\rangle \quad \text{where } c_+ = \sqrt{n + 1}$$

For the ground state, we can write it in

$$|n\rangle = \frac{\hat{a}^\dagger}{\sqrt{n}}|n - 1\rangle = \frac{\hat{a}^\dagger}{\sqrt{n}} \cdot \frac{\hat{a}^\dagger}{\sqrt{n - 1}}|n - 2\rangle = \dots = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle$$

Summary:

1.  $[\hat{a}, \hat{a}^\dagger] = 1$
2.  $\hat{N} = \hat{a}^\dagger\hat{a}, \quad [\hat{N}, \hat{a}] = -\hat{a}, \quad [\hat{N}, \hat{a}^\dagger] = \hat{a}^\dagger$
3.  $\hat{N}|n\rangle = n|n\rangle, \quad \hat{a}|n\rangle = \sqrt{n}|n - 1\rangle, \quad \hat{a}^\dagger|n\rangle = \sqrt{n + 1}|n + 1\rangle \quad (n = 0, 1, 2, \dots)$
4.  $\hat{a}|0\rangle = 0$
5.  $|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle$
6. ON:  $\langle n|m\rangle = \delta_{nm}$ , RI:  $\sum_{n=0}^{\infty} |n\rangle\langle n| = \mathbb{1}$
7.  $\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right), \quad \hat{H}|n\rangle = \hbar\omega\left(n + \frac{1}{2}\right)|n\rangle, \quad \hat{H} = \hbar\omega\left(\hat{N} + \frac{1}{2}\right), \quad \hat{H}|0\rangle = \frac{1}{2}\hbar\omega|0\rangle \quad (\text{zero point energy})$

In vacume, since  $\Delta\hat{x}^2 + \Delta\hat{p}^2 = \hbar^2/4$  is the minimum uncertainty,

$$\langle 0|\hat{x}^2|0\rangle = (\langle 0|\hat{x}|0\rangle)^2 = \Delta\hat{x}^2, \quad \langle 0|\hat{p}^2|0\rangle = (\langle 0|\hat{p}|0\rangle)^2 = \Delta\hat{p}^2$$

which means  $|0\rangle$  is minimum uncertainty state.

Next, discuss wave function of  $|0\rangle$  in  $\hat{x}$ -rep: from  $\langle x|\hat{a}|0\rangle = 0$

$$\langle x|\left(\sqrt{\frac{m\omega}{2\hbar}}x + \sqrt{\frac{\hbar}{2m\omega}}\frac{d}{dx}\right)|0\rangle = 0$$

since  $\psi_0(x) = \langle x|0\rangle$ , take into

$$x\psi_0(x) + \frac{\hbar}{m\omega}\frac{d}{dx}\psi_0(x) = 0 \quad \rightarrow \quad \frac{d\psi_0(x)}{\psi_0(x)} = -\frac{m\omega}{\hbar}x dx \quad \rightarrow \quad \psi_0(x) = A \exp\left\{-\frac{m\omega x^2}{2\hbar}\right\} \xrightarrow{\text{normalization}} A = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}}$$

Expand to  $\psi_n(x)$ ,

$$\begin{aligned} \psi_n(x) &= \langle x|n\rangle = \langle x|\frac{(\hat{a})^n}{\sqrt{n!}}|0\rangle = \langle x|\frac{1}{\sqrt{n!}}\left(\sqrt{\frac{m\omega}{2\hbar}}\hat{x} - i\frac{\hat{p}}{2m\hbar}\right)^n|0\rangle = \frac{1}{\sqrt{2^n n!}}\langle x|\left(\sqrt{\frac{m\omega}{\hbar}}x - \sqrt{\frac{\hbar}{m\omega}}\frac{d}{dx}\right)^n|0\rangle \\ &= \frac{1}{\sqrt{2^n n!}}\langle x|\left(z - \frac{d}{dz}\right)^n|0\rangle = \frac{1}{\sqrt{2^n n!}}\left(z - \frac{d}{dz}\right)^n\psi_0(x) \quad \text{let } z = \alpha x = \sqrt{m\omega/\hbar}x \\ &= \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}}\frac{1}{\sqrt{2^n n!}}\left(z - \frac{d}{dz}\right)^n e^{-z^2/2} \end{aligned}$$

Let

$$\left(z - \frac{d}{dz}\right)^n e^{-z^2/2} = H_n(z)e^{-z^2/2}, \quad \psi_n(x) \sim H_n(\alpha x)e^{-\alpha^2 x^2/2}$$

- $H_n(z)$  is Hermitian polynomial
- $H_n(-z) = (-1)^n H_n(z)$ , when  $n$  is even wave function is even.

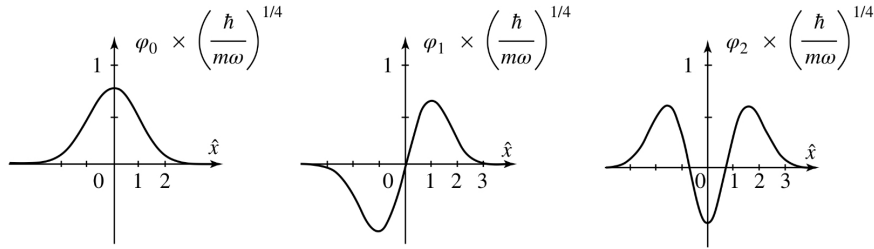


Figure 5.2: Wave function

### A discussion about boundary condition of SSE

Integrate  $\psi''(x) = -\frac{2m}{\hbar^2}(E - V(x))\psi(x)$  of each side,

$$\begin{aligned} \text{l.h.s} &= \int_{x_0-\epsilon}^{x_0+\epsilon} \frac{d}{dx}\psi'(x)dx = \psi'(x)\Big|_{x_0-\epsilon}^{x_0+\epsilon} = \psi'(x_0+\epsilon) - \psi'(x_0-\epsilon) = \Delta x_0 \psi'(x) \\ \text{r.h.s} &= -\frac{2m}{\hbar} \int_{x_0-\epsilon}^{x_0+\epsilon} (E - V(x))\psi(x)dx = -\frac{2m}{\hbar^2}(E - V(\xi))\psi(\xi) \cdot (2\epsilon) \end{aligned}$$

When  $\epsilon \rightarrow 0$ ,  $\Delta x_0 \psi'(x) = 0$ , then  $\psi'(x)$  is continuous. However, if  $V(x)$  is infinite,  $V(x) = -a\delta(x - x_0)$ ,

$$\text{r.h.s} = -\frac{2m}{\hbar^2} \int_{x_0-\epsilon}^{x_0+\epsilon} a\delta(x - x_0)\psi(x)dx = -\frac{2ma}{\hbar^2}\psi(x_0) \Rightarrow \Delta x_0 \psi'(x) = \frac{2m}{\hbar^2}\psi(x_0)$$

To proof  $\psi'(x)$  is continuous, unless  $V(x) \rightarrow \infty$ ,

suppose  $\psi(x)$  is discontinuous, wave function has a break point, and  $\psi'(x) = \delta(x)$ , then

$$\begin{aligned}\langle \hat{T} \rangle &\sim \langle \hat{p}^2 \rangle \sim \int \psi^* \frac{d^2}{dx^2} \psi = \int \psi^* \frac{d}{dx} \psi' dx = \int \psi^* \frac{d}{dx} \delta(x) dx = \int \psi^* d\delta(x) \\ &= \psi^* \delta(x) \Big|_{-\infty}^{\infty} - \int \delta(x) \frac{d\psi^*}{dx} dx = 0 - \int_{-\infty}^{\infty} \delta(x) \delta(x) dx \sim \delta(0) = \infty\end{aligned}$$

### 5.3 Delta Potential

Potential

$$V(x) = -a\delta(x) \quad (a > 0)$$

**When  $E < 0$ : (Bound state)**

The wave function is

$$\psi''(x) = \kappa^2 \psi(x) \Rightarrow \psi(x) = Ae^{\kappa x} + Be^{-\kappa x}, \quad \kappa = \frac{\sqrt{-2mE}}{\hbar}$$

For region I:  $x \rightarrow -\infty$ ,  $\psi_I(x) \rightarrow 0 \Rightarrow \psi_I = Ae^{\kappa x}$ . Similarly in region II,  $\psi_{II}(x) = Be^{-\kappa x}$ .

Consider boundary condition, since  $\psi(x)$  is continuous,  $\psi_I(0) = \psi_{II}(0)$ , then  $A = B$ .

For  $\psi'(x)$ ,

$$\begin{aligned}\lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{\epsilon} \left( -\frac{\hbar^2}{2m} \psi''(x) - a\delta(x)\psi(x) \right) dx &= \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{\epsilon} E\psi(x) dx \\ \Rightarrow -\frac{\hbar^2}{2m} \psi'(x) \Big|_{-\epsilon}^{+\epsilon} - a\psi(0) &= E\psi(0) \cdot 2\epsilon \rightarrow 0 \\ \Rightarrow \psi'_{II}(0) - \psi'_{I}(0) &= -\frac{2ma}{\hbar^2} \psi(0) \Rightarrow -A\kappa - A\kappa = -\frac{2ma}{\hbar^2} A; \quad \psi(0) = A\end{aligned}$$

Therefore,

$$\kappa = \frac{ma}{\hbar^2}, \quad E = -\frac{\hbar^2 \kappa^2}{2m} = -\frac{ma^2}{2\hbar^2}$$

$$\text{Normalization, } \int_{-\infty}^0 |A|^2 e^{2\kappa x} dx + \int_0^{\infty} |A|^2 e^{-2\kappa x} dx = 1 \Rightarrow A = \sqrt{\kappa} = \frac{\sqrt{ma}}{\hbar}.$$

Bound state solution:

$$E = -\frac{ma^2}{2\hbar^2}; \quad \psi(x) = \frac{\sqrt{ma}}{\hbar} \exp\left\{-\frac{\sqrt{ma}}{\hbar}|x|\right\}$$

**When  $E > 0$ : (Scattering state)**

The wave function is

$$\psi''(x) = -k^2 \psi(x) \Rightarrow \begin{cases} \psi_I(x) = Ae^{ikx} + Be^{-ikx} \\ \psi_{II}(x) = Fe^{ikx} + Ge^{-ikx} \end{cases}, \quad k = \frac{\sqrt{2mE}}{\hbar}$$

Since no reflection in region II, then  $G = 0$ .

Consider boundary condition, since  $\psi(x)$  is continuous,  $A + B = F$ . Next, for  $\psi'(x)$ ,

$$\begin{aligned}\psi'_{II}(0) - \psi'_{I}(0) &= -\frac{2ma}{\hbar^2} \psi(0) \Rightarrow ik(F - G) - ik(A - B) = -\frac{2ma}{\hbar^2} (A + B) \\ \Rightarrow F - G &= A(1 + 2i\beta) - B(1 - 2i\beta) \quad \text{where } \beta = \frac{ma}{\hbar^2 k}\end{aligned}$$

Compare the relations, we got

$$B = \frac{i\beta}{1 - i\beta} A, \quad F = \frac{1}{1 - i\beta} A$$

- Reflection coefficient:  $R = \left| \frac{B}{A} \right|^2 = \frac{\beta^2}{1 + \beta^2}$

- Transmission coefficient:  $T = \left| \frac{F}{A} \right|^2 = \frac{1}{1 + \beta^2}$

NOTE:  $\delta$ -well ( $a > 0$ ) &  $\delta$ -barrier  $a < 0$

## 5.4 Barrier Problem

### 5.4.1 Step Barrier

Consider

$$V(x) = \begin{cases} 0, & x < 0 \\ V_0, & x \geq 0 \end{cases}$$

**When  $E > V_0$ :**

$$\begin{aligned} \text{for } x < 0, \quad \psi'' &= -k^2\psi, \quad \psi(x) = Ae^{ikx} + Be^{-ikx}, \quad \text{where } k = \frac{\sqrt{2mE}}{\hbar} \\ \text{for } x \geq 0, \quad \psi'' &= -k'^2\psi, \quad \psi(x) = Ce^{ik'x}, \quad \text{where } k' = \frac{\sqrt{2m(E - V_0)}}{\hbar} \end{aligned}$$

Boundary condition:

$$\left. \begin{aligned} \psi(0^+) &= \psi(0^-) \Rightarrow A + B = C \\ \psi'(0^+) &= \psi'(0^-) \Rightarrow ik(A - B) = ik'C \end{aligned} \right\} \quad \frac{B}{A} = \frac{k - k'}{k + k'}, \quad \frac{C}{A} = \frac{2k}{k + k'} \quad \text{and} \quad A, B, C \in \mathbb{R}$$

**When  $E = V_0$ :**

$$\psi(x) = \begin{cases} 2A \cos(kx), & (x \neq 0) \\ 2A, & (x > 0) \end{cases}$$

Using the probability current density:  $\vec{\nabla} \cdot \vec{j} = 0, \partial\rho/\partial t \equiv 0$

$$\vec{j} = \frac{\hbar}{2mi} (\psi^* \vec{\nabla} \psi - \psi \vec{\nabla} \psi^*) = \frac{1}{m} \text{Re} \left( \psi^* \frac{\hbar}{i} \vec{\nabla} \psi \right)$$

$j_A \leftrightarrow \psi_A, j_B \leftrightarrow \psi_B, j_C \leftrightarrow \psi_C, j_l \leftrightarrow (\psi_A + \psi_B), j_r \leftrightarrow \psi_C$ , take wave function expression into it, we got

$$j_A = \frac{\hbar k}{m} |A|^2, j_B = \frac{\hbar k}{m} |B|^2, j_l = j_A - j_B = \frac{\hbar k}{m} (|A|^2 - |B|^2), j_C = j_r = \frac{\hbar k'}{m} |C|^2$$

since  $dj/dx = 0$ , then  $j_l = j_r$ ,

$$k(|A|^2 - |B|^2) = k'|C|^2$$

- Reflection coefficient:  $R = \frac{j_B}{j_A} = \frac{|B|^2}{|A|^2} = \frac{|k - k'|^2}{|k + k'|^2} \neq 1$
- Transmission coefficient:  $T = \frac{j_C}{j_A} = \frac{k'|C|^2}{k|A|^2} = \frac{4kk'}{(k + k')^2}$
- Reflection+Transmission:  $R + T = 1$

**When  $E < V_0$ :**

$$\begin{aligned} \text{for } x < 0, \quad \psi'' &= -k^2\psi, \quad \psi(x) = Ae^{ikx} + Be^{-ikx}, \quad \text{where } k = \frac{\sqrt{2mE}}{\hbar} \\ \text{for } x \geq 0, \quad \psi'' &= \kappa^2\psi, \quad \psi(x) = Ce^{\kappa x}, \quad \text{where } \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar} \end{aligned}$$

Introduce  $\theta(E)$ , [PICTURE of theta]

$$\theta(E) = \arg(\kappa + ik) = \tan^{-1} \left( \frac{k}{\kappa} \right) = \tan^{-1} \sqrt{\frac{E}{V_0 - E}}, \quad \theta'(E) = \frac{1}{2} \sqrt{\frac{1}{E(V_0 - E)}}$$

From boundary condition got

$$\frac{B}{A} = \frac{k - i\kappa}{k + i\kappa} = -\frac{\kappa + ik}{\kappa - ik} = -e^{2i\theta(E)}, \quad \kappa + ik = \sqrt{\kappa^2 + k^2} e^{i\theta(E)}$$

Probability current:  $j_A = j_B = \frac{\hbar k}{m}(k^2 \kappa^2)$ ,  $j_C = 0 \Rightarrow T = 0, R = 1$ .

For  $x < 0$ ,

$$\psi^2(x) = \left[ A e^{i\theta(E)} \left( e^{-i\theta(E)} e^{ikx} + e^{i\theta(E)} e^{-ikx} \right) \right]^2 = 4A^2 \sin^2(kx - \theta(E))$$

When  $\psi(x_0) = 0$ ,  $kx_0 = \theta(E)$ .

**Tunneling effect** [NEED RECTIFY]

Penetrate depth:

$$d_p \sim \frac{1}{\kappa}$$

**General discussion of SSE**

1. If  $V \in \mathfrak{R}$ , then  $\psi(x)$  can be chosen to be real.

*Proof.*  $\psi'' + \frac{2m}{\hbar^2}(E - V(x))\psi = 0$ ,  $\psi^{*''} + \frac{2m}{\hbar^2}(E - V(x))\psi^* = 0 \Rightarrow \psi, \psi^*$  are both solution;

$\psi_R = \frac{1}{2}(\psi + \psi^*)$ ,  $\psi_I = \frac{1}{2i}(\psi - \psi^*)$  are both solution. □

2. If  $V(-x) = V(x)$ , then  $\psi(x)$  can be chose to be even/odd.

### 5.4.2 Finite Square Barrier

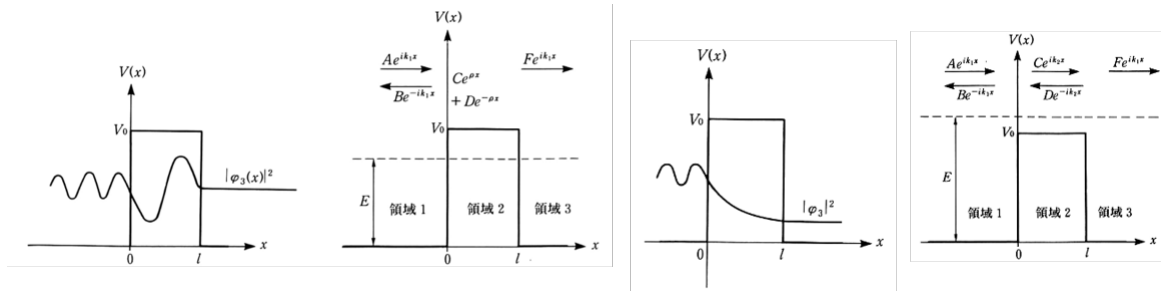


Figure 5.3: Wave function

## Chapter 6

# Central Potential and The Hydrogen Atom

### 6.1 Stationary States of a Particle in a Central Potential

Review basis:

1.  $V(\vec{x}) = V(r) \Rightarrow [\hat{\vec{L}}, \hat{H}] \equiv 0 \Rightarrow [\hat{L}_i, \hat{H}] \equiv 0, [\hat{\vec{L}}^2, \hat{H}] \equiv 0 \Rightarrow \hat{\vec{L}}^2, \hat{L}_z$  are conserved quantities
2. Simultaneous eigenstates with  $\hat{H}$ : CSCO  $\hat{\vec{L}} \leftrightarrow l; \hat{L}_z \leftrightarrow m_l \Rightarrow |l, m_l\rangle$
3. Angular momentum:  $\vec{L} = \vec{x} \times \vec{p}, \hat{L}_i = \varepsilon_{ijk} \hat{x}_j \hat{p}_k$ .
4. Eigenvalue equation for  $\hat{\vec{L}}^2, \hat{L}_z$ :  $\hat{\vec{L}}^2 |l, m_l\rangle = \hbar^2 l(l+1) |l, m_l\rangle; \hat{L}_z |l, m_l\rangle = \hbar m_l |l, m_l\rangle$
5. Orthonormality + Completeness:  $\langle lm | l' m' \rangle = \delta_{ll'} \delta_{mm'}; \sum_n |lm\rangle \langle ml| = \mathbb{1}$

#### 6.1.1 Spherical Harmonic Function

Consider spherical coordinate,

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}, \quad \hat{\vec{L}}^2 = -\hbar^2 \left[ \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]$$

Since  $|l, m_l\rangle$  in  $\vec{x}$ -rep is

$$\langle \vec{x} | l, m_l \rangle = \langle \theta, \phi | l, m_l \rangle = Y_{lm}(\theta, \phi)$$

By solving eigen equations in spherical coordinate,

$$\hat{\vec{L}}^2 Y_{lm}(\theta, \phi) = \hbar^2 l(l+1) Y_{lm}(\theta, \phi), \quad \hat{L}_z Y_{lm}(\theta, \phi) = \hbar m_l Y_{lm}(\theta, \phi)$$

We got <sup>1</sup>

$$Y_{lm}(\theta, \phi) \propto P_l^m(\cos \theta) e^{im\phi} \quad \hat{L}_z e^{im\phi} = \hbar m e^{im\phi}$$

or Spherical Harmonic function:

$$\begin{aligned} Y_{lm}(\theta, \phi) &= (-1)^m \sqrt{\frac{(2l+1)!(l-m)!}{4\pi(l+m)!}} e^{im\phi} P_l^m(\cos \theta) \\ &= \frac{(-1)^l}{2^l l!} \sqrt{\frac{(2l+1)!(l-m)!}{4\pi(l+m)!}} e^{im\phi} (\sin \theta)^m \frac{d^{l-m}}{d(\cos \theta)^{l-m}} (\sin \theta)^{2l} \end{aligned}$$

Need to remember:

---

<sup>1</sup> $P_l^m(\theta, \phi)$ : associated Lagandre function

- $Y_{00}(\theta, \phi) = \frac{1}{\sqrt{4\pi}}$
- $Y_{10}(\theta, \phi) = \sqrt{\frac{3}{4\pi}} \cos \theta = \sqrt{\frac{3}{4\pi}} \left(\frac{z}{r}\right)$
- $Y_{1\pm 1}(\theta, \phi) = \mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\phi} = \mp \sqrt{\frac{3}{8\pi}} \left(\frac{x \pm iy}{r}\right)$

and Orthonormality:

$$\int \langle lm | \vec{x} \rangle \langle \vec{x} | l'm' \rangle d\vec{x} = \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta Y_{lm}^*(\theta, \phi) Y_{l'm'}(\theta, \phi) = \delta_{ll'} \delta_{mm'}$$

### 6.1.2 Radical Wave Function

Wave function of eigen state of Harmonic:

$$\hat{H} |nlm\rangle = E_{nlm} |nlm\rangle, \quad \langle \hat{x} | nlm \rangle = \psi_{nlm}(r, \theta, \phi), \quad \psi_{nlm}(r, \theta, \phi) = R_{nl}(r) Y_{lm}(\theta, \phi)$$

- $R_{nl}(r)$ : radical wave function
- $r \leftrightarrow$  radial;  $\theta, \phi \leftrightarrow$  azimuthal

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(r) = \frac{\hat{p}_r^2}{2m} + \frac{\hat{L}^2}{2mr^2} + V(r)$$

Take  $\hat{H}$  into SSE,

$$\left\{ -\frac{\hbar^2}{2m} \left[ \frac{1}{r} \left( \frac{d^2}{dr^2} r \right) \right] + \frac{l(l+1)\hbar^2}{2mr^2} + V(r) \right\} R_{nl}(r) = E_{nl} R_{nl}(r)$$

Rewrite  $R_{nl}(r)$  and define effective potential as

$$R_{nl}(r) = \frac{1}{r} u_{nl}(r), \quad V_{\text{eff}}(r) = V(r) + \frac{l(l+1)\hbar^2}{2mr^2}$$

Then

$$\left[ -\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + V_{\text{eff}}(r) \right] u_{nl}(r) = E_{nl} u_{nl}(r)$$

- $l(l+1)\hbar^2/2mr^2 \rightarrow$  azimuthal kinetic energy
- $-(\hbar^2/2m) d^2/dr^2 \rightarrow$  radical kinetic energy

If coulomb potential  $V(r) \sim 1/r$ ,  $V_{\text{eff}} \sim -A/r + B/r^2$ . [PICTURE, Veff(r)]

#### Asymptotic analysis

$r \rightarrow 0$ , require that  $|u(r)| \ll N$ , assume that  $V(r) \sim 1/r$  or faster,  $r^2 V(r) \rightarrow 0$ , then SSE become

$$\frac{d^2}{dr^2} u_{nl}(r) = \frac{l(l+1)}{r^2} u_{nl}(r)$$

If  $u_{nl}(r) = r^\alpha$ ,

$$\frac{d^2 u_{nl}}{dr^2} \sim \alpha(\alpha-1)r^{\alpha-2}, \quad \frac{l(l+1)}{r^2} u_{nl} \sim l(l+1)r^{\alpha-2}$$

Therefore,  $\alpha(\alpha-1) = l(l+1)$ , and

$$\begin{cases} \alpha = l+1 & \Rightarrow u_{nl}(r) = r^{l+1} \\ \alpha = -l & \Rightarrow u_{nl}(r) = r^{-l} \end{cases}$$

If  $R_{nl}(r) = r^{-l-1}$ , since  $\nabla^2(1/r) = -4\pi\delta(x)$ , then  $\nabla^2(r^{-(l+1)}) \sim \delta^{(l)}$ ,  $\psi$  is not conserved, give up it.

$R_{nl}(r) = r^l$  is adopted. When  $r \rightarrow 0$ ,  $R_{nl}(r) \rightarrow 0$ ,

$$\psi(r, \theta, \phi) = Nr^l Y_{lm}(\theta, \phi)$$

Normalization:

$$\int_0^{2\pi} d\phi \int_0^\pi \sin\theta d\theta |Y_{lm}(\theta, \phi)|^2 \int_0^\infty r^2 dr |R_{nl}(r)|^2 = 1$$

## 6.2 The Hydrogen Atom

### 6.2.1 Center-of-Mass System: Two Body

Central potential and kinetic energy:

$$\hat{V}(\vec{x}) = \hat{V}(r) \Rightarrow [\hat{\vec{L}}, \hat{H}] = 0$$

$$\hat{T} = \frac{\hat{\vec{p}}^2}{2m} = \frac{\hat{p}_r^2}{2m} + \frac{\hat{L}^2}{2mr^2}$$

Suppose  $m_1, \vec{x}_1, \vec{p}_1$  and  $m_2, \vec{x}_2, \vec{p}_2$ , then

$$\hat{H} = \frac{\hat{\vec{p}}_1^2}{2m} + \frac{\hat{\vec{p}}_2^2}{2m} + V(|\vec{x}_1 - \vec{x}_2|)$$

Define center of mass and other valuables:

$$\vec{X} = \frac{m_1 \vec{x}_1 + m_2 \vec{x}_2}{m_1 + m_2}, \quad \vec{P} = \vec{p}_1 + \vec{p}_2, \quad \vec{x} = \vec{x}_1 - \vec{x}_2, \quad \vec{p} = \alpha \vec{p}_1 - \beta \vec{p}_2$$

And  $\vec{p}_1(\vec{P}, \vec{p})$ ,  $\vec{p}_2(\vec{P}, \vec{p})$ ,  $\vec{x}_1(\vec{X}, \vec{x})$ ,  $\vec{x}_2(\vec{X}, \vec{x})$ , satisfy commutation relations

$$[\hat{x}_{ai}, \hat{x}_{bj}] = [\hat{p}_{ai}, \hat{p}_{bj}] \equiv 0, \quad [\hat{x}_{ai}, \hat{p}_{bj}] = i\hbar \delta_{ab} \delta_{ij}, \quad \begin{array}{l} \text{particle: } a, b = 1, 2 \\ \text{coordinate: } i, j, k = 1, 2, 3 \end{array}$$

Therefore,

$$\alpha = \frac{\mu}{m_1}, \quad \beta = \frac{\mu}{m_2} \quad \text{and} \quad \hat{H} = \frac{\hat{\vec{p}}^2}{2\mu} + \frac{\hat{\vec{P}}^2}{2M} + V(r)$$

- Reduced mass:  $\mu = \frac{m_1 m_2}{m_1 + m_2}$ ; total mass:  $M = m_1 + m_2$
- $r = |\vec{x}_1 - \vec{x}_2| = |\vec{x}|$

Two body  $\rightarrow$  One body, if  $m_1 \ll m_2$ , then  $\mu = m_1$ .

The center of mass and relative SSE are:

$$\begin{aligned} \frac{\hat{\vec{P}}^2}{2M} \Psi_{\text{CM}} &= E_{\text{CM}} \Psi_{\text{CM}} \\ \left[ \frac{\hat{\vec{p}}^2}{2\mu} + V(|\vec{x}|) \right] \Psi_{\text{rel}} &= E_{\text{rel}} \Psi_{\text{rel}} \end{aligned}$$

### 6.2.2 Hydrogen Atom

#### Bohr Model (Semi-Classical)

Potential:

$$V(r) = -\frac{e^2}{r}$$

- Hydrogen-like:  $-Ze^2/r$

Bohr-radius:

$$a_0 = \frac{\hbar^2}{me^2} \sim 0.05\text{nm}$$



- Length scale: since  $\frac{p^2}{2m} \sim \frac{1}{2m} \hbar^2 \nabla^2 \sim \frac{\hbar^2}{ma_0^2}$  and  $V(r) \sim \frac{e^2}{a_0}$ , then  $\frac{e^2}{a_0} = \frac{\hbar^2}{ma_0^2} \Rightarrow a_0 = \frac{\hbar^2}{me^2}$

From  $T = V/2$ , the energy is

$$E_1 = -\frac{e^2}{2a_0} \sim -13.6\text{eV}, \quad E_n = E_1 \cdot \frac{1}{n^2} = -\frac{me^4}{2\hbar^2 n^2} \sim \alpha^2 \cdot (mc^2)$$

where fine structure constant define as (Gauss-unit)

$$\alpha = \frac{e^2}{\hbar c} = \frac{1}{137}$$

- Sommerfield constant:  $\alpha^2 = 1/19000$

Besides,  $E_n \leftrightarrow \psi_{nlm}(r, \theta, \phi)$ , if no spin considered, the number of state are  $\sum_l (2l+1) = n^2$ ; if considered, then  $2n^2$ .

Since under Coloumb field, electron is bounded with nuclei, then  $E < 0$ .  $u(r) = R(r)/r$ , the SSE is

$$\left[ -\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + \frac{l(l+1)\hbar^2}{2mr^2} - \frac{e^2}{r} \right] u(r) = E u(r)$$

Let  $x = 2r/a_0$ , then

$$\left[ -\frac{d^2}{dx^2} + \frac{l(l+1)}{x^2} - \frac{1}{x} \right] u(x) = -\kappa u(x), \quad \kappa = \frac{-E}{2e^2/a_0}$$

- When  $x \rightarrow \infty$ ,

$$\frac{d^2 u}{dx^2} = \kappa u \Rightarrow u \sim e^{\pm \kappa x}$$

Let  $\rho = \kappa x$ , then

$$\left[ -\frac{d^2}{d\rho^2} + \frac{l(l+1)}{\rho^2} - \frac{1}{\kappa\rho} \right] u(\rho) = -u$$

- When  $\rho \rightarrow \infty$ ,  $u \sim e^{-\rho}$
- When  $\rho \rightarrow 0$ ,  $u \sim \rho^{l+1}$

Therefore, we can write  $u(\rho)$  with polynomial function as

$$u(\rho) = e^{-\rho} \rho^{l+1} W(\rho), \quad W(\rho) = \sum_{k=0}^{n_r} a_k \rho^k$$

- $\exists k_{\max}, \forall k > k_{\max}, a_k = 0$
- $n_r$ : number of nodes  $n = n_r + l + 1 \Rightarrow n_r = n - 1, n - 2, \dots, 0$
- Associated Leguered function (polynomial):

$$W(\rho) = L^{2l+1}_{n_r}(\rho) = a_0 \rho^0 + a_1 \rho^1 + \dots + a_{n_r} \rho^{n_r}$$

The solution of SSE for hydrogen atom is:

$$E_n = -\kappa^2 \frac{2e^2}{a_0} = -\frac{1}{(n_r + l + 1)^2} \cdot \frac{e^2}{2a_0} = -\frac{1}{n^2} \cdot E_1 = -13.6\text{eV} \cdot \frac{1}{n^2}$$

$$\Psi_{nlm}(r, \theta, \phi) = N \cdot \left( \frac{r}{a_0} \right)^l L^{2l+1}_{n_r} e^{-r/na_0} Y_{lm}(\theta, \phi) = \sqrt{\frac{2}{(na_0)^3} \frac{(n-l-1)!}{2n(l+1)!}} e^{-\rho} \rho^l L^{2l+1}_{n_r}(\rho) e^{-r/na_0} Y_{lm}(\theta, \phi)$$

$$\text{where } \rho = \frac{r}{na_0}$$

[energy picture]

# Chapter 7

## Approximation Methods

### 7.1 Variational Theory and Helium Atom

#### Raleign-Ritz Variational Principle

Expection value of energy is

$$\langle \hat{H} \rangle = \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} \Rightarrow \delta \langle \hat{H} \rangle - \lambda \delta (\langle \psi | \psi \rangle) = 0$$

where  $\lambda$  is Lagrange multiplier.

#### Ritz Variational Principle

1. Trial wave function  $|\psi(c_i)\rangle$   $i = 1, 2, 3 \dots$  as parameter
2.  $\langle \hat{H} \rangle = \frac{\langle \psi(c_i) | \hat{H} | \psi(c_i) \rangle}{\langle \psi(c_i) | \psi(c_i) \rangle}$
3.  $\delta \langle \hat{H} \rangle = 0 \Rightarrow \sum_i \frac{\partial \langle \hat{H} \rangle}{\partial c_i} \delta c_i = 0 \Rightarrow \frac{\partial}{\partial c_i} \langle \hat{H} \rangle (c_i) = 0 \Rightarrow c_i \Rightarrow \hat{H}, |\psi(c_i)\rangle$

#### Helium Atom

The Harmiltonian of helium atom is

$$\hat{H} = \left( -\frac{\hbar^2}{2m_1} \nabla_1^2 - \frac{2e^2}{r_1} \right) + \left( -\frac{\hbar^2}{2m_2} \nabla_2^2 - \frac{2e^2}{r_2} \right) + \frac{e^2}{r_{12}} = \hat{H}_{01} + \hat{H}_{02} + \hat{H}_z$$

Trivial wavefunction:

$$\Psi(\vec{x}_1, \vec{x}_2) = \Psi_{100}(\vec{x}_1) \Psi_{100}(\vec{x}_2) = \left\{ \frac{1}{\sqrt{\pi}} \left( \frac{Z}{a_0} \right)^{\frac{3}{2}} e^{-\frac{Zr_1}{a_0}} \right\} \left\{ \frac{1}{\sqrt{\pi}} \left( \frac{Z}{a_0} \right)^{\frac{3}{2}} e^{-\frac{Zr_2}{a_0}} \right\} = \frac{Z^3}{\pi a_0^3} e^{-\frac{Z}{a_0}(r_1+r_2)}$$

Take it into the expectation of  $\hat{H}$ ,

$$\langle \hat{H} \rangle = \int d^3 \vec{x}_1 \int d^3 \vec{x}_2 \Psi^*(\vec{x}_1, \vec{x}_2) \hat{H} \Psi(\vec{x}_1, \vec{x}_2) = \frac{e^2 Z^2}{a_0} - \frac{4e^2 Z}{a_0} + \frac{5e^2 Z}{8a_0}$$

As the variational principle,  $\partial \langle \hat{H} \rangle / \partial Z = 0$ , then  $Z = 27/16 \approx 1.69$ , therefore the eigenvalue and eigen functon of ground state are

$$E_{\text{ground}} = \frac{e^2}{a_0} \left( Z^2 - 4Z + \frac{5}{8} Z \right) \cong -2.85 \left( \frac{e^2}{a_0} \right) \approx -77 \text{eV}$$
$$\Psi_{\text{ground}} = \left( \frac{27}{16} \right)^3 \frac{1}{\pi a_0^3} \exp \left\{ -\frac{27}{16} (r_1 + r_2) \right\}$$

- V.S. perturbation:  $E_{\text{ground}} \cong -2.75 \left( \frac{e^2}{a_0} \right) \approx -75 \text{eV}$

## 7.2 Scattering Theory

Background:

- Elastic collision:  $A + B \rightarrow A + B$ , potential scattering  $V(\vec{x}) = V(r)$
- Inelastic collision:  $A + B \rightarrow A^* + B$  (relativistic QM, QFT)
- Reaction:  $A + B \rightarrow C + D + E + \dots$

Quantities:

$$[\rho] = [\rho]^2 = [L]^{-3}, [\vec{j}] = [\rho][T]^{-1}[L] = [L]^{-2}[T]^{-1}$$

- Equation of continuity:  $\partial \rho / \partial t + \vec{\nabla} \cdot \vec{j} = 0$
- Probability density:  $\rho = \psi^* \psi$
- Probability current density:  $\vec{j} = \frac{\hbar}{2mi} \left( \psi^* \vec{\nabla} \psi - \text{c.c.} \right) = \text{Re} \left[ \psi^* \left( -\frac{i\hbar}{m} \vec{\nabla} \right) \psi \right]$

### SSE + Boundary condition

$$\left[ -\frac{\hbar^2}{2m} \nabla^2 + V(r) \right] \psi(\vec{x}) = E \psi(\vec{x}) \quad (E > 0)$$

The wave function can be written as

$$\psi(\vec{x}) = \psi_{\text{incident}} + \psi_{\text{scattering}}$$

- $\psi_{\text{incident}} = e^{ikz}$
- $\psi_{\text{scattering}} = \frac{e^{ikr}}{r} f(\theta, \phi) + \mathcal{O}\left(\frac{1}{r^2}\right)$ 
  - $\frac{e^{ikr}}{r}$ : ideal spherical wave function
  - $f(\theta, \phi)$ : scattering amplitude

### Probability current potential

When  $r \rightarrow \infty$ , the probability current density of scattering and incident is <sup>1</sup>

$$\begin{aligned} \vec{j}_{\text{sca}} &= \hat{\mathbf{e}}_r \left( \frac{\hbar k}{m} \right) \left| \frac{f(\theta, \phi)}{r} \right|^2 = \hat{\mathbf{e}}_r v \left| \frac{f(\theta, \phi)}{r} \right|^2 \\ \vec{j}_{\text{inc}} &= \hat{\mathbf{e}}_z \frac{\hbar k}{m} = v \hat{\mathbf{e}}_z \end{aligned}$$

Since  $\int_V \vec{\nabla} \cdot \vec{j} d^3 \vec{x} = \oint_S \vec{j} \cdot d\vec{S} = 0$ , then

$$\vec{j}_{\text{inc}} \cdot d\vec{\sigma}_{\text{in}} = \vec{j}_{\text{sca}} \cdot d\vec{\sigma}_{\text{sca}}$$

---

<sup>1</sup>Spherical coordinate:  $\vec{\nabla} = \hat{\mathbf{e}}_r \frac{\partial}{\partial r} + \hat{\mathbf{e}}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\mathbf{e}}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}$ ,  $\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$

- When  $r \rightarrow \infty$ ,  $\vec{\nabla} \rightarrow \hat{\mathbf{e}}_r \frac{\partial}{\partial r}$ ,  $\nabla^2 \rightarrow \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial}{\partial r} \right) + \mathcal{O}\left(\frac{1}{r^2}\right) = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r}$

Since  $d\vec{S} = \hat{n} \cdot r^2 d\Omega$ ,  $\int_{\delta\Omega} \vec{j}_{\text{sca}} d\vec{S} = \int_{\delta\Omega} \vec{j}_{\text{sca}} \cdot \hat{n} \cdot r^2 d\Omega = 0$ ,

$$\vec{j}_{\text{sca}} \cdot d\vec{\sigma}_{\text{sca}} = \int_{\delta\Omega} \frac{|f(\theta, \phi)|^2}{r^2} \cdot v \cdot r^2 d\Omega / \delta\Omega = v \hat{e}_z \hat{e}_z dr$$

We get differential scattering cross-section:

$$\sigma(\theta, \phi) = \frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2$$

Suppose the solution as

$$\psi(r, \theta) = \sum_{l=0}^{\infty} a_l P_l(\cos \theta) R_{nl}(r)$$

$$\bullet \text{ Plane wave: } e^{ikz} = e^{ikr \cos \theta} = \sum_{l=0}^{\infty} (2l+1) i^l j_l(kr) P_l(\cos \theta) \xrightarrow{k_r \rightarrow \infty} \sum_{l=0}^{\infty} \frac{(2l+1)}{2ikr} [(-1)^{l+1} e^{-ikr} + e^{ikr}] P_l(\cos \theta)$$

When  $k_r \rightarrow \infty$ ,

$$\begin{aligned} \psi(r, \theta) = \psi_{\text{inc}} + \psi_{\text{sca}} &\rightarrow \sum_{l=0}^{\infty} a_l \frac{(-i)^l e^{-2i\delta_l}}{ir} [(-1)^{l+1} e^{-ikr} + e^{ikr}] P_l(\cos \theta) \\ &+ \sum_{l=0}^{\infty} a_l \frac{(-i)^l e^{-i\delta_l}}{ir} (e^{i\delta_l} - 1) e^{ikr} P_l(\cos \theta) \\ &= e^{ikr} + \frac{e^{ikr}}{r} \sum_{l=0}^{\infty} \left( \frac{2l+1}{2ik} \right) (e^{2i\delta_l-1} - 1) P_l(\cos \theta) \end{aligned}$$

$$\text{Therefore } f(\theta) = \frac{1}{2ik} \sum_{l=0}^{\infty} (2l+1) (e^{2i\delta_l-1} - 1) P_l(\cos \theta) = \sum_{l=0}^{\infty} f_l(\theta)$$

- $l = 0$ : s-wave
- $l = 1$ : p-wave
- $l = 2$ : d-wave
- $\delta_l$ : phase-shift.  $l = 0$ , dominates;  $\delta_0 = -ka$ ,  $\delta_{l>1} \rightarrow (ka)^{2l+1} \rightarrow 0$

$$\delta_l = \frac{-(ka)^{2l+1}}{(2l-1)!!(2l+1)!!}$$

Then we can got the scattering cross-section

$$\sigma = \int_{\Omega} |f(\theta)|^2 d\Omega = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l = \sum_{l=0}^{\infty} \sigma_l$$

### Hard sphere scattering

$$V(r) = \begin{cases} \infty, & (r < a) \\ 0, & (r > a) \end{cases}$$

1.  $ka \ll 1$ ,  $\lambda_{\text{dB}} \gg a$  <sup>2</sup>

$$\text{Then } f(\theta) \cong -a, \frac{d\sigma}{d\Omega} = a^2, \sigma = \int_{\Omega} \frac{d\sigma}{d\Omega} d\Omega = 4\pi a^2,$$

Therefore, the biggest contribution is from  $l = 0$  potential wave (partial wave).

2.  $ka \gg 1$ ,  $\lambda_{\text{dB}} \ll a$

$$e^{2i\delta_l} = e^{-(ika - i(l+1)\pi)} \Rightarrow \delta_l = \frac{l\pi}{2} - ka,$$

???????????????? don't underst and

---


$$^2k = 2\pi/\lambda_{\text{dB}}$$

## 7.3 Perturbation Theory

### 7.3.1 Non-degenerate

If Harmiltonian can be written as

$$\hat{H}(\lambda) = \hat{H}^{(0)} + \lambda \delta \hat{H} \quad (\lambda \in [0, 1])$$

- $\hat{H}^{(0)}$ : origin Harmiltonian, and the origin SSE is

$$\hat{H}^{(0)} \left| \psi_n^{(0)} \right\rangle = E_n^{(0)} \left| \psi_n^{(0)} \right\rangle$$

- $\lambda$ : dimensionless and natural, artificial

and for non-degenerate, for applying perturbation theory, we need

1.  $\hat{H}^{(0)} \left| \psi_n^{(0)} \right\rangle = E_n^{(0)} \left| \psi_n^{(0)} \right\rangle$
2.  $\left| \delta \hat{H} \right| \ll \left| \hat{H}^{(0)} \right| \Rightarrow \begin{cases} \langle \psi_n^{(0)} | \delta \hat{H} | \psi_n^{(0)} \rangle \ll E_n^{(0)} \\ \langle \psi_n^{(0)} | \delta \hat{H} | \psi_n^{(0)} \rangle \ll \left| E_n^{(0)} - E_m^{(0)} \right| \end{cases}$

Consider such SSE states and its solution (target!)

$$\begin{cases} \hat{H}(\lambda) \left| \psi_n(\lambda) \right\rangle = E_n(\lambda) \left| \psi_n(\lambda) \right\rangle \\ \left| \psi_n(\lambda) \right\rangle = \sum_m C_m(\lambda) \left| \psi_m^{(0)} \right\rangle \end{cases} \quad \begin{cases} C_m(\lambda) = \lambda^0 C_m^{(0)} + \lambda^1 C_m^{(1)} + \lambda^2 C_m^{(2)} + \dots \\ E_n(\lambda) = \lambda^0 E_n^{(0)} + \lambda^1 E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \end{cases}$$

- or  $\left| \psi_n \right\rangle = \left| \psi_n^{(0)} \right\rangle + \sum_{m \neq n} C_m \left| \psi_m^{(0)} \right\rangle$
- $\delta \hat{H}_{nk} = \langle \psi_n^{(0)} | \delta \hat{H} | \psi_k^{(0)} \rangle$

Take the terms into SSE, then

$$\left( \hat{H}^{(0)} + \lambda \delta \hat{H} \right) \sum_m C_m(\lambda) \left| \psi_m^{(0)} \right\rangle = E_n(\lambda) \sum_m C_m(\lambda) \left| \psi_m^{(0)} \right\rangle$$

and multiply each side with  $\langle \psi_k^{(0)} |$ , we have

$$\begin{aligned} \text{l.h.s.} &= \left\langle \psi_k^{(0)} \left| \hat{H}^{(0)} \sum_m C_m(\lambda) \left| \psi_m^{(0)} \right\rangle \right\rangle + \left\langle \psi_k^{(0)} \left| \lambda \delta \hat{H} \sum_m C_m(\lambda) \left| \psi_m^{(0)} \right\rangle \right\rangle = E_k^{(0)} C_k(\lambda) + \sum_m \lambda \delta \hat{H}_{km} C_m(\lambda) \\ &= E_k^{(0)} \left( C_k^{(0)} + \lambda C_k^{(1)} + \lambda^2 C_k^{(2)} + \dots \right) + \lambda \sum_m \delta \hat{H}_{km} \left( C_m^{(0)} + \lambda C_m^{(1)} + \lambda^2 C_m^{(2)} + \dots \right) \\ &= E_k^{(0)} C_k^{(0)} + \lambda \left( E_k^{(0)} C_k^{(1)} + \sum_m \delta \hat{H}_{km} C_m^{(0)} \right) + \lambda^2 \left( E_k^{(0)} C_k^{(2)} + \sum_m \delta \hat{H}_{km} C_m^{(1)} \right) + \dots \\ \text{r.h.s.} &= \sum_m \left\langle \psi_k^{(0)} \left| \left( E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \right) \left( C_m^{(0)} + \lambda C_m^{(1)} + \lambda^2 C_m^{(2)} + \dots \right) \right| \psi_m^{(0)} \right\rangle \\ &= E_n^{(0)} C_k^{(0)} + \lambda \left( E_n^{(0)} C_k^{(1)} + E_n^{(1)} C_k^{(0)} \right) + \lambda^2 \left( E_n^{(2)} C_k^{(0)} + E_n^{(1)} C_k^{(1)} + E_n^{(0)} C_k^{(2)} \right) + \dots \end{aligned}$$

Then we got  $\lambda$  series

1.  $\mathcal{O}(\lambda^0)$ :  $E_k^{(0)} C_k^{(0)} = E_n^{(0)} C_k^{(0)} \rightarrow (E_n^{(0)} - E_k^{(0)}) C_k^{(0)} = 0 \Rightarrow \boxed{C_k^{(0)} = \delta_{nk}}$
2.  $\mathcal{O}(\lambda^1)$ :  $E_k^{(0)} C_k^{(1)} + \sum_m \delta \hat{H}_{km} C_m^{(0)} = E_n^{(0)} C_k^{(1)} + E_n^{(1)} C_k^{(0)}$ 
  - for  $k = n$ :  $\sum_m \delta \hat{H}_{nm} \delta_{nm} = E_n^{(1)} \delta_{nn} \Rightarrow \boxed{E_n^{(1)} = \delta \hat{H}_{nn}}$

- for  $k \neq n$ :  $E_k^{(0)} C_k^{(1)} + \sum_m \delta \hat{H}_{km} \delta_{nm} = E_n^{(0)} C_k^{(1)} \rightarrow (E_n^{(0)} - E_k^{(0)}) C_k^{(1)} = \delta \hat{H}_{kn} \Rightarrow$

$$C_k^{(1)} = \frac{\delta \hat{H}_{kn}}{E_n^{(0)} - E_k^{(0)}}$$

3.  $\mathcal{O}(\lambda^2)$ :  $E_k^{(0)} C_k^{(2)} + \sum_m \delta \hat{H}_{km} C_m^{(1)} = E_n^{(2)} C_k^{(0)} + E_n^{(1)} C_k^{(1)} + E_n^{(0)} C_k^{(2)}$

- for  $k = n$ :  $\sum_m \delta \hat{H}_{nm} C_m^{(1)} = \delta \hat{H}_{nn} C_n^{(1)} + E_n^{(2)} \Rightarrow E_n^{(2)} = \sum_{m \neq n} \frac{|\delta \hat{H}_{nm}|^2}{E_n^{(0)} - E_m^{(0)}}$
- for  $k \neq n$ :  $C_k^{(2)} = \sum_{m \neq n} \left[ \frac{\delta \hat{H}_{km} \delta \hat{H}_{mn}}{(E_n^{(0)} - E_k^{(0)})(E_n^{(0)} - E_m^{(0)})} - \frac{\delta \hat{H}_{mn} \delta \hat{H}_{kn}}{(E_n^{(0)} - E_k^{(0)})^2} \right]$  need to be checked.

1-st order and 2nd-order ( $\lambda = 1$ )

$$|\psi_n\rangle \approx |\psi_n^{(0)}\rangle + \sum_{k \neq n} \frac{\delta \hat{H}_{kn}}{E_n^{(0)} - E_k^{(0)}} |\psi_k^{(0)}\rangle, \quad E_n \approx E_n^{(0)} + \delta \hat{H}_{nn} + \sum_{m \neq n} \frac{|\delta \hat{H}_{nm}|^2}{E_n^{(0)} - E_m^{(0)}}$$

\*  $\delta \hat{H}_{nn}$  is diagonal; the others are off-diagonal

### Example of one-dim an harmonic oscillator

Review of the background:

$$\hat{H} = \hat{H}^{(0)} + \delta \hat{H}, \quad \hat{H}^{(0)} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2, \quad \hat{H}^{(0)} |\psi_n^{(0)}\rangle = E_n^{(0)} |\psi_n^{(0)}\rangle$$

the solution are

$$\begin{aligned} \hat{H}^{(0)} &= \hbar \omega \left( \hat{a}^\dagger \hat{a} + \frac{1}{2} \right); \quad E_n^{(0)} = \hbar \omega \left( n + \frac{1}{2} \right), \quad (n = 0, 1, 2, \dots) \\ |\psi_n^{(0)}\rangle &= |n\rangle \quad \text{Occupation number state, Fock state} \\ \hat{a}^\dagger \hat{a} |n\rangle &= \hat{N} |n\rangle = n |n\rangle; \quad \hat{a} |n\rangle = \sqrt{n} |n-1\rangle; \quad \hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle \\ \hat{a} &= \sqrt{\frac{m\omega}{2\hbar}} \hat{x} + \frac{i}{\sqrt{2m\hbar\omega}} \hat{p}; \quad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} - \frac{i}{\sqrt{2m\hbar\omega}} \hat{p} \Rightarrow \hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger) \end{aligned}$$

If there is small magnetic field come in,

$$\delta \hat{H} = -q\mathcal{E}\hat{x} = -\alpha_0 \hbar \omega (\hat{a} + \hat{a}^\dagger)$$

where  $\alpha_0 = \sqrt{\frac{m\omega}{2\hbar}} x_0$ ,  $x_0 = \frac{q\mathcal{E}}{m\omega^2}$ .

Since  $\langle n | \hat{a} | m \rangle = \sqrt{m} \delta_{n,m-1}$ ,  $\langle n | \hat{a}^\dagger | m \rangle = \sqrt{m+1} \delta_{n,m+1}$ ,

$$\begin{aligned} E_n^{(1)} &= \delta \hat{H}_{nn} = \langle \psi_n^{(0)} | \delta \hat{H} | \psi_n^{(0)} \rangle = \langle n | -\alpha_0 \hbar \omega (\hat{a} + \hat{a}^\dagger) | n \rangle \equiv 0 \\ \delta \hat{H}_{kn} &= \langle \psi_k^{(0)} | \delta \hat{H} | \psi_n^{(0)} \rangle = \langle k | -\alpha_0 \hbar \omega (\hat{a} + \hat{a}^\dagger) | n \rangle = (-\alpha_0 \hbar \omega) [\sqrt{n} \delta_{k,n-1} + \sqrt{n+1} \delta_{k,n+1}] \end{aligned}$$

Diagonal:  $\delta\hat{H}_{nn} \equiv 0$ ; off-diagonal:  $\delta\hat{H}_{kn} \neq 0$  ( $k = n \pm 1$ ). When  $k = n \pm 1$

$$E_k^{(2)} = \sum_{k \neq n} \frac{|\hat{H}_{nk}|^2}{E_n^{(0)} - E_k^{(0)}} = \frac{|\hat{H}_{n,n-1}|^2}{\hbar\omega} + \frac{|\hat{H}_{n,n+1}|^2}{-\hbar\omega} = \frac{(-\alpha_0\hbar\omega)^2 n}{\hbar\omega} + \frac{(-\alpha_0\hbar\omega)^2 (n+1)}{-\hbar\omega} = -\alpha_0^2 \hbar\omega$$

$$C_k^{(1)} = \frac{\delta\hat{H}_{kn}}{E_n^{(0)} - E_k^{(0)}} = \frac{(-\alpha_0\hbar\omega)(\sqrt{n}\delta_{k,n-1} + \sqrt{n+1}\delta_{k,n+1})}{(n-k)\hbar\omega}, \quad C_{n-1}^{(1)} = -\alpha_0\sqrt{n}, \quad C_{n+1}^{(1)} = \alpha_0\sqrt{n+1}$$

$$|\psi_n\rangle = |\psi_n^{(0)}\rangle + \sum_{k \neq n} C_k^{(1)} |\psi_k^{(0)}\rangle = |n\rangle + (-\alpha_0\sqrt{n}) |n-1\rangle + (\alpha_0\sqrt{n+1}) |n+1\rangle$$

$$\langle\hat{x}\rangle = \langle\psi_n|\hat{x}|\psi_n\rangle = [\langle n| + \alpha_0(\sqrt{n+1}\langle n+1| - \sqrt{n}\langle n-1|)] \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger) [|n\rangle + \alpha_0(\sqrt{n+1}|n+1\rangle - \sqrt{n}|n-1\rangle)] = \frac{q\mathcal{E}}{m\omega^2}$$

$$P = |q|\langle\hat{x}\rangle - (-|q|)\langle x\rangle = 2|q|\langle\hat{x}\rangle = \frac{2q^2\mathcal{E}}{m\omega} = \chi\mathcal{E} \quad (\text{Polarization})$$

Therefore, in summarize

$$E_n = E_n^{(0)} + E_n^{(1)} + E_n^{(2)} = \hbar\omega\left(n + \frac{1}{2}\right) - \frac{q^2\mathcal{E}^2}{2m\omega^2}$$

$$|\psi_n\rangle = |n\rangle + \alpha_0(\sqrt{n+1}|n+1\rangle - \sqrt{n}|n-1\rangle), \quad \text{where} \quad \alpha_0 = \sqrt{\frac{m\omega}{2\hbar}} \frac{q\mathcal{E}}{m\omega^2}$$

$$\hat{H} = \hat{H}^{(0)} + \delta\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right) - \hbar\omega\alpha_0(\hat{a} + \hat{a}^\dagger) = \hbar\omega\left[(\hat{a}^\dagger - \alpha_0)(\hat{a} - \alpha_0) + \frac{1}{2}\right] - \hbar\omega\alpha_0^2$$

### 7.3.2 Degenerate

For degenerate consideration, similarly, we have

1.  $\hat{H}(\lambda) = \hat{H}^{(0)} + \lambda\delta\hat{H}, \quad (\lambda \in [0, 1])$
2.  $\hat{H}^{(0)}|n\nu\rangle = E_n^{(0)}|n\nu\rangle, \quad (\nu = 1, 2, \dots, \nu_n) \quad * \nu_n \text{ is the degeneracy when in } n\text{'s eigenvalue}$
3. orthonormality+Completeness of  $\{|n\nu\rangle\}$ :

- $\langle n'\nu'|n\nu\rangle = \delta_{nn'}\delta_{\nu\nu'}$
- $\sum_{n\nu} |n\nu\rangle\langle n\nu| = \mathbb{1}$

And the SSE and states

$$\begin{cases} \hat{H}|\psi_k\rangle = E_k|\psi_k\rangle \\ |\psi_k\rangle = \sum_{m\mu} C_{m\mu} |m\mu\rangle \end{cases} \quad \begin{cases} C_{m\mu} = \lambda^0 C_{m\mu}^{(0)} + \lambda^1 C_{m\mu}^{(1)} + \lambda^2 C_{m\mu}^{(2)} + \dots \\ E_k = \lambda^0 E_k^{(0)} + \lambda^1 E_k^{(1)} + \lambda^2 E_k^{(2)} + \dots \end{cases}$$

Notes:

- $C_{m\mu} = \langle m\mu|\psi_k\rangle$
- $\delta\hat{H}_{m\mu,n\nu} = \langle m\mu|\delta\hat{H}|n\nu\rangle$
- $\nu, \mu$  means degeneracy

multiply the SSE with  $|m\mu\rangle$ , then the two side are

$$\begin{aligned}
\text{l.h.s.} &= \langle m\mu | \hat{H}_0 + \lambda \delta \hat{H} | \psi_k \rangle = E_m^{(0)} \langle m\mu | \psi_k \rangle + \sum_{n\nu} \langle m\mu | \lambda \delta \hat{H} | n\nu \rangle \langle n\nu | \psi_k \rangle = E_m^{(0)} C_{m\mu} + \sum_{n\nu} \lambda \delta \hat{H}_{m\mu, n\nu} C_{n\nu} \\
&= E_m^{(0)} \left( \lambda^0 C_{m\mu}^{(0)} + \lambda^1 C_{m\mu}^{(1)} + \dots \right) + \sum_{n\nu} \lambda \delta \hat{H}_{m\mu, n\nu} \left( \lambda^0 C_{n\nu}^{(0)} + \lambda^1 C_{n\nu}^{(1)} + \dots \right) \\
&= E_m^{(0)} C_{m\mu}^{(0)} + \lambda \left( E_m^{(0)} C_{m\mu}^{(1)} + \sum_{n\nu} \delta \hat{H}_{m\mu, n\nu} C_{n\nu}^{(0)} \right) + \dots \\
\text{r.h.s.} &= \langle m\mu | E_k | \psi_k \rangle = E_k C_{m\mu} = \left( \lambda^0 E_k^{(0)} + \lambda^1 E_k^{(1)} + \dots \right) \left( \lambda^0 C_{m\mu}^{(0)} + \lambda^1 C_{m\mu}^{(1)} + \dots \right) \\
&= E_k^{(0)} C_{m\mu}^{(0)} + \lambda \left( E_k^{(0)} C_{m\mu}^{(1)} + E_k^{(1)} C_{m\mu}^{(0)} \right) + \dots
\end{aligned}$$

Then we got  $\lambda$  series:

$$\begin{aligned}
1. \mathcal{O}(\lambda^0): \quad E_m^{(0)} C_{m\mu}^{(0)} &= E_k^{(0)} C_{m\mu}^{(0)} \Rightarrow \boxed{C_{m\mu}^{(0)} = a_\mu \delta_{km}} \\
2. \mathcal{O}(\lambda^1): \quad E_m^{(0)} C_{m\mu}^{(1)} + \sum_{n\nu} \delta \hat{H}_{m\mu, n\nu} C_{n\nu}^{(0)} &= E_k^{(0)} C_{m\mu}^{(1)} + E_k^{(1)} C_{m\mu}^{(0)} \text{ some mistakes here...} \\
\bullet \text{ for } k = m: \quad \sum_{\nu} \delta \hat{H}_{k\mu, n\nu} a_\nu - E_k^{(1)} a_\mu &= 0, \text{ let } \delta \hat{H}_{k\mu, k\nu} = \delta \hat{H}_{\mu\nu} \Rightarrow \sum_{\mu\nu} \left( \delta \hat{H}_{\mu\nu} - E_k^{(1)} \delta_{\mu\nu} \right) a_\nu = 0, \\
\text{therefore} \quad \det \left| \sum_{\mu\nu} \left( \delta \hat{H}_{\mu\nu} - E_k^{(1)} \delta_{\mu\nu} \right) \right|_{\nu_n \times \nu_n} &= 0 \tag{7.1}
\end{aligned}$$

We can got  $E_{k\alpha}^{(1)}$ , ( $\alpha = 1, \dots, \alpha_k$ )

- $\alpha_k = \nu_k$ , then degeneracy completely lifted
- $\alpha_k = 1$ , then degeneracy not broken
- $1 < \alpha_k < \nu_k$ , then partially lifted

Finally, we can got the state in zero-th order, and engery in first order

$$|\psi_{k\alpha}\rangle = \sum_{\alpha} a_{\mu\alpha} |ka\rangle$$

## 7.4 Application to Perturbation

### 7.4.1 Stark Effect

Phenomenon: when there is electric field

$$\vec{E} = \mathcal{E} \hat{e}_z$$

come in to the Hydrogen atom. Normally,  $n = 2, l = 0 (m = 0), l = 1 (m = 0, \pm 1)$ , for four energy line, however, there only three. Denote four state:

$$\begin{aligned}
2S: \quad |200\rangle &\rightarrow |1\rangle; \\
2P_0: \quad |210\rangle &\rightarrow |2\rangle; \quad 2P_1: \quad |211\rangle \rightarrow |3\rangle; \quad 2P_{-1}: \quad |2, 1, -1\rangle \rightarrow |4\rangle;
\end{aligned}$$

Because for degeneracy,  $\nu_n = 4$  Normally SSE is

$$\hat{H}^{(0)} |2lm\rangle = E_2^{(0)} |2lm\rangle, \quad E_n = -\frac{e^2}{2a_0} = -12.6\text{eV}/n^2$$

Then, adding perturbation

$$\lambda \delta \hat{H} = e\mathcal{E}r \cos z = \left( \frac{e\mathcal{E}a_0}{e^2/a_0} \right) \left( \frac{e^2}{a_0^2} r \cos \theta \right) \Rightarrow \lambda = \frac{W_{\text{ext}}}{W_{\text{col}}} \ll 1$$

1.  $W_{\text{ext}} = e\mathcal{E}a_0$ : energy of electric field
2.  $W_{\text{col}} = e^2/a_0$ : energy of self



3.  $e^2/a_0^2$ : Column force

Ask  $\delta H_{\mu\nu}$  ( $\nu, \mu = 1, 2, 3, 4$ ) ( $\delta \hat{H}_{\mu\nu} = \delta \hat{H}_{k\mu, k\nu}$ ,  $k = 2$ ) a  $4 \times 4$  matrix.

Since  $[\hat{L}_z, \hat{z}] = 0$  which means  $\hat{L}_z, \hat{z}$  have simultaneous eigenvalue, and suppose  $|lm\rangle$  is eigenstate of  $\hat{z}$ , then

$$\langle lm' | \hat{z} | lm \rangle \propto \delta_{m'm}$$

which means a selection rule for  $m$ :

$$\Delta m = m - m' = 0$$

Similarly,

$$\langle lm | \delta \hat{H} | lm' \rangle \sim \delta_{l+1, l'} \Rightarrow \Delta l = l' - l = \pm 1$$

Since  $z = r \cos \theta \sim Y_{10}(\theta, \phi)$ , we have already the solution is

$$\cos \theta Y_{lm}(\theta, \phi) \propto a \cdot Y_{l+1, m}(\theta, \phi) + b \cdot Y_{l-1, m}(\theta, \phi) \quad \text{where} \quad Y_{lm}(\theta, \phi) = |lm\rangle$$

Hence,

$$\langle l'm | \cos \theta | lm \rangle \sim a \langle l' | l+1 \rangle + \langle l' | l-1 \rangle$$

Selection rule for  $l$  and  $m$  on  $\hat{z}$ :

$$\Delta l = \pm 1, \quad \Delta m = 0$$

Therefore,  $|1\rangle$  and  $|2\rangle$  have degenerated. NEED PUT PICTURE, NOT SURE THE RESULTS.

$$\langle 1 | \lambda \delta \hat{H} | 2 \rangle = \langle 2 | \lambda \delta \hat{H} | 1 \rangle \quad \text{and} \quad \langle 200 | \lambda \delta \hat{H} | 210 \rangle = \int \psi_{200}^*(\theta, \phi) \lambda \delta \hat{H} \psi_{210}(\theta, \phi) r^2 dr \sin \theta d\phi = -3ea_0\mathcal{E}$$

Then from Eq.(7.1), we have

$$\begin{vmatrix} -E_2^{(1)} & -3ea_0\mathcal{E} & 0 & 0 \\ -3ea_0\mathcal{E} & -E_2^{(1)} & 0 & 0 \\ 0 & 0 & -E_2^{(1)} & 0 \\ 0 & 0 & 0 & -E_2^{(1)} \end{vmatrix} = 0 \Rightarrow E_2^{(1)} = 0, \pm 3a_0e\mathcal{E}$$

Therefore, for we have the state and the corresponding energy are

$$|\psi_k\rangle = \begin{cases} |211\rangle, |21-1\rangle & \rightarrow E_2^{(0)} + 0 \\ \frac{1}{\sqrt{2}}(|200\rangle + |210\rangle) & \rightarrow E_2^{(0)} - 3ea_0\mathcal{E} \\ \frac{1}{\sqrt{2}}(|200\rangle - |210\rangle) & \rightarrow E_2^{(0)} + 3ea_0\mathcal{E} \end{cases}$$

\* The degeneracy is from multi observables have communicated with  $\hat{H}$ , then the observable is conserved and have multi same eigenstates. The perturbation break the symmetry, and dismiss degeneracy, and revised.

### 7.4.2 Charged Particles in Magnetic Field

Review the classical electromagnetic theory, we know that the Hamilton's canonical equations of motion can get Newton-Lorentz's equation: <sup>3</sup>

$$H = \frac{1}{2m} \left( \vec{p} - \frac{q}{c} \vec{A} \right)^2 + q\phi \Rightarrow \begin{cases} m\ddot{\vec{x}} = q \left( \vec{E} + \frac{\vec{v}}{c} \times \vec{B} \right), & A_i = A_i(\vec{x}, t) \\ \vec{E} = -\vec{\nabla}\phi - \frac{1}{c} \frac{d}{dt} \vec{A}, \quad \vec{B} = \vec{\nabla} \times \vec{A}. & \phi = \phi(\vec{x}, t) \end{cases}$$

*Proof.* Since  $H = \frac{1}{2m} \left( p_i - \frac{q}{c} A_i \right) \left( p_i - \frac{q}{c} A_i \right) + q\phi$ , and  $\dot{A}_i = \frac{\partial A_i}{\partial t} + \frac{\partial A_i}{\partial x_j} \frac{dx_j}{dt} = \partial_t A_i + v_j \partial_j A_i$

<sup>3</sup>Einstein convention is used.

then  $\dot{x}_i = \frac{\partial H}{\partial p_i} = \frac{1}{m} \left( p_i - \frac{q}{c} A_i \right) = v_i$ ,  $\dot{p} = -\frac{\partial H}{\partial x_i} = -\frac{1}{m} \left( p_j - \frac{q}{c} A_j \right) \left( -\frac{q}{c} \frac{\partial A_j}{\partial x_i} \right) - q \partial_i \phi = -q \partial_i \phi + \frac{q}{c} v_j \partial_i \dot{A}_j$ ,  
therefore

$$\begin{aligned} m\ddot{x}_i &= \dot{p}_i - \frac{q}{c} \dot{A}_i = -q \partial_i \phi + \frac{q}{c} v_j \partial_i A_j - \frac{q}{c} \partial_t A_i - \frac{q}{c} v_i \partial_j A_j \\ &= q \left( -\partial_i \phi - \frac{1}{c} \partial_t A_i \right) + \frac{q}{c} v_j (\partial_i A_j - \partial_j A_i) = q E_i + q \left( \frac{\vec{v} \times \vec{B}}{c} \right)_i \end{aligned}$$

since  $\left( \vec{v} \times \vec{B} \right)_i = \varepsilon_{ijk} v_j B_k = \varepsilon_{ijk} v_j \varepsilon_{klm} \partial_l A_m = v_j \partial_i A_j - v_j \partial_j A_i$  □

### Non-Relativistic QM (NO-SPIN)

Since  $[\hat{p}_i, A_j(\hat{x})] = [\hat{p}_i, \hat{x}_\alpha] \partial_\alpha A_j = -i\hbar \partial_i A_j$ , then

$$\hat{p}_i A_i = A_i \hat{p}_i - i\hbar \partial_i A_i \quad \Rightarrow \quad \hat{\vec{p}} \cdot \vec{A} = \vec{A} \cdot \hat{\vec{p}} - i\hbar \vec{\nabla} \cdot \vec{A}$$

Use Columb gauge<sup>4</sup>:  $\vec{\nabla} \cdot \vec{A} = 0$ , then  $\hat{\vec{p}} \cdot \vec{A} = \vec{A} \cdot \hat{\vec{p}}$ . And radiation gauge:  $\phi = 0$ , consider  $\left| \frac{q^2}{c^2} \vec{A}^2 \right| \ll \left| \frac{q}{c} \vec{A} \cdot \hat{\vec{p}} \right|$   
then expand  $\hat{H}$ ,

$$\hat{H} = \frac{1}{2m} \left( \hat{\vec{p}}^2 - \frac{q}{c} \hat{\vec{p}} \cdot \vec{A} - \frac{q}{c} \vec{A} \cdot \hat{\vec{p}} + \frac{q^2}{c^2} \vec{A}^2 \right) + q\phi = \frac{\hat{\vec{p}}^2}{2m} - \frac{q}{mc} \vec{A} \cdot \hat{\vec{p}}$$

Assume  $\vec{B} = \text{const}$ , and  $\vec{A} = \frac{1}{2} \vec{B} \times \vec{x}$ , then

$$\begin{aligned} \vec{\nabla} \times \vec{A} &= \frac{1}{2} \varepsilon_{ijk} \partial_j \varepsilon_{klm} B_l x_m = \frac{1}{2} [\partial_j (B_i x_j) - \partial_j (B_j x_i)] = \frac{1}{2} (B_i \delta_{jj} - B_j \delta_{ij}) = B_i \\ \vec{\nabla} \cdot \vec{A} &= \frac{1}{2} \partial_i \varepsilon_{ijk} B_j x_k = \frac{1}{2} \varepsilon_{ijk} B_j \delta_{ik} = \frac{1}{2} \varepsilon_{iji} B_j \equiv 0 \quad (\text{satisfy Columb gauge}) \\ \hat{H}_M &= -\frac{q}{mc} \vec{A} \cdot \hat{\vec{p}} = -\frac{q}{mc} \frac{1}{2} (\vec{B} \times \hat{\vec{x}}) \cdot \hat{\vec{p}} = -\frac{q}{mc} (\hat{\vec{x}} \times \hat{\vec{p}}) \cdot \vec{B} = -\frac{q}{2mc} \hat{\vec{L}} \times \vec{B} = -\hat{\vec{\mu}}_L \end{aligned}$$

- Orbital magnetic moment:  $\vec{\mu}_L = \frac{q}{2mc} \vec{L} = \frac{e}{2mc} \vec{L} = \mu_B \left( \frac{\vec{L}}{\hbar} \right)$ ,  $q = -e$
- Bohr magneton:  $\mu_B = \frac{e\hbar}{2mc}$

Magnetic Harmiltonian of charged partical without spin in magnetic field and its  $z$ -direction:

$$\hat{H}_M = -\frac{q}{2mc} \hat{\vec{L}} \times \vec{B} = -\vec{\mu}_L \cdot \vec{B}, \quad \hat{H}_{M_z} = \mu_B B \left( \frac{L_z}{\hbar} \right)$$

Order of magnitude analysis:

1.  $\hat{H}_E = -q \vec{d} \cdot \vec{E} = ezE$ ,  $\hat{H}_M = \mu_B B (L_z/\hbar) \sim \mu_B B$ ,  
So, if  $|E| \sim |B|$ ,  $\frac{|H_M|}{|H_E|} \sim \frac{\mu_B}{ea_0} = \frac{e\hbar}{2mc} \cdot \frac{1}{ea_0} \sim \frac{\hbar}{mc} \frac{1}{a_0} \sim \frac{\lambda}{a_0} = \alpha = \frac{1}{137}$ 
  - Compton wavelength:  $\lambda = \hbar/mc$ ;  $\alpha$  fine structure constant
2.  $\frac{q^2}{c^2} \vec{A}^2 \sim \frac{q^2}{c^2} (\vec{B} \times \vec{x})^2 \sim \frac{q^2}{c^2} B^2 r^2 = \frac{e^2}{c^2} B^2 a_0^2$   
 $\frac{\left| \frac{q^2}{c^2} \vec{A}^2 \right|}{\frac{q}{mc} \vec{A} \cdot \hat{\vec{p}}} \sim \left( \frac{e^2}{\hbar c} \right) \frac{B a_0^2}{4e} \sim \alpha \left( \frac{B}{e/a_0^2} \right) = \frac{1}{137} \cdot \delta$ 
  - even if  $B < 10^5 \text{Gs}$  very big, it is  $\sim 10^{-4}$
3.  $\vec{\mu}_L = \frac{q}{2mc} \vec{L} \cong g_L \mu_B \frac{\vec{L}}{\hbar} \Leftrightarrow \vec{\mu}_L = \frac{q}{mc} \vec{S} \cong g_S \mu_B \frac{\vec{S}}{\hbar}$  (intrinsic magnetic moment)
  - $g_L \cong 1$ ,  $g_S \cong 2$

<sup>4</sup>Gauge can be chose for convenience:  $\vec{A} \rightarrow \vec{A}' = \vec{A} + \vec{\nabla}\chi$ ,  $\phi \rightarrow \phi' = \phi - \frac{1}{c} \partial_t \chi \Rightarrow \vec{E}' = \vec{E}$ ,  $\vec{B}' = \vec{B}$

**Relativistic QM (SPIN)**

From scalar to spinor, then  $\hat{H} = \frac{\hat{p}^2}{2m} \cdot \mathbb{1} = \frac{1}{2m} (\hat{\sigma} \cdot \hat{p})^2$ , in Magnetic field ( $\phi = 0$ )

$$\hat{H} = \frac{1}{2m} \left[ \hat{\sigma} \cdot \left( \hat{p} - \frac{q}{c} \vec{A} \right) \right]^2 = \frac{1}{\pi} (\hat{\sigma} \cdot \hat{\pi})^2, \quad \hat{\pi} = \hat{p} - \frac{q}{c} \vec{A}$$

Since Pauli matrix  $(\vec{\sigma} \cdot \vec{A})(\vec{\sigma} \cdot \vec{B}) = (\vec{A} \cdot \vec{B})\mathbb{1}_2 + i\vec{\sigma}(\vec{A} \times \vec{B})$  and  $\vec{S} = \frac{\hbar}{2}\vec{\sigma}$ , then

$$\hat{H} = \frac{1}{2m} \left[ \hat{\pi}^2 + i\hat{\sigma} \cdot (\hat{\pi} \times \hat{\pi}) \right] = \frac{\hat{p}^2}{2m} - \frac{q}{2mc} \hat{L} \cdot \vec{B} - \frac{q}{mc} \vec{S} \cdot \vec{B}$$

*Proof.*  $(\hat{\pi} \times \hat{\pi})_i = \varepsilon_{ijk} \hat{\pi}_j \hat{\pi}_k = \frac{1}{2} \varepsilon_{ijk} [\hat{\pi}_j, \hat{\pi}_k] = -\frac{1}{2} \varepsilon_{ijk} \frac{q}{c} ([\hat{p}_j, \hat{A}_k] + [\hat{A}_j, \hat{p}_k]) = \frac{i\hbar q}{2c} \varepsilon_{ijk} (\partial_j A_k - \partial_k A_j) = \frac{i\hbar q}{c} \vec{B}$   $\square$

Introduce

- $\hat{\mu}_L = \frac{q}{2mc} \hat{L} = \frac{-e}{2mc} \hat{L} = -\mu_B \left( \frac{\hat{L}}{\hbar} \right)$
- $\hat{\mu}_S = \frac{q}{mc} \vec{S} = \frac{-e}{mc} \vec{S} = -2\mu_B \left( \frac{\hat{S}}{\hbar} \right)$

Magnetic Hamiltonian of charged particle with spin in magnetic field and its  $z$ -direction:

$$\hat{H}_M = -(\hat{\mu}_L + \hat{\mu}_S) \cdot \vec{B}, \quad \hat{H}_{M_z} = \mu_B B \left( \frac{\hat{L}_z + 2\hat{S}_z}{\hbar} \right)$$

Note:

- $\mu_B$  is energy dimension;  $(\hat{L}_z + 2\hat{S}_z)/\hbar$  is dimensionless
- $[\hat{\pi}_i, \hat{\pi}_j] = \frac{iq\hbar}{c} (\partial_i - \partial_j) A_k = \frac{iq\hbar}{c} \varepsilon_{ijk} B_k \Leftrightarrow \vec{\pi} \times \vec{\pi} = \frac{iq\hbar}{c} \vec{B}$
- $\hat{v}_i = \frac{\hat{\pi}_i}{m} \Rightarrow \hat{v} \times \hat{v} = \frac{iq\hbar}{m^2 c} \vec{B} \Leftrightarrow [\hat{v}_i, \hat{v}_j] = \frac{iq\hbar}{m^2 c} \varepsilon_{ijk} B_k$
- $[\hat{v}_i, \hat{v}^2] = \frac{iq\hbar}{m^2 c} \varepsilon_{ijk} (\hat{v}_j B_k - B_j \hat{v}_k) \Leftrightarrow [\vec{v}, \hat{v}^2] = \frac{iq\hbar}{m^2 c} (\hat{v} \times \vec{B} - \vec{B} \times \hat{v})$

**7.4.3 Zeeman Effect**

1. Phenomenon: 1896

2. QM interpretation:

- (a) Normal Zeeman: Classical Mechanics; Non-relativistic QM
- (b) Anomalous Zeeman: Relativistic QM; Spin: fine structure

**Theoretical analysis of normal Zeeman effect**

Consider  $\vec{B} = B \vec{e}_z$ ,  $\hat{H}_M = \frac{\mu_B B \hat{L}_z}{\hbar}$ , and use perturbation theory

$\hat{H}^{(0)} = \frac{\hat{p}^2}{2m} - \frac{e^2}{r} \rightarrow$  eigenstate is  $|nlm_l\rangle$ , satisfy  $\hat{H}_M \ll E_n^{(0)}$

since CSCO of  $(\hat{H}^{(0)}, \hat{L}^2, \hat{L}_z)$ , then

$$\langle nlm_l | \hat{H}_M | nlm_l \rangle = \frac{\mu_B B}{\hbar} \langle nlm_l | \hat{L}_z | nlm_l \rangle = \mu_B B m_l \Rightarrow E_{nlm_l} = E_n^{(0)} + \mu_B B m_l$$

**Theoretical analysis of anormalous Zeeman effect**

Review Hydrogen atom, eigen state is  $|nlm_l s_l\rangle$ , CSSO  $\left(\hat{H}^{(0)}, \hat{L}^2, \hat{L}, \hat{L}_z, \hat{S}^2, \hat{S}_z\right)$ , good quantum number  $(n, l, m_l, s, m_s)$

For fine structure analysis:

$$\hat{H}^{(0)} = \frac{\hat{p}^2}{2m} - \frac{e^2}{r} + a_r \hat{p}^4 + a_{L-S} \xi(r) \hat{\vec{S}} \cdot \hat{\vec{L}} + \delta \hat{H}$$

\*  $a_r \hat{p}^4$ : relativistic revise

\*  $l = 0 \Rightarrow \delta \hat{H} = 0$

- Eigen state  $|nljm_j\rangle$ , CSSO  $\left(\hat{H}^{(0)}, \hat{L}^2, \hat{S}^2, \hat{J}^2, \hat{J}_z\right)$ , good quantum number  $(n, l, s, j, m_j)$

$$\begin{aligned} \text{Proof. } [\hat{L}_i, V(r)] = 0 &\Rightarrow [\hat{L}_z, \xi(r)] = 0, \quad [\hat{L}_i, \hat{\vec{L}} \cdot \hat{\vec{S}}] = [\hat{L}_i, \hat{L}_j] \hat{S}_j = i\hbar \varepsilon_{ijk} \hat{L}_k \hat{S}_j \neq 0 \\ [\hat{L}^2, \hat{\vec{L}} \cdot \hat{\vec{S}}] &= [\hat{L}^2, \hat{L}_z] \hat{S}_z = 0, \quad [\hat{S}^2, \hat{\vec{L}} \cdot \hat{\vec{S}}] = [\hat{S}^2, \hat{S}_z] \hat{L}_z = 0 \\ [\hat{J}_i, \hat{\vec{L}} \cdot \hat{\vec{S}}] &= [\hat{L}_i, \hat{L}_j \hat{S}_j] + [\hat{S}_i, \hat{L}_j \hat{S}_j] = i\hbar \varepsilon_{ijk} (\hat{L}_k \hat{S}_j + \hat{L}_j \hat{S}_k) = 0 \end{aligned}$$

□

$$|jm\rangle = \sum_{m_1+m_2=m} |j_1 m_1 j_2 m_2\rangle \langle j_1 m_1 j_2 m_2 | jm\rangle = \sum_{m_1+m_2=m} |lm_l sm_s\rangle \langle lm_l sm_s | jm\rangle$$

- $m = -j, \dots, j, \quad j = |l-s|, \dots, |l+s|$

Then eigen state  $|nljm_j\rangle \rightarrow \left(\hat{\vec{J}}^2, \hat{J}_z\right)$  for  $\hat{H}_M = \mu_B B \left(\hat{J}_z + \hat{S}_z\right)/\hbar$

$$\langle nljm_j | \hat{H}_M | nljm_j \rangle = \mu_B B \hbar m_j + \langle nljm_j | \hat{S}_z | nljm_j \rangle$$

From projection theorem (Eq: 7.3),  $\forall \vec{v}$  in arbitrary vector, let  $\vec{v} = \vec{S}$ , dot product  $\hat{\vec{e}}_z$  in both sides, yields  $\vec{v} \cdot \hat{\vec{e}}_z = \hat{S}_z$ ,  $\vec{J} \cdot \hat{\vec{e}}_z = \hat{J}_z$

$$\langle jm | \hat{S}_z | jm \rangle = \frac{\langle jm | (\vec{S} \cdot \vec{J}) \vec{J} | jm \rangle}{\hbar^2 j(j+1)} = \frac{m_j \langle jm | \hat{\vec{S}} \cdot \vec{J} | jm \rangle}{\hbar j(j+1)}$$

Since  $[\hat{J}_i, \hat{S}_i] = [\hat{S}_i, \hat{S}_i] = 0$ ,  $\hat{L}^2 = \hat{J}^2 + \hat{S}^2 - 2\hat{\vec{S}} \cdot \hat{\vec{L}} \Rightarrow \hat{\vec{S}} \cdot \hat{\vec{L}} = \frac{1}{2}(\hat{J}^2 + \hat{S}^2 - \hat{L}^2)$ , then

$$\langle jm | \hat{\vec{S}} \cdot \vec{J} | jm \rangle = \frac{\hbar^2}{2} [j(j+1) + s(s+1) + l(l+1)] = \frac{\hbar^2}{2} \left[ j(j+1) + \frac{3}{4} + l(l+1) \right]$$

The energy is

$$\Delta E = \langle nljm_j | \hat{H}_M | nljm_j \rangle = m_j \left[ 1 + \frac{j(j+1) + \frac{3}{4} + l(l+1)}{2j(j+1)} \right] \mu_B B = \mu_B B m_j g_j$$

Landé  $g$  factor of an atomic level:

$$g_j = 1 + \frac{j(j+1) + l(l+1) + \frac{3}{4}}{2j(j+1)} \quad (7.2)$$

Projection Theorem:

$$\langle jm | \vec{v} | jm \rangle = \frac{\langle jm | (\vec{v} \cdot \vec{J}) \vec{J} | jm \rangle}{\langle jm | \vec{J}^2 | jm \rangle} = \frac{\langle jm | (\vec{v} \cdot \vec{J}) \vec{J} | jm \rangle}{\hbar^2 j(j+1)} \quad (7.3)$$

*Proof.* Since  $[\hat{J}_i, \hat{v}_j] = i\hbar \varepsilon_{ijk} \Leftrightarrow \vec{J} \times \vec{v} + \vec{v} \times \vec{J} = 2i\hbar \vec{v} \Rightarrow [\vec{J}^2, \vec{v}] = 2i\hbar (\vec{v} \times \vec{J} - i\hbar \vec{v}) \Rightarrow \frac{1}{(2i\hbar)^2} [\vec{J}^2, [\vec{J}^2, \vec{v}]] = \left( (\vec{v} \times \vec{J}) \cdot \vec{J} - \frac{1}{2} (\vec{v} \cdot \vec{J}^2 + \vec{J}^2 \cdot \vec{v}) \right); \quad \langle jm | \vec{J}^2 = \hbar^2 j(j+1) \langle jm |$

$$\begin{aligned} \langle jm | [\hat{J}^2, [\hat{J}^2, \hat{v}]] | jm \rangle &= \langle jm | \hat{J}^2 [\hat{J}^2, \hat{v}] - [\hat{J}^2, \hat{v}] \hat{J}^2 | jm \rangle \equiv 0 \quad \Rightarrow \quad [\hat{J}^2, [\hat{J}^2, \hat{v}]] \equiv 0, \text{ therefore} \\ \langle jm | (\hat{v} \cdot \hat{J}) \hat{J} | jm \rangle &= \frac{1}{2} \langle jm | \hat{J}^2 + \hat{v} \hat{J}^2 | jm \rangle = \frac{1}{2} \cdot 2 \langle jm | \vec{v} | jm \rangle j(j+1) \end{aligned}$$

□

????

$$\langle nlm_l m_s | \hat{H}_M | nlm_l m_s \rangle = \mu_B B (m_l + 2m_s)$$

- $m_l + 2m_s$  overlap

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Thanks for warching. You are your best!